



AI-103

Azure AI Apps and Agents Developer Associate

Certification Exam Prep



175 Questions

with Detailed Explanations for each Answer



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Microsoft Foundry & Agent Service



Generative AI, RAG & Prompt Engineering



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CLOUD



AI



BUSINESS

Q1. You are building a speech processing solution in Microsoft Foundry for a customer support platform. The platform will transcribe live phone calls, so that supervisors at your company can view call transcripts and detect issues while the calls are in progress. The call audio will arrive as a continuous stream from the telephony system.

You need to ensure that the call transcripts appear within only a few seconds of the audio stream.

What should you do?

- A. Use text to speech by using a custom neural voice.
- B. Use speech translation to generate the transcripts into multiple languages.
- C. Run a batch transcription job on recorded audio files.
- D. Use real-time speech to text to process streaming audio input.

Answer:

D. Use real-time speech to text to process streaming audio input.

Explanation:

Correct Answer:

D. Use real-time speech to text to process streaming audio input.

Real-time speech to text is specifically designed to transcribe streaming audio with low latency, making it ideal for live scenarios such as customer support phone calls. As audio is received from the telephony system, the Speech service continuously converts speech into text, allowing supervisors to view transcripts within seconds and monitor conversations while they are still in progress. This capability supports live captioning, real-time monitoring, and immediate issue detection, which are key requirements of the scenario.

Incorrect Answers:

A. Use text to speech by using a custom neural voice.

Text to speech converts text into spoken audio. It does not transcribe audio into text.

B. Use speech translation to generate the transcripts into multiple languages.

Speech translation is intended for translating spoken language into another language. The requirement is to generate live transcripts, not translations.

C. Run a batch transcription job on recorded audio files.

Batch transcription processes completed audio recordings and is not suitable for generating transcripts during live phone calls.

Q2. You have a Microsoft Foundry project that contains an agent.

The agent ingests scanned PDF vendor invoices that contain tables and embedded QR codes.

The agent must preserve the PDF layout in the extracted output to ensure that downstream processing can reference sections and tables.

You plan to call Azure Content Understanding in Foundry Tools.

You need to extract content and layout elements and detect QR codes without requiring a language model deployment.

Which built-in analyzer should you use?

- A. prebuilt-documentFieldSchema

- B. prebuilt-read
- C. prebuilt-documentSearch
- D. prebuilt-layout

Answer:

D. prebuilt-layout

Explanation:

Correct Answer:

D. prebuilt-layout

The **prebuilt-layout** analyzer is designed to extract the structural elements of a document while preserving its layout. It identifies text, paragraphs, tables, selection marks, and other layout components, enabling downstream processes to reference the original document structure accurately. It also supports the detection of barcodes and QR codes without requiring a language model deployment. This makes it the best choice for processing scanned PDF vendor invoices that contain tables and embedded QR codes while maintaining the document's layout.

Incorrect Answers:

A. prebuilt-documentFieldSchema

This analyzer focuses on extracting predefined fields from documents rather than preserving the complete document layout or detecting QR codes.

B. prebuilt-read

The **prebuilt-read** analyzer is intended primarily for OCR and text extraction. It does not provide the rich layout analysis needed for preserving document structure.

C. prebuilt-documentSearch

This analyzer is intended for document search and retrieval scenarios rather than extracting layout elements and QR codes for downstream processing.

Q3. You have a Microsoft Foundry project that contains an agent.

The agent accepts user-uploaded screenshots and uses a multimodal chat model. Some screenshots contain potentially malicious embedded text.

You need to prevent a prompt injection attack and ensure that third-party content is treated as lower trust.

How should you configure prompt shields for document attacks?

Prompt shields action:

▼

Disable the shield.
 Set action to block.
 Set action to annotate.

Additional mitigation:

▼

Enable Spotlighting.
 Create a custom blocklist.
 Use optical character recognition (OCR) to extract the text from the images first.

Answer:

Prompt shields action:

▼

Disable the shield.
Set action to block.
 Set action to annotate.

Additional mitigation:

▼

Enable Spotlighting.
 Create a custom blocklist.
 Use optical character recognition (OCR) to extract the text from the images first.

Explanation:

Correct Answers:

Prompt shields action: Set action to block.

Setting the prompt shield action to **block** ensures that any detected prompt injection attack in uploaded screenshots or documents is prevented from reaching the multimodal language model. Since the screenshots may contain malicious embedded text, blocking the request provides the highest level of protection by stopping the attack before it can influence the model's behavior.

Additional mitigation: Enable Spotlighting.

Spotlighting is designed to help the model distinguish between trusted system instructions and untrusted third-party content, such as uploaded screenshots and documents. It marks external content as lower trust, reducing the likelihood that malicious instructions embedded within the content will be followed by the model. This is the recommended mitigation for defending against indirect prompt injection attacks.

Incorrect Answers:

Disable the shield.

Disabling the shield removes protection against prompt injection attacks and leaves the application vulnerable.

Set action to annotate.

Annotate detects and flags potential prompt injection attempts but still allows the request to continue. Since the requirement is to prevent the attack, blocking is the appropriate choice.

Create a custom blocklist.

A custom blocklist filters specific words or phrases but does not protect against prompt injection attacks or establish trust boundaries between trusted and untrusted content.

Use optical character recognition (OCR) to extract the text from the images first.

OCR only extracts text from images. It does not detect or prevent prompt injection attacks, and the extracted text could still contain malicious instructions.

Q4. You are building a web app named App1 that generates responses by using a model deployed to a Microsoft Foundry project named Project1. Before sending the prompts to the model, App1 must retrieve documents by using Azure AI Search. You need to integrate Project1 and App1.

The solution must meet the following requirements:

- Multiple client applications must use the same search configuration.
- A security policy must prevent key-based authentication.
- Administrative effort must be minimized.

What should you do?

- A. Create a custom HTTP connection in Foundry and manually configure Azure AI Search endpoints per application.
- B. Configure an Azure AI Search connection in Project1 and reference the connection in each application.
- C. Call Azure AI Search directly from each application by using Microsoft Entra authentication.
- D. Enable a managed identity for each application and call Azure AI Search directly.

Answer:

B. Configure an Azure AI Search connection in Project1 and reference the connection in each application.

Explanation:**Correct Answer:**

B. Configure an Azure AI Search connection in Project1 and reference the connection in each application.

Creating an Azure AI Search connection in the Microsoft Foundry project centralizes the search configuration so that multiple client applications can reuse the same connection. The connection can use Microsoft Entra ID authentication, satisfying the requirement to prevent key-based authentication. Since the search endpoint and authentication settings are managed once within the Foundry project, administrative effort is minimized while ensuring consistent configuration across all applications.

Incorrect Answers:

A. Create a custom HTTP connection in Foundry and manually configure Azure AI Search endpoints per application.

This requires each application to maintain its own search configuration, increasing administrative overhead and reducing consistency.

C. Call Azure AI Search directly from each application by using Microsoft Entra authentication.

Although this avoids key-based authentication, each application must maintain its own search configuration, which does not meet the requirement for a shared configuration and minimal administration.

D. Enable a managed identity for each application and call Azure AI Search directly.

Managed identities provide secure authentication, but each application still requires separate configuration, increasing administrative effort compared to using a shared Foundry connection.

Q5. You have a Microsoft Foundry project that contains a customer support agent. The agent calls an internal knowledge API tool before generating responses.

Users report the following issues:

- Some requests take more than 15 seconds to complete.
- Some responses are incorrect, even when the knowledge API returns the expected data.

You need to inspect individual agent runs to view the ordered sequence of large language model (LLM) calls, tool invocations, and timing information.

Which observability capability should you use?

- A. token usage
- B. monitoring
- C. safety metrics
- D. tracing

Answer:

D. tracing

Explanation:**Correct Answer:****D. tracing**

Tracing provides detailed visibility into the execution of an individual agent run. It records the complete sequence of operations, including large language model (LLM) calls, tool invocations, prompts, responses, execution order, and timing information. This enables developers to diagnose latency issues, identify where delays occur, and determine whether incorrect responses originate from the model, the tool, or the orchestration logic. Since the requirement is to inspect individual agent runs and view the ordered sequence of events, tracing is the appropriate observability capability.

Incorrect Answers:**A. token usage**

Token usage reports the number of input and output tokens consumed by the model. It does not show the execution flow, tool calls, or timing information.

B. monitoring

Monitoring provides high-level metrics and trends across applications, such as performance and availability, but it does not provide detailed step-by-step execution information for individual agent runs.

C. safety metrics

Safety metrics measure content safety, such as harmful or unsafe outputs, and do not provide information about tool invocations, execution flow, or latency.

Q6. You have a Microsoft Foundry project that contains an agent used by the financial analysts at your company. You need to optimize the agent workflow by providing additional data access and processing capabilities. The solution must meet the following requirements:

- Ensure that the agent can perform calculations during conversations.
- Ensure that the agent can access up-to-date information from public websites.
- Ensure that the agent can retrieve information from documents uploaded directly to the agent.

What should you use for each requirement? To answer, drag the appropriate tools to the correct requirements. Each tool may be used once, more than once, or not at all.

Tools:Access up-to-date information from public websites: Perform calculations during conversations: Retrieve information from documents uploaded directly to the agent: **Answer:****Tools:**Access up-to-date information from public websites: Perform calculations during conversations: Retrieve information from documents uploaded directly to the agent: **Explanation:****Correct Answers:****Access up-to-date information from public websites: Grounding with Bing Search**

Grounding with Bing Search enables an agent to retrieve current information from the public web during a conversation. Since the model's training data is static, Bing Search provides access to the latest information, making it ideal for answering questions about recent events, current facts, or publicly available data.

Perform calculations during conversations: Code interpreter

The Code Interpreter tool allows the agent to execute Python code to perform mathematical calculations, data analysis, statistical operations, and other computational tasks. This enables the agent to produce accurate calculation results instead of relying solely on the language model's reasoning capabilities.

Retrieve information from documents uploaded directly to the agent: File search

File Search indexes documents uploaded directly to the agent and retrieves relevant content during conversations. It enables the agent to answer questions based on the uploaded files using retrieval-augmented generation, without requiring external search services.

Incorrect Answers:**Computer use**

Computer Use is designed to interact with graphical user interfaces and perform actions on applications. It is not intended for calculations, web grounding, or searching uploaded documents.

Microsoft Fabric

Microsoft Fabric is an analytics and data platform used for enterprise data engineering, warehousing, and business intelligence. It is not the appropriate tool for these agent capabilities.

Q7. You have a Microsoft Foundry project named Project1 that contains an agent. The agent uses an OpenAPI 3.0 specification to call an external weather service.

The weather service requires a key to be passed in an HTTP header. The key value is stored as a connection in Project1.

You need to ensure that the key value from the connection is included automatically whenever the OpenAPI tool is invoked.

What should you configure in the OpenAPI specification?

- A. a header parameter defined for each operation
- B. an Azure Key Vault connection
- C. an API key security scheme
- D. a Bearer token security scheme

Answer:

- C. an API key security scheme

Explanation:**Correct Answer:****C. an API key security scheme**

An **API key security scheme** in an OpenAPI 3.0 specification defines how an API key is supplied with requests, such as in an HTTP header, query parameter, or cookie. When the API key value is stored as a connection in Microsoft Foundry, the OpenAPI tool automatically retrieves the key from the connection and includes it in the appropriate HTTP header whenever the tool is invoked. This provides secure authentication without requiring the application or developer to manually include the key in every request.

Incorrect Answers:**A. a header parameter defined for each operation**

A header parameter only defines a request header. It does not automatically bind the value from a Foundry connection or provide standardized API key authentication.

B. an Azure Key Vault connection

Azure Key Vault securely stores secrets, but the question states that the key is already stored as a connection in Project1. The OpenAPI specification still requires an API key security scheme to use that connection automatically.



D. a Bearer token security scheme

A Bearer token security scheme is used for OAuth or JWT bearer tokens, not for API keys passed in HTTP headers.

Q8. You have a Microsoft Foundry project that contains three agents as shown in the following table.

Name	Description
TriageAgent	Classifies incoming customer requests
PolicyAgent	Answers policy questions by searching internal content
ActionAgent	Creates or updates tickets by calling an HTTP API

You need to orchestrate the agents to ensure that the customer requests meet the following requirements:

- Support a deterministic, step-based process that uses conditional branching and shared state across the agents.
- Optionally trigger a ticket action based on the triage result.
- The solution must minimize development effort.

What should you include in the solution?

- A. a workflow
- B. threads and runs without a workflow
- C. a multi-agent group chat session
- D. separate agent runs coordinated in the application code

Answer:

- A. a workflow

Explanation:**Correct Answer:****A. a workflow**

A **workflow** in Microsoft Foundry is designed for deterministic, step-based orchestration of agents. It supports sequential execution, conditional branching, shared state between steps, and optional execution paths. In this scenario, the workflow can first invoke the TriageAgent, then conditionally call the PolicyAgent or ActionAgent based on the triage result, while maintaining shared context throughout the process. Since workflows provide these capabilities out of the box, they minimize development effort and are the best choice for implementing predictable business processes.

Incorrect Answers:**B. threads and runs without a workflow**

Threads and runs manage conversations with agents but do not provide built-in deterministic orchestration, branching, or shared workflow state across multiple agents.

C. a multi-agent group chat session

A multi-agent group chat is intended for collaborative conversations among agents. It is not designed for deterministic, step-based business workflows with conditional branching.

D. separate agent runs coordinated in the application code

Although this approach can implement the required behavior, it requires custom orchestration logic, increasing development effort. The question specifically requires minimizing development effort.

Q9. You have a Microsoft Foundry project that contains a customer support agent. The agent is deployed by using a GitHub Actions workflow.

The organization has the following requirements:

- Every deployment to the production environment must use the latest approved evaluation baseline.
- Deployments must stop automatically if the current evaluation scores regress beyond the configured tolerance.

How should you configure the workflow?

Evaluation comparison:

- Compare against the latest approved baseline
- Compare against the previous workflow run
- Compare against the production deployment logs
- Compare against the repository default branch

If evaluation regression exceeds the configured tolerance:

- Continue deployment and send an alert
- Fail the workflow
- Retry the evaluation automatically
- Lock the target branch

Answer:

Evaluation comparison:

Compare against the latest approved baseline

Compare against the previous workflow run

Compare against the production deployment logs

Compare against the repository default branch

If evaluation regression exceeds the configured tolerance:

Continue deployment and send an alert

Fail the workflow

Retry the evaluation automatically

Lock the target branch

Explanation:

Correct Answers:

Evaluation comparison: Compare against the latest approved baseline

The requirement states that **every deployment to the production environment must use the latest approved evaluation baseline**. Configuring the workflow to compare the current evaluation results against the latest approved baseline ensures that new deployments are validated against the organization's approved quality standard rather than against previous runs or source code changes.

If evaluation regression exceeds the configured tolerance: Fail the workflow

Failing the workflow automatically stops the deployment whenever evaluation scores regress beyond the configured tolerance. This prevents lower-quality models or agents from reaching the production environment and enforces the organization's deployment quality gates.

Incorrect Answers:

Compare against the previous workflow run

The previous workflow run may not represent the latest approved baseline and therefore does not satisfy the requirement.

Compare against the production deployment logs

Production logs contain runtime information and are not used as evaluation baselines for deployment validation.

Compare against the repository default branch

The repository's default branch contains source code, not approved evaluation results for comparison.

Continue deployment and send an alert

This allows a deployment with degraded evaluation scores to reach production, which violates the requirement to stop deployments automatically.

Retry the evaluation automatically

Retrying an evaluation does not address a genuine regression in evaluation scores.

Lock the target branch

Locking the branch prevents code changes but does not stop a deployment based on evaluation regression.

Q10. Your company is piloting a customer support agent in a Microsoft Foundry project name Project1. Project1 is connected to an existing Application Insights resource, and the company's support team reviews runs in the Traces tab.

The Foundry Agent Service is configured to perform the following actions:

Retrieve the Application Insights connection string by calling
`project_client.telemetry.get_application_insights_connection_string()`.
 Call `configure_azure_monitor(connection_string=...)` to enable telemetry.

A separate LangChain service is configured to use OpenTelemetry and has the following configurations:
 Uses `AzureAIOpenTelemetryTracer(connection_string=..., enable_content_recording=False)`

Passes the tracer by using `config= "callbacks":[azure_tracer]` Company policy has the following requirements:

- Telemetry from LangChain and OpenTelemetry must be distinguishable within the same Application Insights resource.
- Secrets and credentials must NOT be stored in prompts, tool arguments, or span attributes.

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Statements	Yes	No
The LangChain service will appear in Traces without configuring a tracer.	☐	☐
Setting different <code>OTEL_SERVICE_NAME</code> values separates the services in Application Insights.	☐	☐
When using <code>enable_content_recording=False</code> , prompts and tool data will be captured in the telemetry.	☐	☐

Answer:

Statements	Yes	No
The LangChain service will appear in Traces without configuring a tracer.	☐	☒
Setting different <code>OTEL_SERVICE_NAME</code> values separates the services in Application Insights.	☒	☐
When using <code>enable_content_recording=False</code> , prompts and tool data will be captured in the telemetry.	☐	☒

Explanation:**Correct Answers:****1. The LangChain service will appear in Traces without configuring a tracer. — No**

LangChain does not automatically send telemetry to Application Insights. To capture LangChain traces, you must configure an OpenTelemetry tracer, such as **AzureAIOpenTelemetryTracer**, and pass it to LangChain through the callback configuration. Without configuring a tracer, LangChain activity will not appear in the Traces view.

2. Setting different OTEL_SERVICE_NAME values separates the services in Application Insights. — Yes

The **OTEL_SERVICE_NAME** resource attribute identifies the originating service for OpenTelemetry telemetry. Assigning different service names allows telemetry from multiple applications or services to be distinguished within the same Application Insights resource, making it easier to filter and analyze traces for each service independently.

3. When using enable_content_recording=False, prompts and tool data will be captured in the telemetry. — No

Setting **enable_content_recording=False** disables recording of sensitive content such as prompts, model responses, and tool inputs or outputs. This helps prevent secrets, credentials, or other sensitive information from being stored in telemetry, aligning with the company's security policy.

Q11. You have a Microsoft Foundry project that contains an agent and an image generation model deployment. The agent generates original images from user-supplied product photos.

You need to ensure that the generated images maintain the product identity and visual characteristics of the provided photo.

What should you do?

- A. Set the input_fidelity parameter to high.
- B. Apply a groundedness detection filter.
- C. Include a prompt and input image in the request.
- D. Decrease the value of the temperature parameter.

Answer:

- A. Set the input_fidelity parameter to high.

Explanation:**Correct Answer:****A. Set the input_fidelity parameter to high.**

The **input_fidelity** parameter controls how closely the generated image preserves the identity and visual characteristics of the input image. Setting **input_fidelity** to **high** ensures that the model strongly adheres to the product's appearance, including its shape, colors, textures, branding, and other visual details, while still allowing the requested creative modifications. Since the requirement is to maintain the product identity and visual characteristics of the provided photo, this is the most appropriate setting.

Incorrect Answers:

B. Apply a groundedness detection filter.

Groundedness detection verifies whether generated content is supported by the provided source or context. It does not control how faithfully an image preserves the characteristics of an input image.

C. Include a prompt and input image in the request.

Although an image generation request requires a prompt and an input image, simply including them does not guarantee that the generated image will preserve the product's identity. The **input_fidelity** parameter controls that behavior.

D. Decrease the value of the temperature parameter.

Lowering the temperature reduces randomness in the model's output, but it does not specifically ensure that the generated image maintains the identity and visual characteristics of the input image.

Q12. You are developing a travel planning agent by using Microsoft Foundry Agent Service.

The solution must meet the following requirements:

- Remember each user's preferred airline and hotel chain across future conversations.
- Ensure that payment card details are discarded after the conversation ends.
- Minimize custom development.

Which implementation should you recommend?

- A. Store both preferences and payment information in persistent agent memory.
- B. Store preferences in persistent agent memory and retain payment information only in orchestration-managed session context.
- C. Store both preferences and payment information in conversation history.
- D. Store payment information in Azure AI Search and preferences in session context.

Answer:

B. Store preferences in persistent agent memory and retain payment information only in orchestration-managed session context.

Explanation:**Correct Answer:**

B. Store preferences in persistent agent memory and retain payment information only in orchestration-managed session context.

Persistent agent memory is designed to store long-term user preferences that should be available across future conversations, such as a user's preferred airline or hotel chain. In contrast, sensitive information like payment card details should not be retained after a conversation ends. Storing payment information only in the orchestration-managed session context ensures that it is available only during the current conversation and is discarded when the session ends. This approach satisfies both requirements while minimizing custom development by using the built-in memory capabilities of Microsoft Foundry Agent Service.

Incorrect Answers:

A. Store both preferences and payment information in persistent agent memory.

Payment card details should not be stored in persistent memory because they must be discarded after the conversation ends.

C. Store both preferences and payment information in conversation history.

Conversation history is not intended for long-term storage of user preferences across future conversations, and it may also retain sensitive payment information longer than required.

D. Store payment information in Azure AI Search and preferences in session context.

Azure AI Search is intended for indexing and retrieving searchable content, not for storing sensitive payment information. In addition, storing preferences in session context would not preserve them across future conversations.

Q13. You have a Microsoft Foundry project that contains an agent. The agent uses tools to retrieve internal content and call external APIs. The agent is configured to let the model decide when to call the tools.

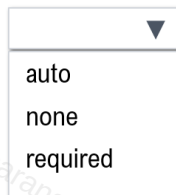
You need to publish the agent for a compliance workflow.

The solution must meet the following requirements:

- Each workflow run must include a retrieval step before generating a response.
- Tool calls must authenticate by using the published agent's own identity.
- Tool access must use an identity isolated from other project resources.
- Tool access must use support audit tracing.

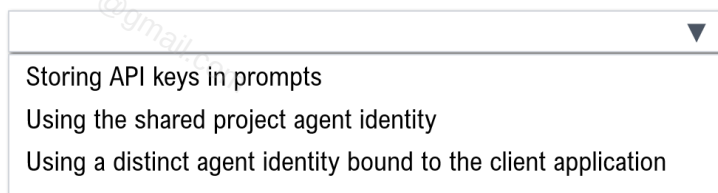
What should you do?

Set tool_choice to:



A screenshot of a dropdown menu for the 'tool_choice' property. The menu is open, showing three options: 'auto', 'none', and 'required'. The 'auto' option is selected and highlighted in blue. Above the dropdown is a faint blue arc graphic.

Configure the tool to authenticate by:



A screenshot of a dropdown menu for configuring tool authentication. The menu is open, showing three options: 'Storing API keys in prompts', 'Using the shared project agent identity', and 'Using a distinct agent identity bound to the client application'. The 'Using a distinct agent identity bound to the client application' option is selected and highlighted in blue.

Answer:

Set tool_choice to:

▼

auto

none

required

Configure the tool to authenticate by:

▼

Storing API keys in prompts

Using the shared project agent identity

Using a distinct agent identity bound to the client application

Explanation:**Correct Answers:****Set tool_choice to: required**

Setting **tool_choice** to **required** ensures that the model must invoke a tool before generating a response. Since the compliance workflow requires **every workflow run to include a retrieval step**, this setting guarantees that the retrieval tool is always called instead of leaving the decision to the model.

Configure the tool to authenticate by: Using a distinct agent identity bound to the client application

Using a **distinct agent identity** provides an isolated identity for the published agent, separate from other project resources. This approach supports least-privilege access, enables comprehensive audit tracing of tool calls, and ensures that all tool invocations are performed using the published agent's own identity. It satisfies the requirements for identity isolation and compliance auditing.

Incorrect Answers:**auto**

The model decides whether to call a tool. This does not guarantee that every workflow run performs the required retrieval step.

none

The model is prevented from calling any tools, making it impossible to perform the mandatory retrieval step.

Storing API keys in prompts

Embedding API keys in prompts is insecure and violates security best practices. It also does not support identity-based authentication or audit tracing.

Using the shared project agent identity

A shared project identity is not isolated from other project resources and does not meet the requirement for dedicated identity isolation for the published agent.

Q14. You have a chat app in a Microsoft Foundry project and an Azure AI Search vectorized index. You need to connect to the index to meet the following requirements:

- Complex questions must retrieve information from multiple chunks.
- Multi-turn conversations must influence retrieval planning.
- Retrievals must run in parallel to reduce latency.

Which retrieval approach should you use?

- A. iterative retrieval
- B. agentic Retrieval Augmented Generation (RAG)
- C. chain of thought
- D. classic Retrieval Augmented Generation (RAG)

Answer:

B. agentic Retrieval Augmented Generation (RAG)

Explanation:

Correct Answer:

B. agentic Retrieval Augmented Generation (RAG)

Agentic Retrieval Augmented Generation (RAG) is designed for advanced retrieval scenarios where an AI agent actively plans and orchestrates the retrieval process. It can decompose complex questions into multiple search queries, retrieve information from multiple chunks in parallel, and use the context of previous conversation turns to improve retrieval planning. This enables more accurate responses for complex, multi-turn conversations while reducing latency through parallel retrieval operations. These capabilities directly satisfy all the requirements in the scenario.

Incorrect Answers:

A. iterative retrieval

Iterative retrieval performs searches sequentially, refining results over multiple steps. While it can improve retrieval quality, it does not inherently support parallel retrieval or conversation-aware retrieval planning.

C. chain of thought

Chain of thought is a reasoning technique used by language models to improve reasoning. It is not a retrieval approach for Azure AI Search.

D. classic Retrieval Augmented Generation (RAG)

Classic RAG performs a single retrieval step before generating a response. It does not perform retrieval planning across multiple conversation turns or execute parallel retrievals for complex queries.

Q15. You have a Microsoft Foundry project that contains an agent. The agent processes scanned employee identification cards from multiple countries. The solution must meet the following requirements:

- Extract structured fields such as employee ID, full name, and expiration date.
- Return the extracted values as a structured schema instead of raw text.
- Minimize custom post-processing.

You plan to call Azure Content Understanding in Foundry Tools.

Which built-in analyzer should you use?

- A. `prebuilt-layout`
- B. `prebuilt-documentSearch`
- C. `prebuilt-documentFieldSchema`
- D. `prebuilt-read`

Answer:

C. `prebuilt-documentFieldSchema`

Explanation:

Correct Answer:

C. `prebuilt-documentFieldSchema`

The **`prebuilt-documentFieldSchema`** analyzer is specifically designed to extract predefined structured fields from documents and return the results in a structured schema. It is ideal for processing employee identification cards because it can identify fields such as employee ID, full name, and expiration date without requiring extensive custom parsing. Since the output is already structured, it minimizes the amount of post-processing required by the application, directly meeting all the stated requirements.

Incorrect Answers:

A. `prebuilt-layout`

The **`prebuilt-layout`** analyzer preserves the document's structure by extracting text, tables, paragraphs, and other layout elements. It does not focus on returning predefined fields as a structured schema.

B. `prebuilt-documentSearch`

The **`prebuilt-documentSearch`** analyzer is intended for document search and retrieval scenarios rather than extracting structured document fields.

D. `prebuilt-read`

The **`prebuilt-read`** analyzer performs OCR and extracts raw text from documents. It does not return structured fields such as employee ID or expiration date without additional post-processing.

Q16. You have a Microsoft Foundry project that contains a deployed ticket-triage agent.

You discover that sometimes the agent responds without calling any tools, even when a tool is required. You need to ensure that the agent calls a tool during execution.

How should you complete the Python code?

To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all.

Values:

```
run_payload = {  
    "assistant_id": agent_id,  
    : ,  
    "metadata": {  
        "scenario": "ticket-triage"  
    }  
}
```

Answer:**Values:**

```
run_payload = {  
    "assistant_id": agent_id,  
    "tool_choice" : "required" ,  
    "metadata": {  
        "scenario": "ticket-triage"  
    }  
}
```

Explanation:**Correct Answers:****First blank: "tool_choice"**

The **tool_choice** parameter controls whether the language model can decide to use a tool or must use one. It determines the model's tool invocation behavior during execution. Since the requirement is to ensure that a tool is always called, the payload must include the **tool_choice** parameter.

Second blank: "required"

Setting **tool_choice** to **required** forces the model to invoke at least one tool before generating a response. This guarantees that the deployed ticket-triage agent does not respond directly without calling the necessary retrieval or action tool.

Incorrect Answers:**"auto"**

The model decides whether to call a tool. It may generate a response without using any tool, which does not satisfy the requirement.

"response_format"

This parameter controls the structure of the model's output, not whether tools are invoked.

"tools"

The **tools** property defines the available tools but does not force the model to use them.

"type"

The **type** field specifies the type of a tool or schema and does not control tool invocation behavior.

Q17. You have a customer support agent built by using the Microsoft Foundry Agent Service. The agent calls an Azure OpenAI model deployment.

During load testing, calls intermittently fail and return an HTTP 429 rate limit exceeded error.

You need to handle throttling to reduce call failures and improve reliability under load. The solution must remain within the service and model limits.

What should you do?

- A. Create a new thread and retry the calls immediately.
- B. Reduce the number of registered tools.
- C. Implement a retry policy that uses exponential backoff and jitter.
- D. Split uploaded content into smaller files.

Answer:

C. Implement a retry policy that uses exponential backoff and jitter.

Explanation:**Correct Answer:**

C. Implement a retry policy that uses exponential backoff and jitter.

An HTTP **429 (Too Many Requests)** response indicates that the service or model deployment has reached its rate limit. The recommended approach is to implement a retry policy that uses **exponential backoff** combined with **jitter**. Exponential backoff progressively increases the delay between retry attempts, while jitter introduces randomness to prevent multiple clients from retrying simultaneously. This reduces contention, improves reliability under load, and allows requests to succeed without exceeding the service's configured limits.

Incorrect Answers:

A. Create a new thread and retry the calls immediately.

Creating a new thread does not bypass rate limits, and immediately retrying requests can make throttling even worse.

B. Reduce the number of registered tools.

The number of registered tools does not affect Azure OpenAI rate limits or resolve HTTP 429 errors.

D. Split uploaded content into smaller files.

Splitting files may reduce the size of individual requests but does not address request throttling or rate-limit exceeded errors.

Q18. You are planning a Microsoft Foundry project named Project1 that will contain several prompt flow applications and agents.

Each application must use the same Azure AI Foundry Model deployment hosted in another Azure resource. You need to recommend a solution that enables all applications in Project1 to reuse the model configuration without duplicating connection settings.

What should you recommend?

- A. Create a connection to the model resource.
- B. Configure a managed private endpoint.
- C. Enable role-based access control (RBAC) on the model deployment.
- D. Configure diagnostic settings for the model deployment.

Answer:

A. Create a connection to the model resource.

Explanation:

Correct Answer:

A. Create a connection to the model resource.

A **connection** in Microsoft Azure AI Foundry provides a centralized way to reference external resources such as model deployments. By creating a connection to the Azure AI Foundry model hosted in another Azure resource, all prompt flow applications and agents within **Project1** can reuse the same model configuration. This eliminates the need to duplicate endpoint URLs, authentication settings, and other connection details across multiple applications, reducing administrative effort and ensuring consistent configuration.

Incorrect Answers:

B. Configure a managed private endpoint.

A managed private endpoint provides secure network connectivity to Azure resources but does not enable applications to share or reuse model connection settings.

C. Enable role-based access control (RBAC) on the model deployment.

RBAC controls who can access the model deployment, but it does not centralize or reuse model configuration across multiple applications.

D. Configure diagnostic settings for the model deployment.

Diagnostic settings collect logs and metrics for monitoring. They do not provide reusable model connection configurations.

Q19. You have a Microsoft Foundry project that contains an agent that answers questions by using Retrieval-Augmented Generation (RAG).

A GitHub Actions workflow automatically deploys updates whenever a pull request is approved.

The organization has the following requirements:

- Validate that responses remain grounded in the retrieved documents before deployment.
- Prevent pull requests from being merged if the evaluation does not satisfy the required quality thresholds.

Evaluation to execute:

▼

Groundedness evaluation
Fluency evaluation
Similarity evaluation
Content Safety evaluation

Pull request policy:

▼

Require the evaluation workflow to succeed before merging
Automatically rerun failed evaluations until they pass
Notify reviewers when evaluations fail
Allow the merge and run evaluations after deployment

Answer:

Evaluation to execute:

▼

Groundedness evaluation
Fluency evaluation
Similarity evaluation
Content Safety evaluation

Pull request policy:

▼

Require the evaluation workflow to succeed before merging
Automatically rerun failed evaluations until they pass
Notify reviewers when evaluations fail
Allow the merge and run evaluations after deployment

Explanation:

Correct Answers:

Evaluation to execute: Groundedness evaluation

Groundedness evaluation verifies whether the agent's responses are supported by the retrieved documents. Since the agent uses Retrieval-Augmented Generation (RAG), this evaluation ensures that answers remain faithful to the retrieved content and do not introduce hallucinations. This directly satisfies the requirement to validate that responses remain grounded before deployment.

Pull request policy: Require the evaluation workflow to succeed before merging

Requiring the evaluation workflow to succeed before merging enforces quality gates in the CI/CD pipeline. If

the groundedness evaluation fails to meet the configured quality thresholds, the pull request cannot be merged, preventing unvalidated changes from being deployed to production.

Incorrect Answers:

Fluency evaluation

Fluency evaluates grammar, readability, and language quality, but it does not verify whether responses are grounded in retrieved documents.

Similarity evaluation

Similarity compares responses against reference answers but does not determine whether the responses are supported by the retrieved documents.

Content Safety evaluation

Content Safety evaluates harmful or unsafe content. It does not measure whether responses are grounded in the retrieved knowledge.

Automatically rerun failed evaluations until they pass

Automatically retrying evaluations does not resolve genuine quality issues and does not enforce deployment quality standards.

Notify reviewers when evaluations fail

Notification alone does not prevent a pull request from being merged.

Allow the merge and run evaluations after deployment

This allows potentially poor-quality changes to reach production, violating the requirement to validate responses before deployment.

Q20. You are creating an image-processing workflow in a Microsoft Foundry project. The workflow must meet the following requirements:

- Generate multiple alternative versions of an existing product image for marketing campaigns.
- Preserve the overall composition, lighting, and subject.
- Use the built-in image generation capabilities without training a custom model.

You need to configure the workflow to generate several stylistic alternatives from the original image.

How should you configure the workflow?

- Enable `image_variation` mode and provide the original image as the input.
- Enable `mask_inpainting` and provide a mask that covers the entire image.
- Enable `text_to_image` mode and describe the original image in the prompt.
- Enable `image_to_image` mode with the strength parameter set to 1.0.

Answer:

A. Enable `image_variation` mode and provide the original image as the input.

Explanation:

Correct Answer:

A. Enable `image_variation` mode and provide the original image as the input.

The **image_variation** mode is specifically designed to generate multiple alternative versions of an existing image while preserving its key visual characteristics. It maintains the overall composition, lighting, subject, and structure of the original image, making it ideal for creating different marketing variations of a product

photo. Since it is a built-in capability of Microsoft Foundry image generation models, it also satisfies the requirement of generating alternatives without training a custom model.

Incorrect Answers:

B. Enable mask_inpainting and provide a mask that covers the entire image.

Mask inpainting is intended to modify selected portions of an image. Using a mask that covers the entire image effectively repaints the whole image and is not designed for generating multiple stylistic variations while preserving the original composition.

C. Enable text_to_image mode and describe the original image in the prompt.

Text-to-image generation creates a new image based solely on the text prompt. Even with a detailed description, it cannot reliably preserve the exact composition, lighting, and subject of the original product image.

D. Enable image_to_image mode with the strength parameter set to 1.0.

With **image-to-image** generation, a **strength** value of **1.0** gives maximum influence to the text prompt and minimal influence to the input image. This makes it unlikely to preserve the original composition and visual characteristics.

Q21. You have a Microsoft Foundry project that contains an agent.

The agent uses a knowledge source built from documents stored in Azure Blob Storage. The documents include digitally scanned PDFs that contain multipage tables.

You have an ingestion job that extracts only plain text, causing loss of table structure, headings, and page-number metadata.

Users frequently ask questions that require the retrieval of specific table rows across the pages.

You need to configure an ingestion job for a Retrieval Augmented Generation (RAG) pipeline that performs optical character recognition (OCR) on scanned PDFs, preserves tables and headings as structure-aware chunks, and stores page-number metadata with each chunk.

How should you configure the ingestion job?

- A. Use advanced data parsing to reingest the documents.
- B. Use OCR and page-level chunking.
- C. Use page-level OCR extraction and store each page as a single chunk.
- D. Use basic parsing and fixed-size chunking.

Answer:

A. Use advanced data parsing to reingest the documents.

Explanation:

Correct Answer:

A. Use advanced data parsing to reingest the documents.

Advanced data parsing is designed for ingestion scenarios involving complex documents such as scanned PDFs. It performs OCR on scanned content, preserves document structure including headings, tables, and sections, creates structure-aware chunks instead of plain-text chunks, and stores useful metadata such as page numbers with each chunk. This enables Retrieval-Augmented Generation (RAG) systems to retrieve specific table rows across multiple pages while maintaining the original document context. Since the current

ingestion process loses table structure and metadata, reingesting the documents with advanced data parsing directly addresses all the stated requirements.

Incorrect Answers:

B. Use OCR and page-level chunking.

Although OCR extracts text from scanned PDFs, page-level chunking stores an entire page as a chunk and does not preserve document structure such as tables and headings as structure-aware chunks.

C. Use page-level OCR extraction and store each page as a single chunk.

Storing each page as a single chunk produces large chunks that make retrieval less precise, especially when users need information from specific table rows across pages.

D. Use basic parsing and fixed-size chunking.

Basic parsing with fixed-size chunking loses document structure, table relationships, headings, and page metadata, making it unsuitable for scanned documents used in RAG.

Q22. A Microsoft Foundry agent helps engineers troubleshoot production systems.

The solution must meet the following requirements:

- Download diagnostic logs from an internal web application that does not expose APIs.
- Analyze the downloaded log files to identify performance bottlenecks.

Which combination of tools should you recommend?

- A. Computer use and Code interpreter
- B. File search and Microsoft Fabric
- C. Grounding with Bing Search and Code interpreter
- D. Computer use and File search

Answer:

A. Computer use and Code interpreter

Explanation:

Correct Answer:

A. Computer use and Code interpreter

This scenario requires two distinct capabilities. **Computer Use** enables the agent to interact with web applications through their graphical user interface, making it possible to navigate the internal web application and download diagnostic log files even when no APIs are available. **Code Interpreter** can then analyze the downloaded log files by executing Python code to identify performance bottlenecks, detect patterns, generate statistics, and visualize results. Together, these tools satisfy both requirements with minimal custom development.

Incorrect Answers:

B. File search and Microsoft Fabric

File Search retrieves information from indexed documents, while Microsoft Fabric is an analytics platform. Neither tool can interact with a web application's user interface to download log files.

C. Grounding with Bing Search and Code interpreter

Grounding with Bing Search retrieves information from public websites. It cannot access an internal web application or download diagnostic logs.

D. Computer use and File search

Computer Use can download the log files, but File Search is intended for retrieving information from indexed documents rather than analyzing log files for performance bottlenecks.

Q23. You have a Microsoft Foundry project that contains an internal Q&A agent.

Users report the following issues when they ask the agent questions:

An increase in the following response: "No relevant information found"

Periodic HTTP 429 rate limit exceeded errors during peak hours

You need to identify whether each issue is caused by model unavailability, resource limits, or inference failures.

What should you do?

Metrics to enable:

Model Availability Rate and Provisioned Utilization
Only Tokens Cache Match Rate
Only Total Requests filtered to status code 200
Time To Response and Total Tokens

Diagnostic log to collect:

AllMetrics
audit
RequestResponse
trace

Answer:

Metrics to enable:

Model Availability Rate and Provisioned Utilization

Only Tokens Cache Match Rate

Only Total Requests filtered to status code 200

Time To Response and Total Tokens

Diagnostic log to collect:

AllMetrics

audit

RequestResponse

trace

Explanation:

Correct Answers:

Metrics to enable: Model Availability Rate and Provisioned Utilization

Model Availability Rate helps determine whether requests are failing because the model deployment is unavailable. **Provisioned Utilization** shows how much of the model's allocated capacity is being consumed, making it possible to identify whether HTTP 429 errors are caused by resource exhaustion during peak usage. Together, these metrics help distinguish between model unavailability and capacity-related throttling.

Diagnostic log to collect: RequestResponse

The **RequestResponse** diagnostic log captures detailed information about inference requests and responses, including HTTP status codes, error details, latency, and failure reasons. This enables developers to determine whether requests failed because of inference errors, throttling, or other runtime issues, making it the appropriate diagnostic log for troubleshooting the reported problems.

Incorrect Answers:

Only Tokens Cache Match Rate

This metric measures cache effectiveness and does not help identify model availability or throttling issues.

Only Total Requests filtered to status code 200

Monitoring only successful requests ignores failed requests, including HTTP 429 errors and inference failures.

Time To Response and Total Tokens

These metrics measure latency and token consumption but do not identify whether failures are caused by model availability or resource limits.

AllMetrics

AllMetrics provides aggregated performance metrics but does not include the detailed request and response information required to diagnose inference failures.

audit

Audit logs record administrative and management operations, not inference request failures.

trace

Trace logs provide execution traces but do not capture the complete inference request and response details needed to analyze HTTP 429 and inference failures.

Q24. You have a Microsoft Foundry project that contains a prompt agent used by a customer support web app. The agent is invoked from a Python service that does NOT run in the Foundry portal. You need to implement end-to-end tracing to capture latency breakdowns and exceptions across agent runs. Which two components can you use?

- A. a Log Analytics workspace
- B. Application Insights
- C. OpenTelemetry
- D. the Azure Monitor Agent
- E. Microsoft Sentinel

Answer:

B. Application Insights

C. OpenTelemetry

Explanation:**Correct Answers:**

B. Application Insights

Application Insights is used to collect, store, and visualize telemetry generated by the application and Microsoft Foundry Agent Service. It provides end-to-end monitoring, including request latency, dependency tracking, exceptions, and distributed traces. This allows developers to analyze agent performance and troubleshoot failures across agent runs.

C. OpenTelemetry

OpenTelemetry is the industry-standard framework for collecting distributed traces, metrics, and logs. Since the Python service runs outside the Foundry portal, it can be instrumented with OpenTelemetry to generate end-to-end traces. These traces can then be exported to Application Insights, providing detailed latency breakdowns and exception tracking across the entire application workflow.

Incorrect Answers:

A. a Log Analytics workspace

A Log Analytics workspace stores and queries monitoring data, but it does not provide application instrumentation or distributed tracing by itself.

D. the Azure Monitor Agent

The Azure Monitor Agent collects telemetry from Azure virtual machines and servers. It is not used to instrument Python applications or capture end-to-end application traces.

E. Microsoft Sentinel

Microsoft Sentinel is a security information and event management solution used for threat detection and security analytics. It is not intended for application performance tracing.



Q25. You have a Microsoft Foundry project that contains a customer support agent. The agent uses multiple tools during execution. Users report that responses are delayed because the agent waits for several independent tool calls to complete.

You need to reduce the overall response time. The solution must be implemented in the application logic. What should you do?

- A. Execute independent tool calls in parallel before generating the response.
- B. Increase the maximum completion tokens.
- C. Replace the language model with a larger deployment.
- D. Enable persistent agent memory.

Answer:

A. Execute independent tool calls in parallel before generating the response.

Explanation:

Correct Answer:

A. Execute independent tool calls in parallel before generating the response.

When multiple tool calls are independent of each other, executing them in parallel significantly reduces the overall response time. Instead of waiting for one tool to finish before starting the next, the application can invoke all independent tools simultaneously and wait for all results before generating the final response. Since the question specifically states that the solution must be implemented in the **application logic**, parallel execution is the recommended approach to improve performance.

Incorrect Answers:

B. Increase the maximum completion tokens.

Increasing the maximum completion tokens allows the model to generate longer responses but does not reduce the latency caused by waiting for tool calls.

C. Replace the language model with a larger deployment.

A larger model generally increases processing time and cost. It does not address delays caused by sequential tool execution.

D. Enable persistent agent memory.

Persistent agent memory stores information across conversations to improve personalization. It does not reduce the execution time of independent tool calls.

Q26. You have a Microsoft Foundry project that contains a customer support agent grounded in internal documentation. After a recent update, users report the following issues:

- Some answers are unsupported by retrieved documents.
- A small number of responses are flagged for policy violations.

You need to evaluate each issue.

Which observability signals should you use for each issue?

To answer, drag the appropriate observability signals to the correct issues. Each observability signal may be used once, more than once, or not at all.

Observability signals:Unsupported responses: Policy violations: **Answer:****Observability signals:**Unsupported responses: Policy violations: **Explanation:****Correct Answers:****Unsupported responses: Groundedness evaluation metrics**

Groundedness evaluation metrics measure whether an agent's responses are supported by the retrieved documents in a Retrieval-Augmented Generation (RAG) solution. If answers are not backed by the retrieved content, these metrics help identify hallucinations and determine whether the response is properly grounded in the knowledge source. Since users report that some answers are unsupported by the retrieved documents, groundedness evaluation metrics are the appropriate observability signal.

Policy violations: Risk and safety metrics

Risk and safety metrics evaluate responses for content policy violations, including harmful, unsafe, or otherwise non-compliant content. They help identify responses that violate organizational or responsible AI policies, making them the appropriate observability signal for monitoring policy violations.

Incorrect Answers:**Latency breakdown traces**

Latency breakdown traces measure the execution time of different stages of an agent run. They are useful for performance analysis but do not evaluate grounding or policy compliance.

Token usage analytics

Token usage analytics track input and output token consumption for cost and usage analysis. They do not determine whether responses are grounded or whether they violate content policies.



Sarang Joshi

Q27. You are creating an image-editing workflow in a Microsoft Foundry project.

The workflow must meet the following requirements:

- Ensure that background objects can be removed by applying a mask-based inpainting edit.
- Preserve the original lighting and style of the edited images.
- Use the built-in image editing controls, NOT a custom model.

You need to ensure that image edits apply exclusively inside the masked area.

How should you configure the workflow?

- Set generation mode to `image_variation` and provide the original image as a reference.
- Enable `text_to_image` mode and a prompt describing the desired background removal.
- Enable `image_to_image` mode and a high-strength value to regenerate the full image based on the prompt.
- Enable `mask_inpainting` and supply both the input image and a mask indicating which part of the image to modify.

Answer:

D. Enable `mask_inpainting` and supply both the input image and a mask indicating which part of the image to modify.

Explanation:

Correct Answer:

D. Enable `mask_inpainting` and supply both the input image and a mask indicating which part of the image to modify.

Mask inpainting is specifically designed for selective image editing. By providing both the original image and a mask, the model modifies only the pixels within the masked region while preserving the rest of the image, including its lighting, style, composition, and subject. This makes it ideal for removing background objects while keeping the remainder of the image unchanged. Since mask inpainting is a built-in image editing capability of Microsoft Foundry, it also satisfies the requirement to avoid training a custom model.

Incorrect Answers:

A. Set generation mode to `image_variation` and provide the original image as a reference.

Image variation generates alternative versions of the entire image. It is not intended for editing only a specific masked region.

B. Enable `text_to_image` mode and a prompt describing the desired background removal.

Text-to-image generates a completely new image from a text prompt. It cannot reliably preserve the original lighting, composition, and style, nor does it support mask-based editing.

C. Enable `image_to_image` mode and a high-strength value to regenerate the full image based on the prompt.

A high **strength** value causes the model to regenerate much of the image based on the prompt, rather than limiting edits to the masked area. As a result, changes may extend beyond the intended region.

Q28. You have a Microsoft Foundry project that contains an agent. The agent uses Azure AI Search in Foundry Tools to retrieve documents from a search index.

The application returns an error indicating that the specified index cannot be found.

You verify that the Azure AI Search service is reachable and authentication succeeds.

You need to configure the request to reference the correct search index.

To what should you set the index property?

- A. The Azure AI Search service name
- B. The Azure AI Search endpoint URL
- C. The Azure AI Search index name
- D. The Microsoft Foundry project name

Answer:

C. The Azure AI Search index name

Explanation:

Correct Answer:

C. The Azure AI Search index name

The **index** property must be set to the **Azure AI Search index name**, which uniquely identifies the search index that contains the documents to be queried. Since the Azure AI Search service is reachable and authentication succeeds, the error indicates that the request is referencing an incorrect or non-existent index. Specifying the correct index name enables the agent to retrieve documents from the intended search index.

Incorrect Answers:

A. The Azure AI Search service name

The service name identifies the Azure AI Search service instance, not the specific index within that service.

B. The Azure AI Search endpoint URL

The endpoint URL specifies the location of the Azure AI Search service. It does not identify which search index should be queried.

D. The Microsoft Foundry project name

The Microsoft Foundry project name identifies the AI project but has no relationship to the Azure AI Search index used for document retrieval.

Q29. You need to recommend a plan to create a customer support agent by using the Microsoft Foundry Agent Service.

The agent must meet the following requirements:

- Retain user preferences across multiple conversations.
- Enable users to provide contextual grounding by directly uploading documents during a chat.

Which Foundry capability should you recommend for each requirement?

To retain user preferences across conversations, use:

Agent memory that uses persistent storage
Conversation history
Orchestration-managed session context

To enable users to provide contextual grounding during chats, use the:

Azure AI Search tool
Code interpreter tool
File search tool

Answer:

To retain user preferences across conversations, use:

Agent memory that uses persistent storage
Conversation history
Orchestration-managed session context

To enable users to provide contextual grounding during chats, use the:

Azure AI Search tool
Code interpreter tool
File search tool

Explanation:

Correct Answers:

To retain user preferences across conversations: Agent memory that uses persistent storage

Persistent agent memory is designed to store long-term user information such as preferences, allowing the agent to remember details across multiple conversations. This capability enables personalized interactions without requiring users to repeat their preferences in every new chat.

To enable users to provide contextual grounding during chats: File search tool

The **File search** tool allows users to upload documents directly during a conversation. The agent indexes these uploaded files and retrieves relevant information from them to ground its responses. This enables contextual question answering based on documents supplied by the user during the chat.

Incorrect Answers:

Conversation history

Conversation history maintains messages within a conversation but is not intended for storing long-term user preferences across multiple conversations.

Orchestration-managed session context

Session context exists only for the duration of the current conversation and is discarded when the session ends.

Azure AI Search tool

Azure AI Search retrieves information from existing Azure AI Search indexes. It is not designed for users to upload documents directly during a chat.

Code interpreter tool

The Code Interpreter executes code and analyzes uploaded files but is not intended to provide document grounding through semantic document retrieval.

Q30. You are developing an image-editing workflow in a Microsoft Foundry project.

The workflow must meet the following requirements:

- Replace the sky in landscape photographs with a sunset.
- Modify only the selected sky region.
- Preserve all foreground objects without regeneration.
- Use the built-in image editing capabilities.

You need to configure the workflow.

How should you configure the workflow?

- Enable `image_variation` mode and provide the original image.
- Enable `mask_inpainting` and provide the original image together with a mask covering the sky.
- Enable `text_to_image` mode and describe the desired landscape.
- Enable `image_to_image` mode with a high-strength value.

Answer:

B. Enable `mask_inpainting` and provide the original image together with a mask covering the sky.

Explanation:**Correct Answer:**

B. Enable `mask_inpainting` and provide the original image together with a mask covering the sky.

Mask inpainting is the built-in image editing capability designed for selective image modifications. By supplying the original image along with a mask covering only the sky, the model applies the requested edit exclusively within the masked region while preserving the rest of the image. This ensures that the foreground objects, lighting, composition, and other unmasked areas remain unchanged. It satisfies all the stated requirements by replacing only the sky with a sunset using the built-in image editing capabilities.

Incorrect Answers:

A. Enable `image_variation` mode and provide the original image.

Image variation generates alternative versions of the entire image rather than modifying only a selected region defined by a mask.

C. Enable `text_to_image` mode and describe the desired landscape.

Text-to-image generates a completely new image from a text prompt and cannot preserve the original foreground objects or composition.

D. Enable image_to_image mode with a high-strength value.

A high-strength value causes much of the image to be regenerated based on the prompt, which can modify areas outside the intended sky region and alter foreground objects.

Q31. You are planning a Microsoft Foundry project named Project1 that will contain multiple agents. Each agent will access the same Azure AI Search resource.

You need to recommend a solution to centrally manage the Azure AI Search credentials within Project1. The solution must be implemented across all the agents.

What should you recommend?

- A. Enable role-based access control (RBAC) for the Azure AI Search resource.
- B. Disable key-based access control on the Azure AI Search resource.
- C. Add a connection to the Azure AI Search resource.
- D. Create a managed private endpoint that connects to the Azure AI Search resource.

Answer:

C. Add a connection to the Azure AI Search resource.

Explanation:

Correct Answer:

C. Add a connection to the Azure AI Search resource.

A **connection** in Microsoft Foundry provides a centralized way to store and manage the endpoint and authentication credentials for external resources such as Azure AI Search. Once the connection is created within **Project1**, all agents in the project can reference the same connection instead of maintaining separate credentials. This simplifies administration, ensures consistent configuration, and eliminates duplication across multiple agents.

Incorrect Answers:

A. Enable role-based access control (RBAC) for the Azure AI Search resource.

RBAC controls access permissions to the Azure AI Search resource but does not centrally manage or share connection credentials across multiple agents.

B. Disable key-based access control on the Azure AI Search resource.

Disabling key-based authentication changes the authentication method but does not provide centralized credential management within the Foundry project.

D. Create a managed private endpoint that connects to the Azure AI Search resource.

A managed private endpoint provides secure network connectivity to the Azure AI Search resource but does not manage or centralize authentication credentials.

Q32. You are deploying a support agent that enables users to upload photos.

You need to automatically classify uploaded images for harmful content. The solution must block content based on severity levels.

What should you do?

- A. Apply keyword scanning to optical character recognition (OCR) output by using Azure Vision in Foundry Tools.
- B. Enable prompt shields.
- C. Use blocklists.
- D. Implement image moderation.

Answer:

D. Implement image moderation.

Explanation:

Correct Answer:

D. Implement image moderation.

Image moderation is designed to analyze uploaded images and classify them for harmful or unsafe content. It evaluates images against predefined safety categories and assigns severity levels, allowing the application to automatically block content that exceeds configured thresholds. This directly satisfies the requirement to classify uploaded images and block them based on severity levels using Microsoft's built-in responsible AI capabilities.

Incorrect Answers:

A. Apply keyword scanning to optical character recognition (OCR) output by using Azure Vision in Foundry Tools.

OCR extracts text from images, but keyword scanning cannot detect harmful visual content such as violence, sexual content, or graphic imagery.

B. Enable prompt shields.

Prompt shields protect against prompt injection attacks and malicious instructions. They do not classify uploaded images for harmful visual content.

C. Use blocklists.

Blocklists filter specific words or phrases in text. They are not intended for image content moderation or severity-based image classification.

Q33. You have a Microsoft Foundry project that contains an agent named PaymentAgent.

PaymentAgent includes a function tool that issues customer refunds by using an external API.

You are creating a workflow in YAML.

You need to ensure that the workflow pauses for human approval and continues with the refund step only after approval is granted.

How should you complete the workflow definition?

steps:

- id: propose_refund
type: agent
agent: PaymentAgent

- id: approval
type:

ask_question
basic_chat
data_transformation

- id: execute_refund
type: agent
agent: PaymentAgent
condition:

approval == "approved"
propose_refund.output != null
true

Answer:

steps:

- id: propose_refund
type: agent
agent: PaymentAgent

- id: approval
type:

ask_question
basic_chat
data_transformation

- id: execute_refund
type: agent
agent: PaymentAgent
condition:

approval == "approved"
propose_refund.output != null
true

Explanation:

Correct Answers:

Type: ask_question

The **ask_question** workflow step pauses execution and waits for human input before continuing. It is designed for Human-in-the-Loop (HITL) scenarios where a user or approver must review and approve an action before the workflow proceeds. Since the requirement is to pause the workflow until a human approves the refund, this is the appropriate step type.

Condition: approval == "approved"

The refund should only be executed after explicit approval is received. Setting the condition to **approval == "approved"** ensures that the **execute_refund** step runs only when the approval step returns the value **approved**, preventing unauthorized refunds.

Incorrect Answers:**basic_chat**

A basic chat step generates conversational responses but does not pause the workflow for structured human approval.

data_transformation

A data transformation step modifies or reshapes data. It is not intended for collecting human approval.

propose_refund.output != null

This condition only checks whether the previous step produced an output. It does not verify that a human approved the refund.

true

This condition always evaluates to true, causing the refund to execute regardless of whether approval was granted.

Q34. You have a Microsoft Foundry project that uses Azure AI Search to ground an agent in internal documentation. After a recent content update, users report that the agent's answers have become less accurate.

You need to identify whether the retrieved content is negatively influencing the model's generated responses. Which observability signal should you review?

- A. indexer status and failure history
- B. latency breakdown traces
- C. prediction drift metrics
- D. groundedness evaluation metrics

Answer:**D. groundedness evaluation metrics****Explanation:****Correct Answer:****D. groundedness evaluation metrics**

Groundedness evaluation metrics measure whether the agent's generated responses are supported by the retrieved documents in a Retrieval-Augmented Generation (RAG) solution. If recent updates to the indexed content cause the retrieved information to become less relevant or incorrect, groundedness metrics help

determine whether the retrieved content is negatively influencing the model's responses. This is the appropriate observability signal for diagnosing answer quality issues in a RAG-based agent.

Incorrect Answers:

A. indexer status and failure history

Indexer status helps determine whether documents were successfully indexed, but it does not evaluate whether the retrieved content is producing accurate or well-grounded responses.

B. latency breakdown traces

Latency breakdown traces measure the execution time of retrieval, tool calls, and model inference. They do not measure response quality or grounding.

C. prediction drift metrics

Prediction drift metrics monitor changes in model predictions over time, typically for machine learning models. They are not used to evaluate whether RAG responses are supported by retrieved documents.

Q35. You have a Microsoft Foundry project that contains an agent used by a legal department. The agent generates answers by using documents retrieved from an Azure AI Search index. Users report that some answers omit important clauses from lengthy legal documents. You need to improve the completeness of responses. The solution must be implemented in the application logic without changing the search index or replacing the deployed model. What should you do?

- A. Increase the number of retrieved document chunks before constructing the prompt.
- B. Enable Prompt Shields for document attacks.
- C. Configure Content Safety to annotate responses.
- D. Use a lower temperature value.

Answer:

A. Increase the number of retrieved document chunks before constructing the prompt.

Explanation:

Correct Answer:

A. Increase the number of retrieved document chunks before constructing the prompt.

The agent uses Retrieval-Augmented Generation (RAG) with documents stored in an Azure AI Search index. If users report that important clauses from lengthy legal documents are being omitted, the most effective solution is to retrieve more relevant document chunks before building the prompt sent to the language model. By increasing the number of retrieved chunks, the application provides the model with a broader and more complete context, improving the likelihood that all relevant legal clauses are included in the generated response. This solution is implemented entirely in the application logic and does not require changes to the search index or the deployed model.

Incorrect Answers:

B. Enable Prompt Shields for document attacks.

Prompt Shields protect against prompt injection attacks and malicious instructions. They do not improve retrieval completeness or include additional document content.

C. Configure Content Safety to annotate responses.

Content Safety evaluates responses for harmful or policy-violating content. It does not improve the completeness of responses generated from retrieved documents.

D. Use a lower temperature value.

Reducing the temperature makes responses more deterministic, but it does not provide the model with additional document context or prevent important clauses from being omitted.

Q36. You have a Microsoft Foundry project that processes procurement documents submitted by suppliers. You need to implement two pipelines by using Azure Content Understanding in Foundry Tools.

The solution must meet the following requirements:

- Include a pipeline named Pipeline1 that supports cost-effective, high-volume processing of standalone PDF invoices.
- Include a pipeline named Pipeline2 that supports cross-document validation by using multi-step reasoning and reference data.

How should you configure each pipeline?

To answer, drag the appropriate configurations to the correct pipelines. Each configuration may be used once, more than once, or not at all.

Configurations:

Multi-file task in pro mode

Multi-file task in standard mode

Single-file task in pro mode

Single-file task in standard mode

Pipeline1:

Pipeline2:

Answer:**Configurations:**

Multi-file task in pro mode

Multi-file task in standard mode

Single-file task in pro mode

Single-file task in standard mode

Pipeline1:

Single-file task in standard mode

Pipeline2:

Multi-file task in pro mode

Explanation:**Correct Answers:****Pipeline1: Single-file task in standard mode**

Pipeline1 is intended for **cost-effective, high-volume processing of standalone PDF invoices**. Since each invoice is processed independently and does not require advanced reasoning or cross-document analysis, a

single-file task in standard mode is the most efficient and economical choice. Standard mode is optimized for straightforward document extraction at scale.

Pipeline2: Multi-file task in pro mode

Pipeline2 must perform **cross-document validation** using **multi-step reasoning** and **reference data**. This requires analyzing multiple related documents together, comparing information across them, and applying advanced reasoning. A **multi-file task in pro mode** is specifically designed for these complex document understanding scenarios.

Incorrect Answers:

Multi-file task in standard mode

Standard mode does not provide the advanced multi-step reasoning capabilities needed for cross-document validation.

Single-file task in pro mode

Although Pro mode supports advanced reasoning, a single-file task cannot analyze relationships across multiple documents and therefore does not meet the cross-document validation requirement.

Q37. You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.

You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.

You need to implement controls to mitigate the risk.

Solution: You configure a prompt shield for user prompts.

Does this meet the goal?

- A. Yes
- B. No

Answer:

B. No

Explanation:

Prompt Shields for **user prompts** protect against direct prompt injection attacks where malicious instructions are entered directly by the user in the prompt. In this scenario, the threat comes from **uploaded images containing hidden or embedded text** that is extracted and supplied to the multimodal model. This is an **indirect prompt injection (document attack)**. To mitigate this risk, you should configure **Prompt Shields for document attacks**. If the content comes from third-party sources, enabling **Spotlighting** provides additional protection by treating that content as lower trust.

Q38. You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.

You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.

You need to implement controls to mitigate the risk.

Solution: You configure image moderation to block unsafe content before processing the images.

Does this meet the goal?

- A. Yes
- B. No

Answer:

B. No

Explanation:

Image moderation is designed to detect and block **unsafe visual content**, such as violence, sexual content, or other harmful imagery based on configured severity levels. However, it does **not** protect against **hidden prompt injection instructions embedded as text within images**. In this scenario, the risk is an **indirect prompt injection attack**, which should be mitigated by using **Prompt Shields for document attacks**. If the images originate from external or third-party sources, **Spotlighting** should also be enabled to treat the extracted content as lower trust.

Q39. You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.

You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.

You need to implement controls to mitigate the risk.

Solution: You configure a prompt shield for documents. .

Does this meet the goal?

- A. Yes
- B. No

Answer:

A. Yes

Explanation:

Prompt Shields for documents are specifically designed to protect against **indirect prompt injection attacks** originating from documents, images, and other external content. In this scenario, users upload images containing **hidden embedded text** that is extracted and passed to the multimodal model. By enabling Prompt Shields for documents, the service analyzes the extracted content and detects attempts to manipulate the model through embedded instructions. This directly mitigates the identified risk. For third-party or untrusted content, enabling **Spotlighting** provides additional protection by treating the extracted content as lower trust.

Q40. You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.

You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.

You need to implement controls to mitigate the risk.

Solution: You configure protected material detection.

Does this meet the goal?

- A. Yes
- B. No

Answer:

B. No

Explanation:

Protected material detection is designed to identify copyrighted or protected content, such as copyrighted text, code, lyrics, or other protected material. Its purpose is to help prevent the generation or reproduction of copyrighted content. It does **not** detect or prevent **indirect prompt injection attacks** where hidden instructions are embedded in uploaded images. To mitigate this scenario, the appropriate control is **Prompt Shields for document attacks**, which analyze extracted content from documents and images for malicious instructions.

Q41. You have a Microsoft Foundry project that ingests scanned PDF invoices stored in Azure Blob Storage. Each invoice contains printed fine items and has a table-based layout.

Extracted results are stored as structured JSON and used as grounding data for an agent in a Retrieval Augmented Generation (RAG) solution.

You need to create a single analyzer that meets the following requirements:

- Extracts the invoice number, invoice date, vendor name, and total amount across varying templates
- Returns confidence scores so that results with confidence below 0.80 can be routed for supervisor review

What should you use?

- A. a Foundry agent that has groundedness guardrails enabled to extract invoice fields and confidence scores
- B. a custom Azure Content Understanding in Foundry Tools analyzer that defines the required fields as the extracted fields and the returned confidence scores for routing
- C. the Azure Content Understanding in Foundry Tools prebuilt-layout analyzer
- D. the Azure Content Understanding in Foundry Tools prebuilt-documentSearch analyzer and search.score from the Azure AI Search results for routing

Answer:

B. a custom Azure Content Understanding in Foundry Tools analyzer that defines the required fields as the extracted fields and the returned confidence scores for routing

Explanation:**Correct Answer:**

B. a custom Azure Content Understanding in Foundry Tools analyzer that defines the required fields as the extracted fields and the returned confidence scores for routing

A **custom Azure Content Understanding analyzer** allows you to define the exact fields to extract, such as **invoice number, invoice date, vendor name, and total amount**, even when invoices use different templates. It also returns **confidence scores** for each extracted field, enabling the application to automatically identify low-confidence results. By routing documents with confidence scores below **0.80** for supervisor review, the solution meets both the extraction and validation requirements while producing structured JSON suitable for a Retrieval-Augmented Generation (RAG) solution.

Incorrect Answers:

A. a Foundry agent that has groundedness guardrails enabled to extract invoice fields and confidence scores

Groundedness guardrails evaluate whether generated responses are supported by retrieved content. They are not designed to extract structured invoice fields or provide extraction confidence scores.

C. the Azure Content Understanding in Foundry Tools prebuilt-layout analyzer

The **prebuilt-layout** analyzer extracts document layout elements such as text, tables, and paragraphs, but it does not extract predefined invoice fields with confidence scores across varying invoice templates.

D. the Azure Content Understanding in Foundry Tools prebuilt-documentSearch analyzer and search.score from the Azure AI Search results for routing

The **search.score** in Azure AI Search represents the relevance of search results, not the confidence of extracted document fields. It cannot be used to determine the accuracy of invoice field extraction.

Q42. A Microsoft Foundry agent retrieves content from external websites by using Grounding with Bing Search.

Security administrators want to:

- Detect indirect prompt injection attempts.
- Review detections before deciding whether to block requests.

Which configuration should you recommend?

- A. Disable Prompt Shields.
- B. Configure Prompt Shields to Block.
- C. Configure Prompt Shields to Annotate and enable Spotlighting.
- D. Configure Content Safety to Block Hate content.

Answer:

C. Configure Prompt Shields to Annotate and enable Spotighting.

Explanation:**Correct Answer:**

C. Configure Prompt Shields to Annotate and enable Spotighting.

This solution satisfies both requirements. Setting **Prompt Shields** to **Annotate** detects indirect prompt injection attempts without automatically blocking requests, allowing security administrators to review

detections before taking action. Enabling **Spotlighting** marks third-party content, such as web content retrieved through **Grounding with Bing Search**, as lower-trust input. This helps the model distinguish trusted instructions from potentially malicious embedded instructions, reducing the risk of indirect prompt injection attacks.

Incorrect Answers:**A. Disable Prompt Shields.**

Disabling Prompt Shields removes protection against indirect prompt injection attacks and does not detect malicious instructions.

B. Configure Prompt Shields to Block.

Block mode automatically rejects detected prompt injection attempts. The requirement is to review detections before deciding whether to block requests.

D. Configure Content Safety to Block Hate content.

Content Safety filters harmful or unsafe content such as hate, violence, or sexual content. It does not detect or mitigate indirect prompt injection attacks.

Q43. You have a Microsoft Foundry project that contains a workflow for a customer support triage process. You have an Ask a question node that stores user responses in a local variable named Var01. You need to create the following Power Fx expressions:

- An if/else condition expression that ensures that Var01 contains a value
- A Send message expression that returns the stored user response in uppercase

How should you configure the expressions?

If/else condition expression:

IsBlank(Local.Var01)
IsEmpty(Local.Var01)
Not(IsBlank(Local.Var01))

Send message expression:

{Local.Var01}
{Upper(Local.Var01)}
{Upper(Var01)}

Answer:

If/else condition expression:

IsBlank(Local.Var01)
IsEmpty(Local.Var01)
Not(IsBlank(Local.Var01))

Send message expression:

{Local.Var01}
{Upper(Local.Var01)}
{Upper(Var01)}

Explanation:**Correct Answers:****If/else condition expression: Not(IsBlank(Local.Var01))**

The requirement is to ensure that **Var01 contains a value** before proceeding. The **IsBlank()** function returns **true** when the variable is empty or has no value. By wrapping it with **Not()**, the expression evaluates to **true only when Var01 contains a value**, making it the correct condition for the If/Else node.

Send message expression: {Upper(Local.Var01)}

The **Upper()** function converts the contents of the local variable to uppercase. Using **{Upper(Local.Var01)}** returns the stored user response in uppercase while correctly referencing the local variable.

Incorrect Answers:**IsBlank(Local.Var01)**

This returns **true when the variable is empty**, which is the opposite of the requirement.

IsEmpty(Local.Var01)

IsEmpty() is intended for tables and collections, not for checking whether a text variable contains a value.

{Local.Var01}

This returns the stored value but does not convert it to uppercase.

{Upper(Var01)}

This does not correctly reference the local variable. The variable should be referenced as **Local.Var01**.

Q44. You have a customer support agent that uses the Microsoft Foundry Agent Service.

Sometimes, customers return to a session days later to continue the same support case, and the agent must resume with the full historical context.

The agent must provide the following:

- Multi-turn continuity within the session Cross-session continuity for the same case
- Access to the full interaction history, including user messages, agent messages, tool calls, and tool outputs

You need to ensure that the agent automatically reloads the complete history on each new turn.

What should you do?

- Create and reuse a conversation by storing the conversation's ID and supplying the ID on subsequent requests.
- Persist only the final model response stored in the client application and prepend the response to future prompts.

C. Enable memory summarization on the agent definition to persist the context automatically.

Answer:

A. Create and reuse a conversation by storing the conversation's ID and supplying the ID on subsequent requests.

Explanation:

Correct Answer:

A. Create and reuse a conversation by storing the conversation's ID and supplying the ID on subsequent requests.

Microsoft Foundry Agent Service maintains the complete conversation history within a **conversation (thread)**. By storing the **conversation ID** and supplying it on subsequent requests, the agent automatically reloads the entire interaction history, including **user messages, agent responses, tool calls, and tool outputs**. This provides both **multi-turn continuity** and **cross-session continuity**, allowing users to resume a support case days later without manually rebuilding the context.

Incorrect Answers:

B. Persist only the final model response stored in the client application and prepend the response to future prompts.

Storing only the final response loses the complete conversation history, including earlier messages, tool calls, and tool outputs.

C. Enable memory summarization on the agent definition to persist the context automatically.

Memory summarization stores condensed information rather than the complete interaction history. It does not guarantee that every user message, agent message, tool call, and tool output will be available when the conversation resumes.

Q45. You have a Microsoft Foundry Agent Service project that contains an HR onboarding agent. Employees frequently pause conversations and continue them several days later.

The solution must meet the following requirements:

- Resume the conversation exactly where it ended.
- Ensure the agent can reference previous tool invocations and uploaded documents.
- Avoid manually reconstructing conversation history.

What should you do?

- A. Create a new thread for each user request and copy previous responses into the prompt.
- B. Persist the thread ID and continue subsequent requests by using the existing thread.
- C. Store conversation summaries in persistent memory and start a new thread for every session.
- D. Save only the final assistant response and include it as context in future requests.

Answer:

B. Persist the thread ID and continue subsequent requests by using the existing thread.

Explanation:

Correct Answer:

B. Persist the thread ID and continue subsequent requests by using the existing thread.

In Microsoft Foundry Agent Service, a **thread** maintains the complete conversation history, including **user messages, assistant responses, tool invocations, tool outputs, and uploaded documents**. By persisting the **thread ID** and reusing it for future requests, the agent automatically resumes the conversation exactly where it left off without requiring the application to rebuild the context. This satisfies all the stated requirements, including preserving uploaded documents and previous tool interactions.

Incorrect Answers:**A. Create a new thread for each user request and copy previous responses into the prompt.**

Creating a new thread loses the complete execution history, including tool invocations and uploaded documents, and requires unnecessary manual reconstruction.

C. Store conversation summaries in persistent memory and start a new thread for every session.

Conversation summaries preserve only condensed context and do not retain the complete history of tool calls, uploaded documents, and detailed interactions.

D. Save only the final assistant response and include it as context in future requests.

Saving only the final response loses earlier user messages, tool executions, and uploaded documents, preventing the agent from fully resuming the conversation.

Q46. You have a Microsoft Foundry project that processes regulatory compliance documents submitted by regional offices.

You need to implement two pipelines by using Content Understanding in Microsoft Foundry Tools.

The solution must meet the following requirements:

- Include a pipeline named Pipeline1 that extracts key compliance information from individual audit reports while minimizing processing costs.
- Include a pipeline named Pipeline2 that analyzes multiple related audit reports together to identify inconsistencies across reporting periods by using advanced reasoning.

How should you configure each pipeline?

To answer, drag the appropriate configurations to the correct pipelines. Each configuration may be used once, more than once, or not at all.

Configurations:

Multi-file task in pro mode

Multi-file task in standard mode

Single-file task in pro mode

Single-file task in standard mode

Pipeline1:

Pipeline2:

Answer:

Configurations:

Multi-file task in pro mode

Multi-file task in standard mode

Single-file task in pro mode

Single-file task in standard mode

Pipeline1: **Single-file task in standard mode**Pipeline2: **Multi-file task in pro mode****Explanation:****Correct Answers:****Pipeline1: Single-file task in standard mode**

Pipeline1 processes **individual audit reports** and the primary goal is to **minimize processing costs**. Since each document is processed independently and does not require advanced reasoning or cross-document analysis, a **single-file task in standard mode** is the most cost-effective configuration. It is optimized for high-volume document extraction without the additional cost of advanced reasoning capabilities.

Pipeline2: Multi-file task in pro mode

Pipeline2 must analyze **multiple related audit reports together** to identify **inconsistencies across reporting periods** using **advanced reasoning**. This requires comparing information across multiple documents and performing multi-step reasoning. A **multi-file task in pro mode** is specifically designed for these advanced cross-document analysis scenarios.

Incorrect Answers:**Multi-file task in standard mode**

Although it processes multiple documents, Standard mode does not provide the advanced reasoning capabilities required to identify inconsistencies across reporting periods.

Single-file task in pro mode

Pro mode supports advanced reasoning, but a single-file task cannot compare multiple related audit reports, so it does not meet the cross-document analysis requirement.

Q47. You have a Microsoft Foundry project that contains a high-traffic agent. After a recent update, operational costs increase significantly.

Monitoring confirms that the volume of user traffic to the agent remains unchanged.

You suspect that changes to the request or response characteristics are causing the increase.

You need to identify whether the additional costs are driven by the model input size, the model output size, or expanded tool usage.

Which observability capability should you use?

- A. latency
- B. evaluation metrics
- C. run success rate
- D. token usage

Answer:**D. token usage****Explanation:****Correct Answer:****D. token usage**

Token usage provides detailed insights into the number of input tokens, output tokens, and tokens consumed during tool calls. Since user traffic has remained unchanged but operational costs have increased, the most likely cause is that requests or responses have become larger or that the agent is using more tools. Token usage metrics allow you to determine whether the increase in cost is driven by larger prompts, longer model responses, or expanded tool usage, making it the correct observability capability for this scenario.

Incorrect Answers:**A. latency**

Latency measures how long requests take to complete. It does not explain increases in token consumption or operational costs.

B. evaluation metrics

Evaluation metrics assess response quality, such as groundedness, fluency, or similarity. They do not measure the factors that drive model costs.

C. run success rate

Run success rate indicates whether agent runs complete successfully. It does not provide information about token consumption or cost increases.

Q48. You have a Microsoft Foundry project that contains an Azure Storage account used to store documents for Retrieval-Augmented Generation (RAG).

The agents authenticate by using managed identities.

The agents must be able to read blob data but must not upload, modify, or delete files.

Which role-based access control (RBAC) role should you assign to the managed identities?

- A. Storage Blob Data Owner
- B. Storage Blob Data Contributor
- C. Storage Blob Data Reader
- D. Contributor

Answer:**C. Storage Blob Data Reader****Explanation:****Correct Answer:****C. Storage Blob Data Reader**

The **Storage Blob Data Reader** role grants **read-only access** to blob data in an Azure Storage account. It allows managed identities to read and download blobs, which is exactly what Retrieval-Augmented Generation (RAG) agents need to retrieve documents. At the same time, it prevents uploading, modifying, or deleting blobs, satisfying the principle of least privilege and meeting the security requirements.

Incorrect Answers:**A. Storage Blob Data Owner**

This role provides full control over blob data, including reading, writing, deleting, and managing permissions, which exceeds the required permissions.

B. Storage Blob Data Contributor

This role allows reading, creating, modifying, and deleting blob data. It grants write permissions that are not required.

D. Contributor

The Contributor role provides management access to Azure resources but does not specifically grant blob data access. It also grants far more permissions than are needed for read-only document retrieval.

Q49. You have a Microsoft Foundry project that contains an agent.

You use a GitHub Actions workflow for CI/CD.

You need to configure the workflow to automatically evaluate the agent when a pull request (PR) is created and prevent branches from merging if the evaluation results do NOT meet the defined thresholds.

How should you configure the workflow?

Authentication method:

- A personal access token (PAT)
- A user-assigned managed identity
- An Azure Login action that uses OpenID Connect (OIDC)

If the evaluation results are NOT met, configure the workflow to:

- Lock the target branch
- Send an alert
- Fail

Answer:

Authentication method:

- A personal access token (PAT)
- A user-assigned managed identity
- An Azure Login action that uses OpenID Connect (OIDC)**

If the evaluation results are NOT met, configure the workflow to:

- Lock the target branch
- Send an alert
- Fail**

Explanation:**Correct Answers:****Authentication method: An Azure Login action that uses OpenID Connect (OIDC)**

The recommended authentication method for GitHub Actions is **Azure Login with OpenID Connect (OIDC)**. OIDC enables GitHub Actions to authenticate securely to Azure without storing long-lived credentials such as secrets or personal access tokens. It follows Microsoft's recommended security best practices and is the preferred authentication mechanism for CI/CD workflows deploying Microsoft Foundry resources.

If the evaluation results are NOT met, configure the workflow to: Fail

Configuring the workflow to **Fail** ensures that the GitHub Actions workflow stops when evaluation thresholds are not met. Combined with GitHub branch protection rules that require the workflow to succeed before merging, this prevents pull requests from being merged until the evaluation passes, satisfying the stated requirement.

Incorrect Answers:**A personal access token (PAT)**

A PAT is a long-lived credential that requires manual management and is not the recommended authentication method for Azure resources.

A user-assigned managed identity

Managed identities are intended for Azure-hosted resources and cannot be used directly by GitHub-hosted runners.

Lock the target branch

Locking a branch prevents all updates to the branch rather than preventing merges based on evaluation results.

Send an alert

Sending an alert notifies administrators but does not stop the pull request from being merged.

Q50. You have a Microsoft Foundry project that contains an agent. The agent uses Azure Speech in Foundry Tools. You fine-tune a baseline speech to text model for the en-us locale and publish the model.

The agent calls the Speech to text REST API and returns an error message indicating that the project ID is invalid. You need to set the project property to the correct ID.

To what should you set the project property?

- A. the project URL
- B. the custom speech project ID
- C. the project ID
- D. the custom speech endpoint URL

Answer:

B. the custom speech project ID

Explanation:

Correct Answer:

B. the custom speech project ID

When using a **custom Speech-to-Text model** that has been trained and published in Azure Speech, the Speech-to-Text REST API expects the **project** property to contain the **Custom Speech project ID**. This unique identifier associates the request with the correct custom speech project and its published models. If an invalid or incorrect ID is provided, the service returns an error indicating that the project ID is invalid.

Incorrect Answers:

A. the project URL

The REST API expects a project ID, not the URL of the project.

C. the project ID

The generic Microsoft Foundry project ID is different from the **Custom Speech project ID** required by the Speech service.

D. the custom speech endpoint URL

The endpoint URL specifies where the request is sent, but it is not used as the value of the project property.

Q51. You have a Microsoft Foundry project named Project1.

Project1 contains an application that processes PDF vendor invoices.

You need to configure Azure Document Intelligence in Foundry Tools to generate a Markdown output that preserves the sections and table structure of the PDFs. The solution must minimize development effort.

What should you do?

- A. Configure output=figures when you analyze the PDF.
- B. Configure content=markdown when you analyze the document.
- C. Increase the confidence threshold.
- D. Set the output_content_format=ContentFormat.MARKDOWN value.

Answer:

D. Set the output_content_format=ContentFormat.MARKDOWN value.

Explanation:

Correct Answer:

D. Set the output_content_format=ContentFormat.MARKDOWN value.

Azure Document Intelligence supports returning document content in **Markdown** format by setting the **output_content_format** parameter to **ContentFormat.MARKDOWN**. This preserves the document's logical

structure, including headings, sections, paragraphs, lists, and tables, making it ideal for Retrieval-Augmented Generation (RAG) and downstream processing. Since this is a built-in capability, it minimizes development effort while producing well-structured Markdown output.

Incorrect Answers:

A. Configure output=figures when you analyze the PDF.

The **figures** output extracts figures and images from the document. It does not generate Markdown or preserve the complete document structure.

B. Configure content=markdown when you analyze the document.

There is no **content=markdown** parameter in the Azure Document Intelligence API for specifying Markdown output.

C. Increase the confidence threshold.

Increasing the confidence threshold affects the acceptance of extracted content but does not change the output format to Markdown.

Q52. A legal research agent must satisfy the following requirements:

- Search contracts uploaded by users during an active conversation.
- Search millions of historical court decisions stored in a centralized enterprise index.

Which combination should you recommend?

- A. File Search for both requirements
- B. Azure AI Search for both requirements
- C. File Search for uploaded contracts and Azure AI Search for historical court decisions
- D. Microsoft Fabric for uploaded contracts and File Search for historical court decisions

Answer:

C. File Search for uploaded contracts and Azure AI Search for historical court decisions

Explanation:

Correct Answer:

C. File Search for uploaded contracts and Azure AI Search for historical court decisions

The **File Search** tool is designed to work with documents that users upload during an active conversation. It automatically indexes the uploaded files and enables the agent to retrieve relevant information from those documents without requiring a separate indexing process. This makes it ideal for searching contracts uploaded during the current chat.

Azure AI Search is designed for large-scale enterprise search scenarios. It can index and search millions of documents, supports advanced indexing, vector search, hybrid search, semantic ranking, and scalable retrieval. This makes it the appropriate choice for searching a centralized repository containing millions of historical court decisions.

Incorrect Answers:

A. File Search for both requirements

File Search is intended for documents uploaded during a conversation. It is not designed to index and search millions of enterprise documents.

B. Azure AI Search for both requirements

Azure AI Search is excellent for enterprise repositories but is not intended for temporary user-uploaded documents within an active conversation, where File Search provides a simpler and more efficient solution.

D. Microsoft Fabric for uploaded contracts and File Search for historical court decisions

Microsoft Fabric is an analytics platform rather than a conversational document retrieval tool, and File Search is not intended for searching millions of historical enterprise documents.



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Q53. You have a Python application named App1 that integrates with a Microsoft Foundry project named Project1.

You need to ensure that App1 meets the following requirements:

- Authenticates by using a Microsoft Entra managed identity
- Sends prompts to a deployed model by using the Azure OpenAI Responses API

How should you complete the Python code?

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
credential =  ()

project_client = AIProjectClient(
    endpoint="https://contosoai.services.ai.azure.com/api/projects/project1",
    credential=credential,
)

with project_client.get_openai_client() as openai_client:
    response = openai_client.responses. (

        model="trail-guide-chat",
        input="Create a 3-day hiking itinerary near Seattle.",
    )
print(response.output_text)
```

Answer:

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
credential =  ()

project_client = AIProjectClient(
    endpoint="https://contosoai.services.ai.azure.com/api/projects/project1",
    credential=credential,
)

with project_client.get_openai_client() as openai_client:
    response = openai_client.responses. (

        model="trail-guide-chat",
        input="Create a 3-day hiking itinerary near Seattle.",
    )
print(response.output_text)
```

Explanation:

Correct Answers:**Credential: DefaultAzureCredential**

DefaultAzureCredential enables the application to authenticate by using **Microsoft Entra ID**, including a **managed identity** when the application is running in Azure. It automatically selects the appropriate authentication mechanism based on the execution environment, making it the recommended credential for applications that use managed identities.

Responses API method: create

The Azure OpenAI **Responses API** uses the `responses.create()` method to send prompts to a deployed model and generate responses. This is the standard method for creating a new model response through the Responses API.

Incorrect Answers:**AzureKeyCredential**

This uses an API key rather than Microsoft Entra managed identity authentication.

ClientSecretCredential

Although it uses Microsoft Entra ID, it requires storing a client secret instead of using a managed identity.

compact

This is not a valid method for sending requests to the Responses API.

retrieve

This retrieves an existing response rather than creating a new one.

Q54. You have a Microsoft Foundry project that generates product marketing images from text prompts. After publishing several images, the legal team at your company identifies a competitor's logo on a sign in the background of an image.

You need to remove only the logo, while preserving the rest of the image.

What should you do?

- A. Apply a mask-based inpainting edit to the part of the image that contains the logo.
- B. Increase the prompt guidance strength.
- C. Modify the original prompt to exclude brand names.
- D. Rerun the prompt by using a different random seed.

Answer:

A. Apply a mask-based inpainting edit to the part of the image that contains the logo.

Explanation:**Correct Answer:**

A. Apply a mask-based inpainting edit to the part of the image that contains the logo.

Mask-based inpainting is designed to modify only a selected region of an existing image. By providing a mask that covers the competitor's logo, the image generation model replaces only that specific area while preserving the rest of the image, including the background, lighting, composition, and surrounding objects. This is the most precise way to remove unwanted content without regenerating the entire image.

Incorrect Answers:**B. Increase the prompt guidance strength.**

Increasing prompt guidance influences how closely the generated image follows the prompt, but it does not selectively remove an existing logo from an image.

C. Modify the original prompt to exclude brand names.

Changing the prompt may reduce the chance of logos appearing in future generations, but it does not edit the existing image.

D. Rerun the prompt by using a different random seed.

Using a different random seed generates a completely new image and does not guarantee that only the logo will be removed while preserving the rest of the image.

Q55. You have a Microsoft Foundry project that contains an agent.

The agent processes scanned maintenance reports that are uploaded by field technicians.

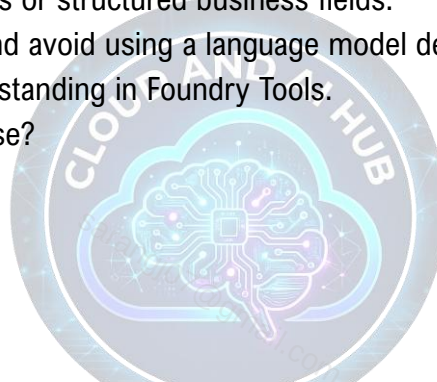
The solution must meet the following requirements:

- Extract all readable text from each document.
- Ignore the document layout, including tables and paragraph positions.
- Do not identify key-value pairs or structured business fields.
- Minimize processing costs and avoid using a language model deployment.

You plan to call Azure Content Understanding in Foundry Tools.

Which built-in analyzer should you use?

- A. `prebuilt-documentFieldSchema`
- B. `prebuilt-layout`
- C. `prebuilt-read`
- D. `prebuilt-documentSearch`

**Answer:****C. prebuilt-read****Explanation:****Correct Answer:****C. prebuilt-read**

The **prebuilt-read** analyzer is designed to perform **optical character recognition (OCR)** and extract all readable text from documents and images. It does not attempt to preserve document layout, detect tables, identify key-value pairs, or extract structured business fields. Because it focuses only on text extraction and does not require a language model deployment, it is the most cost-effective option for this scenario.

Incorrect Answers:**A. prebuilt-documentFieldSchema**

This analyzer extracts structured business fields and returns them according to a predefined schema, which is not required.

B. prebuilt-layout

The prebuilt-layout analyzer preserves document structure, including tables, paragraphs, and layout information, which the requirements specifically state should be ignored.

D. prebuilt-documentSearch

This analyzer is intended for semantic document search scenarios rather than simple OCR-based text extraction.

Q56. You have a Microsoft Foundry project that contains a sales assistant agent. The agent retrieves product information from an Azure AI Search index and generates responses by using a large language model.

After deploying a new version of the agent, users report the following issues:

- The average response time has increased significantly when users ask complex questions.
- Monthly model inference costs have increased even though the number of user requests has remained unchanged.

You need to evaluate each issue.

Which observability signals should you use for each issue?

To answer, drag the appropriate observability signals to the correct issues. Each observability signal may be used once, more than once, or not at all.

Observability Signals:

Groundedness evaluation metrics

Latency breakdown traces

Risk and safety metrics

Token usage analytics

Increased response time:

Increased inference costs:

Answer:**Observability Signals:**

Groundedness evaluation metrics

Latency breakdown traces

Risk and safety metrics

Token usage analytics

Increased response time:

Latency breakdown traces

Increased inference costs:

Token usage analytics

Explanation:**Correct Answers:**

Increased response time: Latency breakdown traces

Latency breakdown traces provide detailed timing information for every stage of an agent run, including Azure AI Search retrieval, tool invocations, prompt construction, and large language model inference. Since response times have increased for complex questions, latency traces help identify which component is introducing the delay and where optimization is needed.

Increased inference costs: Token usage analytics

Token usage analytics show the number of input tokens, output tokens, and token consumption during each request. Because user traffic has remained unchanged while inference costs have increased, token usage

analytics can determine whether the increase is caused by larger prompts, longer model responses, or increased retrieval context.

Incorrect Answers:

Groundedness evaluation metrics

Groundedness metrics determine whether responses are supported by retrieved documents. They do not diagnose latency or inference costs.

Risk and safety metrics

Risk and safety metrics monitor content safety issues and policy violations. They do not provide information about performance or token consumption.

Q57. You have an application named App1 that uses Azure Speech in Foundry Tools to transcribe live calls. Transcript segments often contain both English and Spanish.

App1 sends each segment to Azure Translator in Foundry Tools to translate to another language.

Sometimes, mixed-language segments result in incomplete or incorrect translations.

You need to reduce translation errors. The solution must ensure that the entire transcript is translated successfully.

What should you do before sending the segments to Translator?

- A. Use document translation to translate the entire transcript as a single document.
- B. Split the mixed-language segments into single-language segments and translate each segment separately.
- C. Enable automatic language detection for the translation request.
- D. Specify English as the source language in the translation request for all the segments.

Answer:

B. Split the mixed-language segments into single-language segments and translate each segment separately.

Explanation:

Correct Answer:

B. Split the mixed-language segments into single-language segments and translate each segment separately.

Azure Translator works best when each request contains text in a **single source language**. Mixed-language input can reduce translation accuracy because the service expects a consistent source language for each translation request. By splitting transcript segments into separate English and Spanish portions before sending them to Translator, each segment can be translated accurately, ensuring that the entire transcript is translated successfully.

Incorrect Answers:

A. Use document translation to translate the entire transcript as a single document.

Document Translation is intended for translating complete documents such as PDFs or Office files, not live streaming transcript segments.

C. Enable automatic language detection for the translation request.

Automatic language detection identifies the dominant language in a request. It does not reliably handle segments containing multiple languages.

D. Specify English as the source language in the translation request for all the segments.

Specifying English as the source language causes Spanish portions of the transcript to be translated incorrectly or ignored.

Q58. You have a Microsoft Foundry project that contains an agent used by a customer service application. Following a recent deployment, users report that responses take significantly longer to generate. Monitoring shows that the number of requests per minute has remained constant. You need to determine whether the increased response time is caused by document retrieval, tool execution, or model inference.

Which observability capability should you use?

- A. Evaluation metrics
- B. Latency
- C. Run success rate
- D. Token usage

Answer:

B. Latency

Explanation:

Correct Answer:

B. Latency

Latency observability provides a detailed breakdown of the time spent during each stage of an agent run, including **document retrieval**, **tool execution**, and **model inference**. Since the request volume has remained constant but response times have increased, latency analysis enables you to determine exactly which stage is responsible for the slowdown and target the appropriate optimization.

Incorrect Answers:

A. Evaluation metrics

Evaluation metrics measure response quality, such as groundedness, fluency, and similarity. They do not identify performance bottlenecks.

C. Run success rate

Run success rate indicates whether agent executions succeed or fail, but it does not show where time is being spent during execution.

D. Token usage

Token usage measures the number of input and output tokens consumed. It helps analyze inference costs but does not identify whether delays are caused by document retrieval, tool execution, or model inference.

Q59. You have a Microsoft Foundry project that processes engineering drawings stored in Azure Blob Storage.

Azure AI Content Safety must retrieve images by using blob URLs to evaluate user-submitted images. The solution must meet the following requirements:

- Follow the principle of least privilege.
- Avoid using storage account keys or shared access signatures (SAS).
- Allow Content Safety to read images stored in Azure Blob Storage.

What should you configure for each requirement?

Authentication:

▼

Storage account access keys

Shared Access Signature (SAS)

System-assigned managed identity

User-assigned managed identity

Azure RBAC role:

▼

Storage Blob Data Reader

Storage Blob Data Contributor

Storage Queue Data Contributor

Owner

Answer:

Authentication:

▼

Storage account access keys

Shared Access Signature (SAS)

System-assigned managed identity

User-assigned managed identity

Azure RBAC role:

▼

Storage Blob Data Reader

Storage Blob Data Contributor

Storage Queue Data Contributor

Owner

Explanation**Correct Answers****Authentication: System-assigned managed identity**

A **system-assigned managed identity** enables Azure AI Content Safety to authenticate securely to Azure Blob Storage without using storage account keys or SAS tokens. This satisfies the requirement to avoid shared secrets while leveraging Microsoft Entra ID authentication.

Azure RBAC role: Storage Blob Data Reader

The **Storage Blob Data Reader** role grants **read-only** access to blob data. This follows the **principle of least privilege**, allowing Content Safety to retrieve images for evaluation without permitting uploads, modifications, or deletions.

Incorrect Answers:**Storage account access keys**

These provide full access to the storage account and violate the requirement to avoid storage account keys.

Shared Access Signature (SAS)

Although SAS can provide limited access, the requirement explicitly states to **avoid using SAS**.

User-assigned managed identity

A user-assigned managed identity could work technically, but unless there's a requirement to share the identity across multiple resources, a **system-assigned managed identity** is the simpler and recommended choice.

Storage Blob Data Contributor

This role grants read, write, and delete permissions, exceeding the required read-only access.

Storage Queue Data Contributor

This role applies to Azure Storage Queues, not Blob Storage.

Owner

This grants full management permissions and violates the principle of least privilege.

Q60. You have a Microsoft Foundry project that contains an agent and an image generation model deployment.

The agent creates lifestyle marketing images for clothing products.

Users provide a photograph of a jacket and request new scenes in different environments.

The solution must ensure that the generated images continue to depict the same jacket while allowing the background, lighting, and setting to change.

What should you do?

- A. Include the original image as a reference image in the generation request.
- B. Reduce the max_tokens parameter.
- C. Enable groundedness detection.
- D. Configure Content Safety to annotate the generated images.

Answer:

A. Include the original image as a reference image in the generation request.

Explanation**Correct Answer:****A. Include the original image as a reference image in the generation request.**

Providing the original jacket photo as a **reference image** (image-to-image or reference-guided generation) allows the image generation model to preserve the product's visual characteristics, such as its color, texture, pattern, and design, while generating new scenes with different backgrounds, lighting, and environments. This is the recommended approach for maintaining product identity across generated images.

Incorrect Answers:**B. Reduce the max_tokens parameter.**

The **max_tokens** parameter applies to text generation models and controls the maximum length of the generated text. It has no effect on image generation or visual consistency.

C. Enable groundedness detection.

Groundedness detection evaluates whether generated text is supported by provided source material. It does not help preserve the appearance of an object across generated images.

D. Configure Content Safety to annotate the generated images.

Content Safety is used to detect and annotate potentially harmful or unsafe content. It does not influence how faithfully the model preserves the jacket's appearance.

Q61. You have an Azure Speech in Foundry Tools resource that hosts a custom speech to text model deployed to a custom endpoint. An agent uses the endpoint to perform real-time speech recognition. You are approaching the expiration date of the custom speech to text model. What is the expected behavior when the model expires?

- A. Speech recognition requests will return a 4xx error until a new custom model is deployed.
- B. Speech recognition requests will continue to use the expired custom model until the model is removed manually.
- C. Speech recognition requests will fall back to the most recent base model for the same locale.
- D. The custom model will be deleted automatically when the model expires.

Answer:**C. Speech recognition requests will fall back to the most recent base model for the same locale.****Explanation****Correct Answer:****C. Speech recognition requests will fall back to the most recent base model for the same locale.**

In Azure Speech, when a **custom Speech-to-Text model expires**, it is no longer used for inference. Requests sent to the custom endpoint **automatically fall back to the latest available base model for the same locale**. This ensures that speech recognition continues without interruption, although the accuracy improvements provided by the custom model are no longer available.

Incorrect Answers:**A. Speech recognition requests will return a 4xx error until a new custom model is deployed.**

This is incorrect. The endpoint does not stop functioning or return client errors simply because the custom model expires. Instead, it falls back to the corresponding base model.

B. Speech recognition requests will continue to use the expired custom model until the model is removed manually.

This is incorrect. Once the custom model expires, it is no longer used for inference.

D. The custom model will be deleted automatically when the model expires.

This is incorrect. Expiration prevents the model from being used, but it is not automatically deleted.

Q62. You have a technical support agent that uses the Microsoft Foundry Agent Service.

Customers often upload log files while troubleshooting issues. A support case can remain open for several weeks, and different support engineers may continue the conversation during that time.

The solution must meet the following requirements:

- Maintain the complete conversation history across multiple sessions.
- Preserve uploaded files, tool calls, tool outputs, and agent responses.
- Minimize application code.

What should you do?

- A. Store the thread ID after the initial conversation and provide the same thread ID when continuing the support case.
- B. Store the generated responses in Azure AI Search and prepend them to every new prompt.
- C. Enable persistent agent memory and allow the agent to reconstruct previous conversations automatically.
- D. Store conversation summaries in a database and include the summaries in each new request.

Answer:**A. Store the thread ID after the initial conversation and provide the same thread ID when continuing the support case.****Explanation****Correct Answer:****A. Store the thread ID after the initial conversation and provide the same thread ID when continuing the support case.**

In Microsoft Foundry Agent Service, a **thread** automatically stores the entire conversation state, including user and assistant messages, uploaded files, tool calls, tool outputs and conversation context.

By persisting the **thread ID** and reusing it for subsequent requests, the agent can resume the conversation exactly where it left off, even if different support engineers continue the case weeks later. This approach also **minimizes application code**, since the Agent Service manages the conversation history automatically.

Incorrect Answers:**B. Store the generated responses in Azure AI Search and prepend them to every new prompt.**

This requires manually reconstructing the conversation and does not preserve uploaded files, tool calls, or tool outputs.

C. Enable persistent agent memory and allow the agent to reconstruct previous conversations automatically.

Persistent agent memory is intended for storing long-term user preferences or facts, not complete conversation history with files, tool invocations, and execution state.

D. Store conversation summaries in a database and include the summaries in each new request.

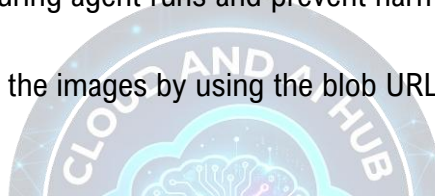
Conversation summaries lose important details such as uploaded files, tool outputs, and the complete execution history. They also require additional application logic.

Q63. You have a Microsoft Foundry project that contains a customer support agent built by using the Foundry Agent Service.

The agent uploads user-provided screenshots to Azure Storage through a ticketing tool and receives a blob URL for additional reasoning.

You need to use image moderation during agent runs and prevent harmful content from being returned during runs.

Azure AI Content Safety must access the images by using the blob URL. The solution must follow the principle of least privilege.



What should you configure for Content Safety?

Guardrails:

- Select Tool call and set Action to Block.
- Select User input and Output and set Action to Annotate.
- Select User input and Tool response and set Action to Annotate.
- Select User input, Output, Tool response, and Tool call and set Action to Block.

Storage access:

- Storage account access keys
- A user-assigned identity that is assigned the Storage Queue Data Contributor role
- A system-assigned managed identity that is assigned the Storage Blob Data Reader role
- A system-assigned managed identity that is assigned the Storage Blob Data Contributor role

Answer:

What should you configure for Content Safety?

Guardrails:

- Select Tool call and set Action to Block.
- Select User input and Output and set Action to Annotate.
- Select User input and Tool response and set Action to Annotate.
- Select User input, Output, Tool response, and Tool call and set Action to Block.**

Storage access:

- Storage account access keys
- A user-assigned identity that is assigned the Storage Queue Data Contributor role
- A system-assigned managed identity that is assigned the Storage Blob Data Reader role**
- A system-assigned managed identity that is assigned the Storage Blob Data Contributor role

Explanation**Correct Answers**

Guardrails: Select User input, Output, Tool response, and Tool call and set Action to Block.

The requirement is to **prevent harmful content from being returned during agent runs**. Setting the Content Safety guardrails to **Block** ensures that harmful content is stopped rather than merely annotated. Applying the guardrails to **User input, Output, Tool response, and Tool call** provides end-to-end protection throughout the agent execution pipeline.

Storage access: A system-assigned managed identity that is assigned the Storage Blob Data Reader role.

Azure AI Content Safety must retrieve images by using the **blob URL**. The solution must also follow the **principle of least privilege**.

A **system-assigned managed identity** eliminates the need for storage account keys or SAS tokens, and the **Storage Blob Data Reader** role grants only the permissions required to read the images—nothing more.

Incorrect Answers

Select Tool call and set Action to Block.

This protects only tool calls and does not inspect user input, tool responses, or model output.

Select User input and Output and set Action to Annotate.

Annotate only detects harmful content but does not prevent it from being returned.

Select User input and Tool response and set Action to Annotate.

This also annotates rather than blocks and does not protect all stages of the pipeline.

Storage account access keys.

These provide excessive permissions and do not follow Microsoft's recommended identity-based authentication.

A user-assigned identity assigned the Storage Queue Data Contributor role.

This role applies to Azure Storage Queues, not Blob Storage.

A system-assigned managed identity assigned the Storage Blob Data Contributor role.

Contributor grants write and delete permissions, exceeding the requirement to only read images.

Q64. You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

You need to improve response completeness. The solution must be implemented in the logic of the application code before responses are returned.

What should you do?

- A. Add a retry evaluation before the responses are returned.
- B. Decrease the value of the max_tokens parameter.
- C. Switch to Retrieval Augmented Generation (RAG).
- D. Replace the model with a smaller deployment.

Answer:

A. Add a retry evaluation before the responses are returned.

Explanation

Correct Answer:

A. Add a retry evaluation before the responses are returned.

The requirement is to **improve response completeness** and to implement the solution **in the application logic before responses are returned**.

A **retry evaluation** enables the application to assess whether the generated summary is sufficiently complete. If the evaluation determines that important information is missing, the application can automatically regenerate or refine the response before returning it to the user. This approach improves completeness without changing the underlying model or retrieval architecture.

Incorrect Answers:

B. Decrease the value of the max_tokens parameter.

Reducing **max_tokens** limits the maximum response length, making summaries **less complete**, not more.

C. Switch to Retrieval Augmented Generation (RAG).

The question states that the agent already **generates summaries from retrieved policy documents**, which implies a retrieval-based workflow is already in use. Switching to RAG would not address the stated requirement.

D. Replace the model with a smaller deployment.

A smaller model generally has lower reasoning and summarization capabilities and is less likely to improve response completeness.

Q65. You have an app named App1 that uses a Microsoft Foundry chat model deployment. App1 accepts free-form text entered directly by users. Some users intentionally submit prompts that attempt to bypass system instructions, reveal hidden prompts, or manipulate tool execution. You need to detect these direct prompt injection attempts before the model processes the user's input. What should you use?

- A. Image moderation
- B. Prompt Shields for documents
- C. Protected material text
- D. Prompt Shields for user prompts

Answer:

D. Prompt Shields for user prompts

Explanation

Correct Answer:

D. Prompt Shields for user prompts

Prompt Shields for user prompts are specifically designed to detect and mitigate **direct prompt injection attacks**, where users intentionally try to override or ignore system instructions, reveal hidden prompts, manipulate tool execution and change the behavior of the model.

Since App1 accepts **free-form text entered directly by users**, Prompt Shields for user prompts is the correct protection mechanism.

Incorrect Answers:

A. Image moderation

Image moderation detects harmful or inappropriate content in images. It does not analyze text prompts or detect prompt injection attacks.

B. Prompt Shields for documents

Prompt Shields for documents protect against **indirect prompt injection**, where malicious instructions are embedded in external content such as documents, web pages, or retrieved files. They do not protect against attacks entered directly by users.

C. Protected material text

Protected material detection identifies copyrighted or protected text in model outputs. It does not detect prompt injection attempts.

Q66. You have a Microsoft Foundry project that contains an agent that answers employee questions by using Retrieval-Augmented Generation (RAG).

After deployment, users report the following issues:

- The retrieved documents are relevant, but the generated answers do not address the user's question.
- Responses generated by the agent are significantly longer than expected, increasing inference costs.

You need to determine which diagnostic capability to use for each issue.

Which diagnostic capability should you use?

To answer, drag the appropriate diagnostic capabilities to the correct issues. Each diagnostic capability may be used once, more than once, or not at all.

Diagnostic capabilities:

Relevance Evaluation

Completion token analytics

Groundedness evaluation

Prompt traces

Latency timeline

Answers do not address the user's question:

Responses are longer than expected:

Answer:**Diagnostic capabilities:**

Relevance Evaluation

Completion token analytics

Groundedness evaluation

Prompt traces

Latency timeline

Answers do not address the user's question:

Relevance Evaluation

Responses are longer than expected:

Completion token analytics

Explanation**Correct Answers****Answers do not address the user's question: Relevance Evaluation**

The retrieved documents are already **relevant**, so the retrieval stage is working correctly. The problem is that the **generated answer does not address the user's question**. **Relevance Evaluation** measures whether the model's response is relevant to the user's query, making it the appropriate diagnostic capability.

Responses are longer than expected: Completion token analytics

Longer-than-expected responses directly increase inference costs because more **completion tokens** are generated. **Completion token analytics** helps identify excessive output token usage and determine whether long model responses are driving the higher costs.

Incorrect Answers**Groundedness evaluation**

Groundedness checks whether the response is supported by the retrieved documents. In this scenario, the documents are already relevant. The issue is that the response does not answer the user's question, making **Relevance Evaluation** the better choice.

Prompt traces

Prompt traces help inspect prompts and model interactions for debugging but do not directly evaluate response relevance or output token usage.

Latency timeline

Latency timeline measures execution time across retrieval, tool execution, and inference. It is useful for diagnosing performance issues, not response quality or inference costs.

Q67. You have a Microsoft Foundry project named Project1 that contains the following: An OpenAPI tool that calls an external API
 A project connection named Connection1 that stores the API key of the external API
 When an agent calls the OpenAPI tool, the API returns a 401 unauthorized error, and traces show that the API key header is NOT being sent.
 You need to ensure that the OpenAPI tool automatically includes the API key from Connection1 on all requests.
 What should you do?

- A. Enable identity passthrough so that the tool uses the Microsoft Entra token of the caller.
- B. Add the API key header manually to the OpenAPI specification.
- C. Configure the tool to use the default connection of Project1.
- D. Connect the tool to Connection1.

Answer:

D. Connect the tool to Connection1.

Explanation

Correct Answer:

D. Connect the tool to Connection1.

In Microsoft Foundry, **project connections** securely store credentials such as API keys. When an OpenAPI tool is **associated with a project connection**, the Agent Service automatically retrieves the API key from the connection and injects it into the request according to the **API key security scheme** defined in the OpenAPI specification.

Since the traces show that the API key header is **not being sent**, the OpenAPI tool is not using **Connection1**. Connecting the tool to **Connection1** ensures the API key is automatically included in every request.

Incorrect Answers

A. Enable identity passthrough so that the tool uses the Microsoft Entra token of the caller.

Identity passthrough is used for Microsoft Entra ID authentication, not for APIs that require an API key.

B. Add the API key header manually to the OpenAPI specification.

Manually adding the header hardcodes authentication behavior and does not use the secure API key stored in **Connection1**. The recommended approach is to use a project connection together with an API key security scheme.

C. Configure the tool to use the default connection of Project1.

The question specifies that the API key is stored in **Connection1**. Using a default connection does not guarantee that the correct credentials will be supplied.

Q68. You have a Microsoft Foundry project that contains a customer support agent.

After deploying a new prompt template, administrators observe that Azure OpenAI costs have increased significantly, even though the number of user requests has remained constant.

You need to determine whether the prompt or generated responses are consuming more tokens than expected.

Which observability signal should you review?

- A. Groundedness evaluation metrics
- B. Retrieval evaluation metrics
- C. Token usage analytics
- D. Agent execution timeline

Answer:

C. Token usage analytics

Explanation

Correct Answer:

C. Token usage analytics

Token usage analytics provides detailed information about the number of **input tokens (prompt tokens)** and **output tokens (completion tokens)** consumed by each request.

Since the **number of user requests has remained constant**, and **Azure OpenAI costs have increased significantly**, the most likely cause is that either the new prompt template is sending **more input tokens**, or the model is generating **more output tokens**.

Token usage analytics allows you to identify exactly whether the increase is due to larger prompts, longer responses, or both.

Incorrect Answers

A. Groundedness evaluation metrics

Groundedness metrics determine whether responses are supported by retrieved documents. They do not measure token consumption or explain increased model costs.

B. Retrieval evaluation metrics

Retrieval evaluation metrics assess the quality of document retrieval, such as whether relevant documents were retrieved. They do not show prompt or completion token usage.

D. Agent execution timeline

The execution timeline shows the duration of different stages of an agent run, such as retrieval, tool execution, and model inference. It helps diagnose performance issues, not token consumption or inference costs.

Q69. You have a Microsoft Foundry project that contains a customer support agent. The agent uses Azure AI Search as a knowledge source to answer questions by using Retrieval-Augmented Generation (RAG).

After a recent deployment, users report the following issues:

- The agent occasionally generates responses that are not supported by the retrieved documents.
- Support engineers want the agent to return a fallback response instead of generating an answer when insufficient grounding evidence is available.

You need to identify the evaluation metric and the recommended configuration to address the issues.

What should you configure?

Evaluation metric:

▼
Coherence
Fluency
Groundedness
Similarity

Recommended configuration:

▼
Increase the model temperature.
Reject responses that do not meet the groundedness threshold.
Increase the maximum completion tokens.
Fine-tune the model by using the retrieved documents.

Answer:

Evaluation metric:

▼
Coherence
Fluency
Groundedness
Similarity

Recommended configuration:

▼
Increase the model temperature.
Reject responses that do not meet the groundedness threshold.
Increase the maximum completion tokens.
Fine-tune the model by using the retrieved documents.

Explanation

Correct Answers

Evaluation metric: Groundedness

The issue states that the agent **occasionally generates responses that are not supported by the retrieved documents**.

This is exactly what the **Groundedness** evaluation metric measures. It determines whether the generated response is supported by the retrieved knowledge source rather than containing hallucinations or unsupported claims.

Recommended configuration: Reject responses that do not meet the groundedness threshold.

The second requirement is **Return a fallback response instead of generating an answer when insufficient grounding evidence is available**.

The appropriate configuration is to **reject responses that fail the groundedness threshold**. When a response is not sufficiently grounded, the application can return a fallback message such as *"I couldn't find enough information to answer that question."* instead of returning a potentially hallucinated response.

Incorrect Answers

Increase the model temperature.

A higher temperature makes responses more creative and less deterministic, increasing the likelihood of unsupported answers.

Increase the maximum completion tokens.

Increasing completion tokens only allows longer responses. It does not improve grounding or reduce hallucinations.

Fine-tune the model by using the retrieved documents.

RAG is designed to ground responses using retrieved documents at inference time. Fine-tuning is unnecessary for this requirement and does not ensure that every response is grounded in the retrieved content.

Q70. You have a Microsoft Foundry project that contains an agent.

Your organization needs to detect suspicious sign-in activity, investigate security incidents, and correlate alerts generated from Microsoft Foundry, Microsoft Defender, and Microsoft Entra ID.

The solution must provide a centralized security operations platform.

Which two components should you use?

- A. Microsoft Sentinel
- B. Log Analytics workspace
- C. Azure Monitor Agent
- D. Application Insights
- E. OpenTelemetry

Answer:

A. Microsoft Sentinel

B. Log Analytics workspace

Explanation**Correct Answers**

A. Microsoft Sentinel

Microsoft Sentinel is Microsoft's cloud-native **SIEM (Security Information and Event Management)** and **SOAR (Security Orchestration, Automation, and Response)** platform.

It enables security teams to detect suspicious sign-in activity, investigate security incidents, correlate alerts across multiple services and automate incident response

Since the requirement is for a **centralized security operations platform**, Microsoft Sentinel is the correct choice.

B. Log Analytics workspace

A **Log Analytics workspace** serves as the data repository for Sentinel.

It collects logs from Microsoft Foundry, Microsoft Defender, Microsoft Entra ID, Azure resources and other connected data sources

Sentinel uses the Log Analytics workspace to query, correlate, and investigate security events.

Incorrect Answers

C. Azure Monitor Agent

Azure Monitor Agent collects telemetry from Azure VMs and servers, but it is **not** the centralized security platform or incident correlation engine required in this scenario.

D. Application Insights

Application Insights monitors application performance, availability, requests, dependencies, and exceptions. It is intended for application observability, not security operations.

E. OpenTelemetry

OpenTelemetry provides a vendor-neutral framework for collecting traces, metrics, and logs for application observability. It does not provide SIEM capabilities or security incident correlation.

Q71. You have a Microsoft Foundry project that contains a model deployment.

You have an application that calls the deployment by using the Azure OpenAI v1 API and

DefaultAzureCredential. The developers at your company receive HTTP 403 errors when they send inference requests, even after running az login.

You need to ensure that the developers can perform model inference. The solution must follow the principle of least privilege.

Which role-based access control (RBAC) role should you assign to the developers?

- A. Cognitive Services User
- B. Cognitive Services OpenAI User
- C. Contributor
- D. Cognitive Services Data Reader

Answer:

B. Cognitive Services OpenAI User

Explanation

Correct Answer:

B. Cognitive Services OpenAI User

The application authenticates by using **DefaultAzureCredential**, which relies on **Microsoft Entra ID** for authentication. Although the developers have successfully authenticated by running **az login**, they receive

HTTP 403 Forbidden errors because they do not have sufficient permissions to perform inference on the Azure OpenAI model deployment.

The **Cognitive Services OpenAI User** role grants the minimum permissions required to invoke Azure OpenAI models. It allows developers to perform inference requests without granting unnecessary management permissions, thereby following the principle of least privilege.

Incorrect Answers:

A. Cognitive Services User

This role provides access to Azure AI services in general but does not specifically grant the permissions required to perform inference against Azure OpenAI model deployments.

C. Contributor

The Contributor role grants broad management permissions, including creating, modifying, and deleting Azure resources. This exceeds the permissions required for developers who only need to perform model inference.

D. Cognitive Services Data Reader

This role provides read-only access to resource data and configuration. It does not allow users to invoke Azure OpenAI model deployments.

Q72. You have an app named App1 that uses a Microsoft Foundry multimodal model deployment. App1 runs optical character recognition (OCR) on uploaded images and appends the OCR output to the prompt as additional context.

Some uploaded images contain embedded text.

You need to prevent potentially malicious instructions from being processed by the model.

What should you use?

- A. image moderation
- B. prompt shields for documents
- C. protected material text
- D. prompt shields for user prompts

Answer:

B. Prompt Shields for documents

Explanation

Correct Answer:

B. Prompt Shields for documents

The application performs **OCR on uploaded images** and appends the extracted text to the prompt before sending it to the multimodal model. If an attacker embeds malicious instructions within the image, those instructions become part of the prompt after OCR. This is known as an **indirect prompt injection attack**, also referred to as a **document attack**.

Prompt Shields for documents are specifically designed to detect and mitigate indirect prompt injection attacks originating from documents, images, and other external content. They analyze the extracted content before it reaches the model and help prevent malicious embedded instructions from influencing the model's behavior.

Incorrect Answers:**A. Image moderation**

Image moderation detects harmful visual content such as violence, sexual content, or graphic imagery. It does not detect malicious instructions embedded as text within images.

C. Protected material text

Protected material text identifies copyrighted or protected text. It does not protect against prompt injection attacks.

D. Prompt Shields for user prompts

Prompt Shields for user prompts protect against **direct prompt injection**, where malicious instructions are typed directly into the user's prompt. They do not protect against instructions extracted from uploaded documents or images.

Q73. You need to recommend a solution for a legal research agent by using the Microsoft Foundry Agent Service.

The solution must meet the following requirements:

- Retain attorney preferences across multiple conversations.
- Enable the agent to retrieve reference material from documents uploaded during a conversation.

Which Foundry capability should you recommend for each requirement?

To retain attorney preferences across conversations, use:

Agent memory that uses persistent storage
Conversation history
Orchestration-managed session context
Prompt cache

To retrieve uploaded documents, use:

File search tool
Code interpreter tool
Computer use tool
Azure AI Translator

Answer:

To retain attorney preferences across conversations, use:

Agent memory that uses persistent storage
Conversation history
Orchestration-managed session context
Prompt cache

To retrieve uploaded documents, use:

File search tool
Code interpreter tool
Computer use tool
Azure AI Translator

Explanation**Correct Answers**

The correct answers are **Agent memory that uses persistent storage** and **File search tool**.

To retain attorney preferences across multiple conversations, the agent must use **persistent agent memory**. Unlike conversation history or session context, persistent memory stores user preferences across sessions, allowing the agent to remember long-term information such as preferred legal terminology, citation style, or document formatting preferences.

To retrieve reference material from documents uploaded during a conversation, the **File search tool** is the appropriate capability. It indexes uploaded files and enables the agent to search their contents during the conversation, making it ideal for legal research scenarios where users upload contracts, case files, or other reference documents.

Incorrect Answers:

Conversation history is incorrect because it maintains the messages within a conversation or thread but is not intended for storing long-term user preferences across multiple conversations.

Orchestration-managed session context is incorrect because it exists only for the duration of the current session and is discarded afterward.

Prompt cache is incorrect because it is an optimization feature that reduces latency and cost by caching repeated prompts. It does not store user preferences.

Code interpreter tool is incorrect because it is designed to execute Python code and analyze data rather than search uploaded documents.

Computer use tool is incorrect because it automates interactions with graphical user interfaces and is unrelated to document retrieval.

Azure AI Translator is incorrect because it translates text and documents but does not retrieve information from uploaded files.

Q74. You have a Microsoft Foundry project named Project1.

Project1 contains an application that processes scanned engineering drawings.

You need to configure Azure Document Intelligence in Foundry Tools to extract the document content while preserving tables, headings, paragraphs, and reading order. The extracted content will be passed directly to a large language model for summarization.

The solution must minimize custom post-processing.

What should you do?

- A. Configure output=figures when analyzing the document.
- B. Set output_content_format=ContentFormat.MARKDOWN.
- C. Configure string_index_type=UnicodeCodePoint.
- D. Increase the OCR resolution before processing.

Answer:

B. Set output_content_format=ContentFormat.MARKDOWN.

Explanation**Correct Answer:**

B. Set output_content_format=ContentFormat.MARKDOWN.

The application needs to extract content from scanned engineering drawings while preserving **tables, headings, paragraphs, and reading order**. The extracted content will be passed directly to a large language model, so the output should be structured and require minimal post-processing.

Setting **output_content_format=ContentFormat.MARKDOWN** instructs Azure Document Intelligence to return the extracted content in **Markdown** format. Markdown preserves the logical structure of the document, including headings, tables, paragraphs, and reading order, making it ideal for Retrieval-Augmented Generation (RAG) and LLM summarization scenarios.

Incorrect Answers:**A. Configure output=figures when analyzing the document.**

This option extracts figures and images from the document. It does not preserve the overall document structure or produce structured text suitable for an LLM.

C. Configure string_index_type=UnicodeCodePoint.

This setting controls how text offsets are represented in the response. It does not affect the extracted content or preserve tables, headings, or reading order.

D. Increase the OCR resolution before processing.

Increasing OCR resolution may improve text recognition quality for low-quality scans, but it does not produce structured output or reduce the need for custom post-processing.

Q75. You have a Microsoft Foundry project that contains an Azure AI Search resource used to ground multiple agents.

Developers at your company use Microsoft Entra ID authentication to query the search index from a custom application. They receive HTTP 403 (Forbidden) errors when executing search queries.

You need to grant the minimum permissions required for developers to query the search index.

Which role-based access control (RBAC) role should you assign?

- A. Search Service Contributor
- B. Search Index Data Reader
- C. Contributor
- D. Reader

Answer:**B. Search Index Data Reader****Explanation****Correct Answer:****B. Search Index Data Reader**

The developers are using **Microsoft Entra ID authentication** to query an **Azure AI Search index**. An HTTP **403 (Forbidden)** error indicates that authentication has succeeded, but the developers do not have sufficient permissions to perform search queries.

The **Search Index Data Reader** role grants the minimum permissions required to query documents in an Azure AI Search index. It follows the principle of least privilege by allowing users to execute search operations without granting permissions to modify the search service or indexes.

Incorrect Answers:**A. Search Service Contributor**

This role allows users to create, modify, and manage Azure AI Search resources and indexes. It grants far more permissions than are required for simply querying the index.

C. Contributor

The Contributor role provides broad Azure resource management permissions, including creating, updating, and deleting resources. This exceeds the permissions needed to perform search queries.

D. Reader

The Reader role allows users to view Azure resource metadata but does not grant permission to query data stored in the search index.

Q76. You have a Microsoft Foundry project that contains an agent used by an insurance company.

You need to enhance the agent by using built-in tools. The solution must meet the following requirements:

- Allow the agent to analyze claim spreadsheets uploaded during conversations and generate charts.
- Allow the agent to interact with a legacy claims management website that does not expose APIs.
- Allow the agent to answer questions by using policy documents uploaded directly to the current conversation.

What should you use for each requirement?

To answer, drag the appropriate tools to the correct requirements. Each tool may be used once, more than once, or not at all.

Tools:

Analyze uploaded spreadsheets and generate charts:

Interact with the legacy claims management website:

Retrieve information from uploaded policy documents:

Answer:**Tools:**

Analyze uploaded spreadsheets and generate charts:

Interact with the legacy claims management website:

Retrieve information from uploaded policy documents:

Explanation**Correct Answers:**

Analyze uploaded spreadsheets and generate charts: Code Interpreter

The **Code Interpreter** tool is designed to analyze uploaded files such as spreadsheets. It can execute Python code to process claim data, perform calculations, and generate charts or other visualizations during the conversation.

Interact with the legacy claims management website: Computer Use

The **Computer Use** tool enables an agent to interact with applications and websites that do not expose APIs. It can navigate web pages, click buttons, enter text, and perform other user interface actions, making it ideal for legacy web applications.

Retrieve information from uploaded policy documents: File Search

The **File Search** tool indexes documents uploaded during the current conversation and allows the agent to retrieve relevant information from those files when answering user questions.

Incorrect Answers:

Grounding with Bing Search

This tool retrieves information from public websites on the internet. It cannot search documents uploaded during a conversation or interact with internal legacy applications.

Microsoft Fabric

Microsoft Fabric is an analytics platform for enterprise data workloads. It is not intended for analyzing uploaded spreadsheets during a chat, interacting with websites, or retrieving information from conversation-uploaded documents.

Q77. You have a Microsoft Foundry project that contains an agent. The agent has a Model Context Protocol (MCP) tool that queries a knowledge base stored in Azure AI Search. Some agent runs return answers from the base model without invoking the knowledge base, which results in responses without grounded citations.

You are provided with the following code snippet that runs the agent.

```
run = project_client.agents.runs.create_and_process(
    thread_id=thread.id,
    agent_id=agent.id
)
```

You need to add the correct tool _choice parameter to the code to deterministically force the agent to invoke the MCP tool on each run.

What should you add?

- A. tool_choice= "required"
- B. tool_choice= "auto"
- C. tool_choice= "type": "knowledge_base"
- D. tool_choice = "type": "mcp"

Answer:

D. tool_choice={"type": "mcp"}

Explanation

Correct Answer:

D. tool_choice={"type": "mcp"}

The requirement is to **deterministically force the agent to invoke the MCP tool** on every run. Setting **tool_choice={"type":"mcp"}** explicitly instructs the agent to use the **Model Context Protocol (MCP)** tool instead of allowing the model to decide whether to call a tool. This guarantees that the agent queries the Azure AI Search knowledge base through the MCP tool before generating a response, ensuring grounded answers with citations.

Incorrect Answers:

A. tool_choice="required"

This forces the agent to call **a tool**, but it does not guarantee that the **MCP tool** will be selected if multiple tools are available.

B. tool_choice="auto"

This is the default behavior. The model decides whether to call a tool, which is exactly the problem described in the scenario because it sometimes answers directly from the base model.

C. tool_choice={"type":"knowledge_base"}

There is no supported **knowledge_base** tool type for the tool_choice parameter. Since the knowledge base is exposed through an MCP server, the correct tool type is **mcp**.

Q78. A Microsoft Foundry agent invokes three custom tools. Administrators report that users occasionally experience response delays. They need to determine which tool contributes the most to the total response time. Which diagnostic capability should they review?

- A. Token usage analytics
- B. Tool execution traces
- C. Groundedness evaluation
- D. Conversation history

Answer:

B. Tool execution traces

Explanation

Correct Answer:

B. Tool execution traces

The administrators need to determine **which of the three custom tools contributes the most to the overall response time**. **Tool execution traces** provide detailed diagnostics for each tool invocation, including when a tool starts, how long it executes, whether it succeeds or fails, and how much time each tool adds to the overall agent run. This makes it possible to identify the slowest tool and pinpoint the source of response delays.

Incorrect Answers:

A. Token usage analytics

Token usage analytics measures the number of input and output tokens consumed by the language model. It helps analyze inference costs but does not identify which tool is causing latency.

C. Groundedness evaluation

Groundedness evaluation measures whether the generated response is supported by retrieved data. It evaluates response quality rather than tool performance or execution time.

D. Conversation history

Conversation history records the messages exchanged during a conversation. It does not provide timing or performance information for individual tool executions.

Q79. You need to recommend a solution for a customer service agent by using the Microsoft Foundry Agent Service.

The agent must meet the following requirements:

- Execute Python code to analyze spreadsheets uploaded by users during conversations.
- Access current weather conditions from public websites without requiring users to upload documents.

Which Foundry capability should you recommend for each requirement?

To analyze uploaded spreadsheets, use:

Code Interpreter Tool
File Search Tool
Computer Use Tool
Azure AI Search Tool

To retrieve current weather information, use:

Grounding with Bing Search
File Search Tool
Computer Use Tool
Microsoft Fabric tool

Answer:

To analyze uploaded spreadsheets, use:

Code Interpreter Tool
File Search Tool
Computer Use Tool
Azure AI Search Tool

To retrieve current weather information, use:

Grounding with Bing Search
File Search Tool
Computer Use Tool
Microsoft Fabric tool

Explanation

Correct Answers:**To analyze uploaded spreadsheets: Code Interpreter Tool**

The **Code Interpreter Tool** enables the agent to execute Python code on user-uploaded files. It can analyze spreadsheets, perform calculations, manipulate data, and generate charts or other visualizations during the conversation.

To retrieve current weather information: Grounding with Bing Search

Grounding with Bing Search allows the agent to retrieve current, real-time information from public websites, including weather conditions, without requiring users to upload documents. It is specifically designed to ground responses using live web content.

Incorrect Answers:**File Search Tool**

The File Search Tool retrieves information from files uploaded during the conversation. It cannot access live information from public websites such as current weather.

Computer Use Tool

The Computer Use Tool automates interactions with graphical user interfaces and websites. It is intended for operating applications or websites that require user interface interactions, not for retrieving grounded public web information.

Azure AI Search Tool

Azure AI Search queries enterprise search indexes. It is not intended for analyzing uploaded spreadsheets or retrieving current public weather information.

Microsoft Fabric Tool

Microsoft Fabric is an enterprise analytics platform and is not used to retrieve live weather information during agent conversations.

Q80. You have a Microsoft Foundry project that contains an agent used by an internal HR application. Following a prompt update, administrators notice that users are reporting more incomplete and unsupported answers. The application continues to process requests successfully, and response times remain unchanged. You need to determine whether the quality of the generated responses has degraded.

Which observability capability should you use?

- A. Token usage
- B. Latency
- C. Evaluation metrics
- D. Run success rate

Answer:**C. Evaluation metrics****Explanation****Correct Answer:****C. Evaluation metrics**

The issue is that users are receiving **more incomplete and unsupported answers**, indicating that the **quality of the generated responses has degraded**. Since the application continues to process requests successfully and response times have not changed, the problem is related to response quality rather than performance or reliability.

Evaluation metrics measure qualitative aspects of model outputs, such as **groundedness, relevance, completeness, coherence, and fluency**. These metrics help determine whether a prompt update has negatively affected the quality of the generated responses.

Incorrect Answers:

A. Token usage

Token usage measures the number of input and output tokens consumed by the model. It is useful for analyzing inference costs but does not evaluate response quality.

B. Latency

Latency measures how long requests take to complete and helps diagnose performance bottlenecks. The question states that response times remain unchanged, so latency is not the issue.

D. Run success rate

Run success rate indicates whether agent runs complete successfully or fail. Since the application continues to process requests successfully, this metric does not identify degraded response quality.

Q81. You have a Microsoft Foundry project that serves a high-volume chat app. Most requests are simple FAQs, but some require advanced reasoning.

You need to reduce costs and latency for common queries, without degrading the quality of the responses to complex questions.

What should you do?

- A. Route all the requests to a smaller model.
- B. Use a model cascade that routes the requests to different models.
- C. Increase the value of the max_tokens parameter for all the requests.
- D. Route all the requests to the most capable model.

Answer:

B. Use a model cascade that routes the requests to different models.

Explanation

Correct Answer:

B. Use a model cascade that routes the requests to different models.

The application receives a mix of **simple FAQ requests** and **complex reasoning requests**. A **model cascade** intelligently routes each request to the most appropriate model based on its complexity.

Simple FAQ queries can be handled by a **smaller, lower-cost, lower-latency model**, while complex reasoning tasks are routed to a **more capable model**. This reduces overall inference costs and improves response times without sacrificing the quality of responses for more challenging questions.

Incorrect Answers:

A. Route all the requests to a smaller model.

Using only a smaller model would reduce costs and latency, but complex reasoning tasks could receive lower-quality responses.

C. Increase the value of the max_tokens parameter for all the requests.

Increasing max_tokens allows longer responses but also increases inference costs and does not improve routing or latency.

D. Route all the requests to the most capable model.

Using the largest model for every request maintains response quality but increases both costs and latency for simple FAQ requests.

Q82. A financial reporting application generates JSON that is consumed by downstream automation. Occasionally, responses contain unexpected fields that cause processing failures. Which approach should you recommend?

- A. Lower the model temperature.
- B. Use structured outputs with a JSON schema.
- C. Increase the maximum completion tokens.
- D. Fine-tune the model.

Answer:**B. Use structured outputs with a JSON schema.****Explanation****Correct Answer:****B. Use structured outputs with a JSON schema.**

The application generates **JSON that is consumed by downstream automation**, so the output must conform to a predictable structure. If the model occasionally returns unexpected fields, downstream processing can fail.

Structured outputs with a JSON schema constrain the model to generate responses that match a predefined schema. This ensures that only the expected fields are returned, improving reliability and making the output suitable for automated processing.

Incorrect Answers:**A. Lower the model temperature.**

Lowering the temperature makes responses more deterministic, but it does not guarantee that the generated JSON will conform to a required schema or prevent unexpected fields.

C. Increase the maximum completion tokens.

Increasing the maximum completion tokens allows the model to generate longer responses. It does not enforce a specific JSON structure and may even increase the likelihood of unnecessary content.

D. Fine-tune the model.

Fine-tuning can improve model behavior for specific tasks, but it does not provide the same guarantee of schema compliance as structured outputs. It is also more complex and unnecessary for this scenario.

Q83. You need to recommend a solution for a procurement agent by using the Microsoft Foundry Agent Service.

The solution must meet the following requirements:

- Answer questions by using supplier contracts uploaded by users during conversations.
- Generate charts that summarize purchasing trends from uploaded Excel workbooks.

Which Foundry capability should you recommend for each requirement?

To answer questions from uploaded contracts, use:

▼

File Search Tool

Code Interpreter Tool

Grounding with Bing Search

Computer Use Tool

To generate charts from Excel workbooks, use:

▼

Code Interpreter Tool

File Search Tool

Azure AI Search Tool

Microsoft Fabric Tool

Answer:

To answer questions from uploaded contracts, use:

▼

File Search Tool

Code Interpreter Tool

Grounding with Bing Search

Computer Use Tool

To generate charts from Excel workbooks, use:

▼

Code Interpreter Tool

File Search Tool

Azure AI Search Tool

Microsoft Fabric Tool

Explanation

Correct Answers:

To answer questions from uploaded contracts: **File Search Tool**

The **File Search Tool** is designed to retrieve information from documents uploaded during a conversation. It indexes the uploaded supplier contracts and enables the agent to search them efficiently when answering user questions.

To generate charts from Excel workbooks: **Code Interpreter Tool**

The **Code Interpreter Tool** can execute Python code to analyze uploaded Excel workbooks, perform calculations, summarize purchasing trends, and generate charts and visualizations during the conversation.

Incorrect Answers:

Grounding with Bing Search

This tool retrieves information from public websites. It cannot search user-uploaded contracts or analyze Excel workbooks.

Computer Use Tool

The Computer Use Tool is intended for interacting with graphical user interfaces and legacy applications. It is not designed for document retrieval or spreadsheet analysis.

Azure AI Search Tool

Azure AI Search is used to query enterprise search indexes. The requirement is to use contracts uploaded during the current conversation, making the File Search Tool the appropriate choice.

Microsoft Fabric Tool

Microsoft Fabric is an enterprise analytics platform. It is not the built-in tool used by Foundry agents to analyze uploaded Excel files during a conversation.

Q84. You have a Microsoft Foundry project that contains an agent.

You need to process mixed-format documents that contain scanned text, tables, and multicolumn layouts.

The extracted content must preserve the document structure and be converted into the Markdown format for downstream reasoning.

What should you configure first?

- A. an Azure Language in Foundry Tools text analysis model deployment
- B. a generative chat completion request
- C. an Azure OpenAI Responses API call that uses a multimodal model
- D. an Azure Content Understanding in Foundry Tools analyzer

Answer:

D. An Azure Content Understanding in Foundry Tools analyzer

Explanation

Correct Answer:

D. An Azure Content Understanding in Foundry Tools analyzer

The requirement is to process **mixed-format documents** containing scanned text, tables, and multi-column layouts while **preserving the document structure** and converting the extracted content into **Markdown** for downstream reasoning.

The first step is to configure an **Azure Content Understanding in Foundry Tools analyzer**. Content Understanding provides document analysis capabilities that extract text and preserve structural elements such as headings, tables, paragraphs, and reading order. It also supports returning the extracted content in **Markdown** format, making it ideal for Retrieval-Augmented Generation (RAG) and LLM reasoning scenarios.

Incorrect Answers:

A. An Azure Language in Foundry Tools text analysis model deployment

Azure Language performs natural language processing tasks such as sentiment analysis, entity recognition, and key phrase extraction. It does not extract document layouts or preserve document structure.

B. A generative chat completion request

A chat completion request generates responses from an LLM but does not perform structured document extraction or layout analysis.

C. An Azure OpenAI Responses API call that uses a multimodal model

A multimodal model can interpret document images, but it is not the recommended service for extracting structured document content while preserving tables, headings, reading order, and generating Markdown. Azure Content Understanding is specifically designed for this purpose.

Q85. You have a Microsoft Foundry project.

You plan to build a customer support solution that contains an agent. The solution must meet the following requirements:

Provide accurate, context-aware responses grounded in internal product documentation stored in Azure AI Search. Require deep, multi-step reasoning across long contexts.

Generate detailed natural language responses.

Which type of model should you use to power the agent?

- A. a multimodal model
- B. a small language model (SLM)
- C. a key phrase extraction model
- D. a large language model (LLM)

Answer:

D. A large language model (LLM)

Explanation

Correct Answer:

D. A large language model (LLM)

The customer support agent must provide **accurate, context-aware responses** grounded in internal product documentation stored in **Azure AI Search**, perform **deep, multi-step reasoning across long contexts**, and generate **detailed natural language responses**.

A **Large Language Model (LLM)** is specifically designed for these requirements. LLMs can process large context windows, perform sophisticated reasoning, integrate Retrieval-Augmented Generation (RAG) with Azure AI Search, and generate high-quality, detailed responses suitable for customer support scenarios.

Incorrect Answers:

A. A multimodal model

Multimodal models are designed to process multiple input types such as text, images, and audio. The scenario only involves text-based product documentation and does not require multimodal capabilities.

B. A small language model (SLM)

Small Language Models are optimized for lower cost and latency but have more limited reasoning capabilities and smaller context windows. They are not the best choice for deep, multi-step reasoning over long contexts.

C. A key phrase extraction model

A key phrase extraction model identifies important phrases within text. It does not perform reasoning, retrieval, or generate natural language responses.

Q86. You have a Microsoft Foundry project that contains a legal research agent.

The solution must meet the following requirements:

- Research recently published court decisions available on public websites.
- Search contracts that attorneys upload during a conversation.
- Log in to a third-party compliance portal and download supporting evidence for the current case.

What should you use for each requirement?

To answer, drag the appropriate tools to the correct requirements. Each tool may be used once, more than once, or not at all.

Tools:

Research recent court decisions:

Search uploaded contracts:

Download evidence from the compliance portal:

Answer:**Tools:**

Research recent court decisions:

Search uploaded contracts:

Download evidence from the compliance portal:

Explanation**Correct Answers:****Research recent court decisions: Grounding with Bing Search**

The **Grounding with Bing Search** tool enables the agent to retrieve up-to-date information from public websites. Since recently published court decisions are available on public websites, this is the appropriate tool for accessing current legal information.

Search uploaded contracts: File Search

The **File Search** tool indexes documents uploaded during the current conversation and allows the agent to retrieve relevant information from those contracts when answering questions.

Download evidence from the compliance portal: Computer Use

The **Computer Use** tool allows an agent to interact with websites through their graphical user interfaces. It can log in to third-party portals, navigate pages, click buttons, and download documents, making it the correct choice for interacting with a compliance portal that requires user interface automation.

Incorrect Answers:

Code Interpreter

Code Interpreter is designed for executing Python code to analyze data and files. It is not intended for searching public websites, retrieving uploaded documents, or automating interactions with web portals.

Microsoft Fabric

Microsoft Fabric is an enterprise analytics platform and is not used for web search, document retrieval during conversations, or browser automation

Q87. You have a Microsoft Foundry project that contains an agent.

The knowledge source for the agent is a set of scanned PDF troubleshooting guides stored in Azure Blob Storage. The guide pages contain two-column layouts and tables.

You use Azure Content Understanding in Foundry Tools to process the PDFs.

You plan to ingest the processed content into an index for Retrieval Augmented Generation (RAG) and store extracted fields for downstream automation.

Stakeholders must be able to verify where each extracted field value came from in the original PDF and route low-reliability extractions for manual review.

You need to ensure that the Content Understanding document analyzer output includes a per-field confidence score and source grounding to locations within the source document.

What should you do?

- A. Set enableSegment to true.
- B. Provide labeled samples.
- C. Enable estimateFieldSourceAndConfidence.
- D. Configure the analyzer to use generative extraction for all fields.

Answer:

C. Enable estimateFieldSourceAndConfidence.

Explanation

Correct Answer:

C. Enable estimateFieldSourceAndConfidence.

The requirements are return a **confidence score for each extracted field**, provide **source grounding** so stakeholders can identify where each extracted value originated in the original PDF and route **low-confidence extractions** for manual review.

The **estimateFieldSourceAndConfidence** option is specifically designed for this purpose. When enabled, Azure Content Understanding estimates both **Per-field confidence scores**, allowing low-confidence values to be identified for human review and **Source locations** within the original document, enabling traceability back to the PDF content.

This directly satisfies both requirements.

Incorrect Answers:**A. Set enableSegment to true.**

This enables document segmentation but does not provide per-field confidence scores or source grounding information.

B. Provide labeled samples.

Labeled samples help train or improve a custom analyzer, but they do not by themselves cause the analyzer to return confidence scores and source locations for extracted fields.

D. Configure the analyzer to use generative extraction for all fields.

Generative extraction can improve extraction flexibility, but it does not guarantee that field confidence scores and source grounding information are included in the output.

Q88. A Microsoft Foundry agent answers questions from an Azure AI Search index.

Administrators observe the following:

- Average prompt size has doubled.
- Azure OpenAI costs have increased significantly.
- Response quality remains unchanged.

Which action should you recommend first?

- A. Replace the model with GPT-4.1.
- B. Retrieve fewer document chunks before constructing the prompt.
- C. Increase the chunk overlap during indexing.
- D. Increase the model context window.

Answer:**B. Retrieve fewer document chunks before constructing the prompt.****Explanation****Correct Answer:****B. Retrieve fewer document chunks before constructing the prompt.**

The scenario states that **average prompt size has doubled, azure OpenAI costs have increased significantly and response quality remains unchanged.**

This indicates that the application is sending more retrieved content to the model than necessary. Since response quality has not improved, the additional document chunks are increasing token consumption and inference costs without providing any benefit.

Reducing the number of retrieved document chunks before constructing the prompt decreases the input token count, lowering Azure OpenAI costs while maintaining the same response quality.

Incorrect Answers:**A. Replace the model with GPT-4.1.**

Changing the model does not address the root cause, which is the increased prompt size. Costs may even increase depending on the selected model.

C. Increase the chunk overlap during indexing.

Increasing chunk overlap creates more redundant content in the search index. This can further increase prompt size and token usage, making the problem worse.

D. Increase the model context window.

A larger context window allows the model to accept longer prompts but does not reduce the number of input tokens or lower inference costs.

Q89. You have a Microsoft Foundry project that contains an agent deployed to multiple environments. The organization has adopted a passwordless authentication strategy. You need to configure the GitHub Actions workflow to meet the following requirements:

- Eliminate long-lived credentials.
- Authenticate securely to Azure resources.

How should you configure the workflow?

Authentication method:

▼

Azure Login action that uses OpenID Connect (OIDC)
Personal access token (PAT)
Storage account access key
Publish Profile

Credential management:

▼

Store credentials as GitHub repository secrets
Configure workload identity federation
Rotate PAT every 30 days
Use a shared administrator account

Answer:

Authentication method:

▼

Azure Login action that uses OpenID Connect (OIDC)
Personal access token (PAT)
Storage account access key
Publish Profile

Credential management:

▼

Store credentials as GitHub repository secrets
Configure workload identity federation
Rotate PAT every 30 days
Use a shared administrator account

Explanation**Correct Answers:**

Authentication method: Azure Login action that uses OpenID Connect (OIDC)

The organization has adopted a **passwordless authentication strategy** and wants to **eliminate long-lived credentials**. The Azure Login GitHub Action supports **OpenID Connect (OIDC)**, allowing GitHub Actions to obtain short-lived Microsoft Entra tokens without storing secrets such as client secrets or passwords.

Credential management: Configure workload identity federation

Workload identity federation establishes trust between GitHub Actions and Microsoft Entra ID. It enables GitHub workflows to exchange GitHub-issued OIDC tokens for Azure access tokens, eliminating the need for long-lived credentials while providing secure authentication.

Incorrect Answers:

Personal access token (PAT)

PATs are long-lived credentials and do not support a passwordless authentication strategy.

Storage account access key

Storage account keys are long-lived secrets and violate the requirement to eliminate long-lived credentials.

Publish Profile

A publish profile contains long-lived deployment credentials and is not recommended for secure GitHub Actions authentication.

Store credentials as GitHub repository secrets

Although repository secrets protect stored credentials, they are still long-lived secrets and do not satisfy the passwordless requirement.

Rotate PAT every 30 days

Regular rotation improves security but still relies on long-lived credentials instead of passwordless authentication.

Use a shared administrator account

This violates security best practices and the principle of least privilege.

Q90. You have an application that processes scanned PDF invoices. The invoices have varied layouts and include multipage tables.

You have a pipeline that uses optical character recognition (OCR) and extracts totals and invoice numbers. The results are often incorrect because the document structure is ignored.

You need to implement a solution that provides OCR, layout analysis, and template-generalizing field extraction. The solution must NOT require training a custom model. The solution must minimize administrative effort.

What should you include in the solution?

- A. Azure Language in Foundry Tools
- B. Azure Content Understanding in Foundry Tools
- C. an Azure Machine Learning model

Answer:

B. Azure Content Understanding in Foundry Tools

Explanation

Correct Answer:

B. Azure Content Understanding in Foundry Tools

The solution must process **scanned PDF invoices** using OCR, understand **document layout**, including **multipage tables**, extract fields such as **invoice numbers** and **totals** across **varying invoice templates**, **not require training a custom model** and **minimize administrative effort**.

Azure Content Understanding in Foundry Tools is designed specifically for these scenarios. It combines **OCR** to extract text from scanned documents, **Layout analysis** to preserve tables, reading order, and document structure and **Template-generalizing field extraction**, allowing it to extract common business fields across different document layouts without requiring a custom trained model.

This makes it the ideal solution with minimal administration.

Incorrect Answers:

A. Azure Language in Foundry Tools

Azure Language provides text analytics capabilities such as entity recognition, sentiment analysis, and summarization. It does not perform OCR or understand document layouts.

C. An Azure Machine Learning model

A custom Azure Machine Learning model would require development, training, deployment, and ongoing maintenance, which contradicts the requirements to avoid custom model training and minimize administrative effort.

Q91. You are creating an agent workflow in a Microsoft Foundry project to support natural voice interactions. The agent must receive continuous audio input, convert the input into text for reasoning, and then return spoken responses to a user.

The workflow must meet the following requirements:

- Support turn-taking dynamics, where the agent begins to generate the speech output before the user finishes speaking.
- Operate with low latency to maintain conversational experience.

You need to enable both speech to text and text to speech in a real-time agent interaction.

What should you do?

- A. Use batch transcription to convert the audio input and return text responses from the agent.
- B. Use real-time speech to text for incoming audio and text to speech for agent responses.
- C. Use an embeddings model to encode the audio, and then decode the audio into text and speech.
- D. Use speech translation to convert the audio into another language and return the translated text.

Answer:

B. Use real-time speech to text for incoming audio and text to speech for agent responses.

Explanation

Correct Answer:

B. Use real-time speech to text for incoming audio and text to speech for agent responses.

The solution must accept **continuous audio input**, convert speech to text for the agent's reasoning, return **spoken responses**, support **low-latency**, **real-time conversations** and support **turn-taking dynamics**, where the agent can begin responding before the user has completely finished speaking.

The Azure Speech service in Foundry Tools provides **real-time Speech-to-Text (STT)** and **Text-to-Speech (TTS)** capabilities that are specifically designed for conversational AI. These services enable streaming audio, low latency, and natural turn-taking, making them the correct choice for voice agents.

Incorrect Answers:

A. Use batch transcription to convert the audio input and return text responses from the agent.

Batch transcription is intended for processing prerecorded audio files. It introduces higher latency and cannot support real-time conversations or spoken responses.

C. Use an embeddings model to encode the audio, and then decode the audio into text and speech.

Embedding models generate vector representations for similarity search and retrieval. They do not perform speech recognition or speech synthesis.

D. Use speech translation to convert the audio into another language and return the translated text.

Speech translation is designed for multilingual translation scenarios. The requirement is real-time speech recognition and speech synthesis, not language translation.

Q92. You have a Microsoft Foundry project that processes purchase orders stored in Azure Blob Storage. Each purchase order has a different layout depending on the supplier.

The extracted information will be stored as structured JSON and used by a procurement agent.

You need to create a single analyzer that meets the following requirements:

- Extract the purchase order number, supplier name, requested delivery date, and total value from all supported layouts.
- Return confidence scores for each extracted field.
- Minimize post-processing in the application.

What should you use?

- A. A custom Azure Content Understanding analyzer in Foundry Tools that defines the required fields as extracted fields.
- B. The Azure Content Understanding prebuilt-read analyzer.
- C. The Azure Content Understanding prebuilt-layout analyzer.
- D. Azure AI Search with vector search enabled.

Answer:

A. A custom Azure Content Understanding analyzer in Foundry Tools that defines the required fields as extracted fields.

Explanation

Correct Answer:

A. A custom Azure Content Understanding analyzer in Foundry Tools that defines the required fields as extracted fields.

The requirements are extract **purchase order number**, **supplier name**, **requested delivery date**, and **total value**, support **multiple supplier layouts** with a **single analyzer**, return **confidence scores** for each extracted field and produce **structured JSON** while minimizing application post-processing.

A **custom Azure Content Understanding analyzer** allows you to define the required business fields as **extracted fields**. It can generalize across different document layouts, automatically returns structured JSON, and includes **per-field confidence scores**, making it the best fit for downstream procurement agents.

Incorrect Answers:

B. The Azure Content Understanding prebuilt-read analyzer.

The **prebuilt-read** analyzer performs OCR to extract raw text. It does not identify business fields such as purchase order number or supplier name, nor does it produce structured field extraction.

C. The Azure Content Understanding prebuilt-layout analyzer.

The **prebuilt-layout** analyzer identifies document structure, including tables, paragraphs, and reading order. However, it does not extract specific business fields into structured JSON.

D. Azure AI Search with vector search enabled.

Azure AI Search is designed for indexing and retrieving documents. It does not perform document field extraction or produce structured JSON from purchase orders.

Q93. You have a Microsoft Foundry project that contains a deployed chat model.

You have a Python service that sends API requests to the model. The service is integrated with an automated validation system that compares generated outputs against approved response patterns.

Stakeholders report that small wording differences are causing validation mismatches.

You need to update the request parameters to improve output stability. The solution must maximize reasoning quality.

How should you complete the Python code?

```
message = client.messages.create(
    model="deployment-name",
    messages=[
        {
            "role": "user",
            "content": "Summarize the release notes in 3 bullet points."
        }
    ],
    max_tokens=800,
    temperature=
    ,
    thinking={"type": "enabled"},
    output_config={"effort":
    }
)
```

Dropdown menu for temperature: 0, 1, 2

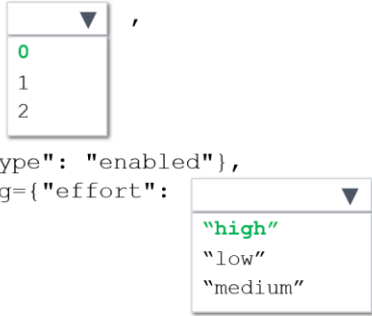
Dropdown menu for effort: "high", "low", "medium"

Answer:

```

message = client.messages.create(
    model="deployment-name",
    messages=[
        {
            "role": "user",
            "content": "Summarize the release notes in 3 bullet points."
        }
    ],
    max_tokens=800,
    temperature=
    thinking={"type": "enabled"},
    output_config={"effort":
    )

```


Explanation**Correct Answers:****Temperature: 0**

The application uses an automated validation system that compares generated responses against approved response patterns. Setting the **temperature to 0** makes the model's output as deterministic as possible, reducing wording variations between identical requests and improving output stability.

Output effort: "high"

The requirement also states to **maximize reasoning quality**. Setting **output_config={"effort": "high"}** instructs the reasoning model to spend more effort generating accurate, well-reasoned responses while still producing deterministic output due to the low temperature.

Incorrect Answers:**Temperature: 1**

A temperature of **1** increases randomness and creativity, making responses less consistent and more likely to fail automated validation.

Temperature: 2

A temperature of **2** produces even greater variability and is the least suitable option for deterministic outputs.

Output effort: "low"

Low effort reduces reasoning depth and may produce lower-quality responses.

Output effort: "medium"

Medium effort provides a balance between speed and reasoning quality but does not maximize reasoning as required.

Q94. You have a Microsoft Foundry project that contains an agent. The agent uses Azure AI Search as the retriever.

You plan to ingest PDF into an Azure AI Search index to ensure that the agent can ground responses in texts in both documents and embedded images.

Users require citations that link to the source files.

You need to ensure that during indexing, the images are extracted into a structure that can be used as input for the built-in optical character recognition (OCR) skill.

Which indexing approach should you use?

- A. an indexer to extract image data into a normalized_images collection
- B. a Shaper skill to restructure the OCR input
- C. a skillset to run the OCR skill directly against the content field of the index
- D. the outputFieldMappings parameter to write image data to a searchable field

Answer:

A. An indexer to extract image data into a normalized_images collection

Explanation

Correct Answer:

A. An indexer to extract image data into a normalized_images collection`

The requirements are index PDF documents stored for RAG, extract text from both the document text and **embedded images**, use the **built-in OCR skill** and preserve source information for citations.

In Azure AI Search, the **indexer** can crack PDF documents and extract embedded images into the **normalized_images** collection. This collection is the expected input for the built-in **OCR skill**, which then extracts text from the images. The extracted text can be indexed alongside the document text while maintaining references to the source documents for citations.

This is the recommended indexing approach for processing images embedded within documents.

Incorrect Answers:

B. A Shaper skill to restructure the OCR input.

A **Shaper skill** reshapes or combines enriched data during skillset execution. It does not extract images from documents or create the normalized_images collection required by the OCR skill.

C. A skillset to run the OCR skill directly against the content field of the index.

The **OCR skill** operates on **image inputs**, not on the document's text (content) field. Images must first be extracted into normalized_images.

D. The outputFieldMappings parameter to write image data to a searchable field.

outputFieldMappings maps outputs from the enrichment pipeline into index fields. It does not extract embedded images or prepare them for OCR.

Q95. You have a Microsoft Foundry project that generates real estate marketing images from text prompts. After publishing several images, you discover that one image contains a vehicle parked in the driveway. The property owner requests that only the vehicle be removed while preserving the house, landscaping, lighting, and shadows.

You need to update the image.

What should you do?

- A. Apply a mask-based inpainting edit to the area containing the vehicle.
- B. Regenerate the image by using a lower temperature value.
- C. Modify the original prompt to specify an empty driveway and generate a new image.
- D. Create an image variation from the original image.

Answer:

A. Apply a mask-based inpainting edit to the area containing the vehicle.

Explanation

Correct Answer:

A. Apply a mask-based inpainting edit to the area containing the vehicle.

The requirement is to remove **only the vehicle** while preserving everything else in the image, including the house, landscaping, lighting, and shadows.

Mask-based inpainting is specifically designed for targeted image editing. You create a mask over the vehicle, and the model regenerates only the masked region while leaving the rest of the image unchanged. This preserves the overall composition and appearance of the property.

Incorrect Answers:

B. Regenerate the image by using a lower temperature value.

Lowering the temperature affects the randomness of image generation but does not selectively remove an object from an existing image.

C. Modify the original prompt to specify an empty driveway and generate a new image.

Generating a new image may change the house, landscaping, lighting, camera angle, and other details. It does not guarantee that only the vehicle will be removed.

D. Create an image variation from the original image.

Image variations generate similar images but do not provide precise control over removing a specific object while preserving the rest of the scene.

Q96. You have a Microsoft Foundry project that contains a customer support agent.

The agent invokes several tools to retrieve order information, customer profiles, and shipping status before generating a response.

Users report that response times have increased significantly after a recent deployment.

You need to identify which stage of the agent execution is contributing most to the increased response time.

Which observability signal should you review?

- A. Groundedness evaluation metrics
- B. Latency breakdown traces
- C. Content Safety metrics
- D. Relevance evaluation metrics

Answer:

B. Latency breakdown traces

Explanation

Correct Answer:

B. Latency breakdown traces

The agent performs multiple stages before generating a response retrieving order information, retrieving customer profiles, retrieving shipping status and model inference.

Users report that **response times have increased** after a deployment. To determine **which stage of the agent execution is causing the delay**, you should review **Latency breakdown traces**.

Latency breakdown traces provide detailed timing information for each step in the agent execution pipeline, such as document retrieval, tool execution, model inference, and response generation. This enables you to pinpoint the exact stage contributing most to the increased response time.

Incorrect Answers:

A. Groundedness evaluation metrics

Groundedness metrics measure whether the generated response is supported by the retrieved data. They evaluate response quality, not performance or latency.

C. Content Safety metrics

Content Safety metrics report detections of harmful or unsafe content. They do not identify performance bottlenecks or execution delays.

D. Relevance evaluation metrics

Relevance metrics evaluate how well the retrieved documents match the user's query. They help assess retrieval quality, not execution time.

Q97. You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.

You need to improve response completeness.

Solution: You increase the value of the `max_tokens` parameter.

Does this meet the goal?

- A. Yes
- B. No

Correct Answer:

A. Yes

The issue is that the required regulatory clauses **are already present in the retrieved content**, but the generated responses **omit** some of them.

Increasing the **max_tokens** parameter allows the model to generate longer responses. If the omissions are caused by the response being truncated due to an insufficient completion token limit, increasing `max_tokens` improves response completeness and enables the model to include all the required clauses.

Therefore, this solution **meets the goal**.

Q98. You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.

You need to improve response completeness.

Solution: You add a reflection pass that regenerates the response if the required clauses are missing.

Does this meet the goal?

- A. Yes
- B. No

Correct Answer:

A. Yes

The retrieved policy documents already contain the required regulatory clauses, but the generated summaries sometimes omit them. This indicates that the retrieval step is working correctly, while the generation step is not producing a complete response.

Adding a **reflection pass** allows the application to evaluate the generated response before returning it. If the required clauses are missing, the application can regenerate the response with instructions to include the omitted information. This improves the completeness of the final answer without changing the retrieval process.

Therefore, this solution **meets the goal**.

Q99. You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.

You need to improve response completeness.

Solution: You increase the value of the temperature parameter.

Does this meet the goal?

A. Yes

B. No

Correct Answer:

B. No

The problem is that the agent **omits required regulatory clauses**, even though those clauses are already present in the retrieved policy documents. This is a response completeness issue, not a creativity issue. Increasing the **temperature** parameter makes the model's output **more random and creative**. It does not improve completeness or make the model more likely to include all required clauses. In fact, a higher temperature can make responses less deterministic and may increase the likelihood of omitting important information or introducing inconsistencies.

Therefore, increasing the temperature **does not meet the goal**.

Q100. You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.

You need to improve response completeness.

Solution: You run an evaluation flow that scores responses for completeness and blocks responses that fall below a defined threshold.

Does this meet the goal?

A. Yes

B. No

Correct Answer:

B. No

The proposed solution runs an **evaluation flow** that scores responses for completeness and **blocks** responses that do not meet a defined threshold.

While this helps **detect** incomplete responses and prevents low-quality answers from being returned to users, it **does not improve the completeness of the responses themselves**. The agent will simply reject responses that score poorly, rather than generating a more complete answer.

Therefore, the solution does **not** meet the goal of improving response completeness.

To improve completeness, the application must take corrective action, such as increasing the maximum completion tokens if responses are being truncated, adding a reflection or retry pass that regenerates incomplete responses and adjusting the retrieval strategy if insufficient context is being provided.

An evaluation flow alone only measures quality. Unless it is combined with a retry or regeneration mechanism, it cannot make responses more complete.

Q101. You have a Microsoft Foundry project that contains a Retrieval Augmented Generation (RAG) chat solution used by customer support agents.

You are adding an automated pre-production evaluation step to a CI/CD pipeline named Pipeline1. The evaluation will run against a labeled test dataset that contains support questions and the expected grounding context.

You need to ensure that Pipeline1 fails if unsupported content or a retrieval mismatch exceeds a defined threshold:

- responses include claims not supported by the retrieved source content
- retrieved source content does not align with the labeled expected context

Which two built-in evaluators should you use in Pipeline1?

- A. Retrieval
- B. Fluency
- C. Coherence
- D. Groundedness
- E. Response Completeness

Correct Answer:

- A. Retrieval
- D. Groundedness

Explanation:

Correct Answer:

A. Retrieval

The Retrieval evaluator measures how well the system retrieves relevant documents compared to the expected grounding context in the labeled dataset. It ensures that the retrieved content aligns with the expected context. This directly satisfies the requirement to detect when retrieved source content does not match the labeled expected context and allows the pipeline to fail when mismatches exceed the threshold.

D. Groundedness

The Groundedness evaluator verifies that the generated response is fully supported by the retrieved source content. It detects hallucinations or unsupported claims in the response. This directly addresses the requirement to fail the pipeline when responses include claims not grounded in the retrieved data.

Incorrect Answers:

B. Fluency

Fluency evaluates grammar and readability, not correctness or grounding.

C. Coherence

Coherence measures logical flow, not retrieval accuracy or factual alignment.

E. Response Completeness

Response Completeness checks coverage of the answer, not grounding or retrieval correctness.

Q102. You have a Microsoft Foundry project that contains a support-ticket triage agent built by using the Foundry Agent Service.

The agent uses tool to classify the ticket type and sot the ticket priority.

Sometimes, the same support case continues across multiple sessions over several days.

You need to persist state by using a durable ID to ensure that the agent can automatically reuse the full interaction history. The solution must preserve previous user messages, tool calls and tool outputs across turns and sessions.

Which runtime component should you use?

- A. output item
- B. agent
- C. conversation
- D. response

Correct Answer:

C. conversation

Explanation:

Correct Answer:

C. conversation

A conversation is the runtime component that maintains state across multiple turns and sessions using a durable identifier. It preserves the full interaction history, including user messages, tool calls, and tool outputs. This allows the agent to resume context seamlessly across sessions, which is required when support cases span multiple days.

Incorrect Answers:

A. output item

An output item represents a single response or piece of output, not a persistent state container.

B. agent

An agent defines behavior and orchestration logic but does not persist interaction history across sessions.

D. response

A response is a single execution result and does not store or manage conversation history.

Q103. You have a Microsoft Founcy project that contains a Retrieval Augmented Generation (RAG) solution. You need to run a pre-production evaluation by using labeled CSV dataset that contains the query, context, response and ground truth.

The evaluation must measure the following:

- Whether responses address the user query
- Whether responses are supported by the provided context

- Whether responses contain sensitive or proprietary information

Which AI quality evaluation metrics should you use?

Answer Area

To measure whether the responses are supported by the provided context and address the user query:

▼

Coherence and Fluency
 GPT similarity and F1 score
Groundedness and Relevance
 Groundedness and ROUGE score

To measure whether responses contain sensitive or proprietary information:

▼

Hateful and unfair content
 Indirect attack
Protected material
 Violent content

Answer:

Answer Area

To measure whether the responses are supported by the provided context and address the user query:

▼

Coherence and Fluency
 GPT similarity and F1 score
Groundedness and Relevance
 Groundedness and ROUGE score

To measure whether responses contain sensitive or proprietary information:

▼

Hateful and unfair content
 Indirect attack
Protected material
 Violent content

Explanation:

Correct Answer:

Groundedness and Relevance

Groundedness evaluates whether the generated response is supported by the provided context, ensuring there are no hallucinations. Relevance measures whether the response appropriately addresses the user query. Together, these metrics satisfy the requirement to verify that responses both answer the user's question and are grounded in the retrieved context.

Protected material

Protected material detection identifies whether the response contains sensitive or proprietary information. This directly satisfies the requirement to detect if responses include confidential or restricted data.

Incorrect Answers:

Coherence and Fluency

These measure readability and language quality, not grounding or relevance to the query.

GPT similarity and F1 score

These compare generated responses with ground truth but do not verify grounding against context or detect sensitive content.

Groundedness and ROUGE score

ROUGE measures overlap with reference text, not whether the response addresses the query or is relevant.

Hateful and unfair content

This detects toxicity or bias, not sensitive or proprietary information.

Indirect attack

This identifies prompt injection or adversarial inputs, not sensitive data leakage.

Violent content

This detects harmful or violent outputs, not proprietary or sensitive information.

Q104. You plan to configure an evaluation in Microsoft Foundry for a Retrieval Augmented Generation (RAG) chat app. You need to provide scores for groundedness, relevance, and harmful content categories. Which two evaluation categories can you use?

- A. risk and safety metrics
- B. fluency evaluator
- C. similarity evaluators
- D. AI quality (NLP) metrics
- E. AI quality (AI assisted) metrics



Correct Answer:

- A. risk and safety metrics
- E. AI quality (AI-assisted) metrics

Explanation:

Correct Answer:

A. risk and safety metrics

Risk and safety metrics are used to evaluate harmful content categories such as hate, violence, self-harm, and other unsafe outputs. This directly satisfies the requirement to assess harmful content categories in the RAG responses.

E. AI quality (AI-assisted) metrics

AI-assisted quality metrics use an LLM evaluator to assess qualitative aspects such as **groundedness** and **relevance**. These metrics do not rely on simple statistical comparison but instead use model-based judgment to determine whether the response is supported by the provided context and whether it addresses the user query. This makes them the correct choice for evaluating groundedness and relevance in a RAG scenario.

Incorrect Answers:

B. fluency evaluator

Fluency evaluates grammar and readability only, not groundedness, relevance, or harmful content.

C. similarity evaluators

Similarity evaluators compare responses with ground truth using overlap or semantic similarity, but do not evaluate grounding or safety risks.

D. AI quality (NLP) metrics

NLP metrics such as BLEU, ROUGE, and F1 rely on statistical comparison with reference text and do not assess groundedness, relevance, or harmful content using model reasoning.

Q105. You have a Microsoft Foundry project that uses Azure Content Understanding in Foundry Tools to analyze marketing videos.

Video segmentation is enabled.

You need to configure an analyzer to output a generated JSON field that describes the color scheme of each video segment.

How should you configure the analyzer?

Values**Answer Area**Field value type: Field method: **Answer:****Values****Answer Area**Field value type: Field method: **Explanation:****Correct Answer:**

Field value type → **string**

A string field type is used when the output is **descriptive text**. In this scenario, the requirement is to describe the color scheme of each video segment. This is not structured multi-field JSON with defined subfields, but rather a **natural language or textual description**, which maps to a string.

Field method → generate

The generate method is used when the analyzer must **create new content using AI** rather than extract it directly from the source. Since the color scheme must be inferred from visual content and described, it requires generative capability.

Incorrect Answers:**group**

Group is used for structured JSON with predefined subfields. The question asks for a generated description, not a structured schema with multiple named fields.

classify

Classify is used when assigning predefined labels, not generating descriptive output.

table

Table is used for structured tabular extraction, not descriptive generation.

Q106. You have a Microsoft Foundry project that contains an agent. The agent uses two tools to perform the following actions:

- Use Azure AI Search to retrieve answers from a private product documentation index.
- Use the web search tool to retrieve public information on the internet.

You need to ensure that for a specific run, the agent deterministically retrieves information only from the internet.

To what should you set tool_choice?

- A. "type": "bing_grouding"
- B. "type": "azure_ai-search"
- C. "auto"
- D. "required"

Correct Answer:

- A. "type": "bing_grounding"

Explanation:**Correct Answer:**

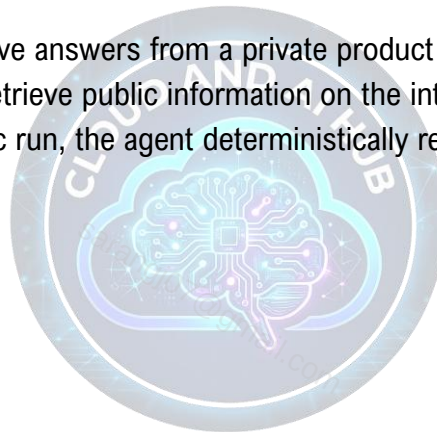
- A. "type": "bing_grounding"

This is correct because setting tool_choice explicitly to bing_grounding forces the agent to deterministically use only the web search tool for that run. It ensures that responses are generated strictly from internet sources and prevents the agent from calling Azure AI Search or any other tool.

Incorrect Answers:

- B. "type": "azure_ai-search"

This forces the agent to use the private index, which is the opposite of the requirement.



C. “auto”

Auto allows the agent to choose tools dynamically, so it does not guarantee internet-only retrieval.

D. “required”

Required forces tool usage but does not specify which tool, so it does not ensure deterministic selection of web search.

Q107. You have a Microsoft Foundry project that contains a customer support application.

You create an evaluation named Run1 that has the following configurations:

- Includes risk and safety metrics
- Includes the protected material evaluation
- Includes harmful content metrics that use a medium severity threshold

You create an evaluation named Run2 that has the following configurations:

- Includes risk and safety metrics
- Includes the protected material evaluation
- Includes harmful content metrics that use a high severity threshold

You run both evaluations against a dataset named DB1 and receive the following results:

- Content harm defect rate of Run1: 12%
- Content harm defect rate of Run2: 4%
- Protected material evaluation of Run1: 6%
- Protected material evaluation of Run2: 6%

You start a fine-tuning job by using DB1. The job fails during automatic RAI checks for multiple content harm types. You discover that the content filtering configuration is set to high severity.

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Answer Area

Statements	Yes	No
Changing the content filtering configuration to low severity will resolve the fine-tuning job issues.	☐	☐
The difference between the 12% and 4% content harm defect rate is consistent with the different severity thresholds used in Run1 and Run2.	☐	☐
The identical 6% protected material evaluation values across Run1 and Run2 indicate that this metric is unaffected by the change in the severity threshold.	☐	☐

Answer:

Answer Area

Statements	Yes	No
Changing the content filtering configuration to low severity will resolve the fine-tuning job issues.	☐	☒
The difference between the 12% and 4% content harm defect rate is consistent with the different severity thresholds used in Run1 and Run2.	☒	☐
The identical 6% protected material evaluation values across Run1 and Run2 indicate that this metric is unaffected by the change in the severity threshold.	☒	☐

Statement 1 --> No

This is incorrect because lowering the severity threshold only reduces how strictly harmful content is flagged, but it does not remove or fix the harmful content present in the dataset. The fine-tuning job fails due to actual content violations detected during RAI checks, so the dataset itself must be cleaned rather than just changing the filtering level.

Statement 2 --> Yes

This is correct because a medium severity threshold flags more content as harmful compared to a high severity threshold, resulting in a higher defect rate (12% vs 4%). This difference directly reflects how varying severity levels impact content harm detection sensitivity.

Statement 3 --> Yes

This is correct because protected material evaluation is independent of harmful content severity thresholds. The identical 6% values across both runs confirm that changing severity levels affects only harmful content metrics and not protected material detection.

Q108. You have a Microsoft Foundry project that contains a customer support agent built on a deployed chat model. The agent responses are validated by using an automated testing system that compares generated answers to stored expected outputs. Identical prompts must return consistent response to prevent automated test failures.

You need to reduce response variability, without modifying the prompt or reducing factual accuracy. What should you do for the model?

- A. Increase the max_tokens parameter.
- B. Remove stop sequences from the requests.
- C. Decrease the temperature parameter.
- D. Increase the temperature parameter.

Correct Answer:

C. Decrease the temperature parameter

Explanation:

Correct Answer:

C. Decrease the temperature parameter

This is correct because temperature controls randomness in model output. Lowering the temperature makes responses more deterministic and consistent for identical prompts, which is essential for automated testing scenarios without impacting factual accuracy.

Incorrect Answers:

A. Increase the max_tokens parameter

This only affects response length, not variability or consistency.

B. Remove stop sequences from the requests

Stop sequences control where responses end, not how deterministic they are.

D. Increase the temperature parameter

Higher temperature increases randomness, leading to more variability and inconsistent outputs.



Sarang Joshi

Q109. You have a Microsoft Foundry project that contains two agents named PolicyWriter and RskReviewer. PolicyWriter generates draft updates for customer policies, and RiskReviewer reviews the drafts.

In the visual builder, you need to create a workflow that meets the following requirements:

- Finalizes low-risk updates without manual intervention
- Ensures predictable execution across the agents
- Requires user approval for highs updates

What should you configure?

Answer Area

Orchestration pattern:

▼

The sequential template that passes outputs node by-node

The group chat template to dynamically route control between the agents

The human-in-the-loop template that pauses execution of the workflow for input

Approval checkpoints:

▼

Add a Basic chat node.

Add a Condition statement.

Add an Ask a question node.

Answer:

Answer Area

Orchestration pattern:

▼

The sequential template that passes outputs node by-node

The group chat template to dynamically route control between the agents

The human-in-the-loop template that pauses execution of the workflow for input

Approval checkpoints:

▼

Add a Basic chat node.

Add a Condition statement.

Add an Ask a question node.

Explanation:

Correct Answer:

Orchestration pattern → The sequential template that passes outputs node by-node

This is correct because the requirement is for predictable execution across agents. A sequential pattern ensures PolicyWriter runs first and then RiskReviewer processes the output in a fixed, deterministic flow, which avoids dynamic routing and ensures consistency.

Approval checkpoints → Add an Ask a question node

This is correct because high-risk updates require explicit user approval. An Ask a question node introduces a human-in-the-loop checkpoint, allowing the workflow to pause and request approval only when needed.

Incorrect Answers:

The group chat template to dynamically route control between the agents

This introduces non-deterministic behavior, which violates the requirement for predictable execution.

The human-in-the-loop template that pauses execution of the workflow for input

This would pause all executions, not just high-risk cases, making it inefficient for low-risk updates.

Add a Basic chat node

This does not provide approval capability.

Add a Condition statement

This can route logic but cannot capture explicit user approval.

Q110. You are developing prompts for a Microsoft Foundry project that classifies incoming support tickets by category. You need to improve accuracy by showing the model how correct classifications look, without retaining the model or storing knowledge permanently.

Which prompt engineering approach should you use?

- A. Retrieval Augmented Generation (RAG)
- B. zero-shot learning
- C. chain of thought
- D. few-shot learning

Correct Answer:

D. few-shot learning

Explanation:

Correct Answer:

D. few-shot learning

This is correct because few-shot learning improves classification accuracy by providing a few labeled examples directly in the prompt. This shows the model what correct outputs look like without retraining or permanently storing knowledge, aligning perfectly with the requirement.

Incorrect Answers:

A. Retrieval Augmented Generation (RAG)

RAG retrieves external data but does not demonstrate example classifications within the prompt.

B. zero-shot learning

Zero-shot provides no examples, so it does not help guide the model toward more accurate classifications.

C. chain of thought

Chain-of-thought improves reasoning steps, not classification accuracy through examples.

Q111. You have a web app named App1 that processes user prompts by integrating with a Microsoft Foundry project named Project1.

App1 performs the following actions:

- Sends prompts directly to a model by using the Azure OpenAI Responses API
- Invokes the Azure AI Content Safety tool by using a Foundry connection within the same request

You need to configure end-to-end visibility into each step of the request workflow.

What should you do?

- A. Enable logging by using the client SDK for Content Safety.
- B. Enable logging by using Foundry Local.
- C. Enable application tracing in Project1.
- D. Route requests through the Azure OpenAI endpoint.

Correct Answer:

C. Enable application tracing in Project1.

Explanation:

Correct Answer:

C. Enable application tracing in Project1

This is correct because application tracing provides end-to-end observability across the entire request flow, including model calls and tool invocations like Content Safety. It captures each step in the workflow, enabling full visibility into how the request is processed within Foundry.

Incorrect Answers:

A. Enable logging by using the client SDK for Content Safety

This only provides visibility into Content Safety calls, not the complete workflow.

B. Enable logging by using Foundry Local

Foundry Local is used for local development and debugging, not full production tracing.

D. Route requests through the Azure OpenAI endpoint

This does not provide tracing across integrated tools or end-to-end workflow visibility.

Q112. You have a Microsoft Foundry project that contains an agent named Agent1.

Agent runs successful, but Foundry Control Plane does NOT display values for error rates, runs, and token usage, and the Traces tab is empty.

You need to ensure that Found Control Plane displays the appropriate values for Agent1. What should you do?

- A. Update Agent1 to a new version.
- B. Restart Agent from Foundry Control Plan
- C. Assign to a Log Analytics workspace to Agent1.
- D. Enable Application Insights for Agent1.

Correct Answer:

D. Enable Application Insights for Agent1.

Explanation:

Correct Answer:

D. Enable Application Insights for Agent1

This is correct because metrics such as error rates, runs, token usage, and traces are collected through Application Insights. Enabling it integrates telemetry and observability, allowing the Foundry Control Plane to display detailed runtime metrics and traces for the agent.

Incorrect Answers:

A. Update Agent1 to a new version

Updating the version does not enable telemetry or monitoring capabilities.

B. Restart Agent from Foundry Control Plane

Restarting does not address missing telemetry configuration.

C. Assign to a Log Analytics workspace to Agent1

Log Analytics alone does not provide the required application-level tracing and metrics integration for Foundry agents.

Q113. You have a Microsoft Foundry project that contains an agent. The agent uses a stored access key to retrieve secrets from an Azure key vault, which violates a keyless-credentials requirement. You need to ensure that the agent can retrieve the secrets. The solution must follow the principle of least privilege. What should you configure?

Answer Area

Managed identity scope:

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▼

Enable a system-assigned managed identity at the Foundry level.
 Enable a system-assigned managed identity at the project level.
 Create a service principal and store the principal's client secret.

Key Vault authorization method:

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▼

Add an API key to application settings.
 Add a Key Vault access policy for the secrets.
 Assign the Key Vault Secrets User role to the managed identity.

Answer:

Answer Area

Managed identity scope:

Enable a system-assigned managed identity at the Foundry level.
Enable a system-assigned managed identity at the project level.
 Create a service principal and store the principal's client secret.

Key Vault authorization method:

Add an API key to application settings.
 Add a Key Vault access policy for the secrets.
Assign the Key Vault Secrets User role to the managed identity.

Explanation:

Correct Answer:

Managed identity scope → Enable a system-assigned managed identity at the project level

This is correct because the agent runs within the Foundry project, so enabling a system-assigned managed identity at the project level ensures secure, keyless authentication scoped specifically to that project, aligning with least privilege.

Key Vault authorization method → Assign the Key Vault Secrets User role to the managed identity

This is correct because RBAC-based access using the Secrets User role allows the agent to retrieve secrets without exposing keys, while granting only the minimum required permissions.

Incorrect Answers:

Enable a system-assigned managed identity at the Foundry level

This grants broader access than required and violates least privilege.

Create a service principal and store the principal's client secret

This reintroduces credential management, which violates the keyless requirement.

Add an API key to application settings

This uses secrets instead of eliminating them.

Add a Key Vault access policy for the secrets

Access policies are legacy and less aligned with RBAC-based least privilege compared to role assignments.

Q114. You have a Microsoft Foundry project that contains an agent.

The agent uses Azure Content Understanding in Foundry Too to process vendor onboarding packets. The packs include digital PDFs that contain tables and hyperlinks.

The extracted content is indexed for search and provided to a downstream agent in the Markdown format.

You need to generate a Markdown output that has a layout and a semantic structure optimized for Retrieval Augmented Generation (RAG) workflows.

Which built-in analyzer should you use?

A. prebuilt-documentFieldSchema

- B. prebuilt-documentSearch
- C. prebuilt-read
- D. prebuilt-layout

Correct Answer:

B. prebuilt-documentSearch

Explanation:

Correct Answer:

B. prebuilt-documentSearch

This is correct because prebuilt-documentSearch is optimized for RAG scenarios, producing structured Markdown output with semantic organization suitable for indexing and retrieval. It preserves layout elements like tables and links while formatting content for efficient downstream search and grounding.

Incorrect Answers:

A. prebuilt-documentFieldSchema

Used for structured field extraction, not for generating RAG-optimized Markdown output.

C. prebuilt-read

Extracts raw text without preserving layout or semantic structure needed for RAG.

D. prebuilt-layout

Captures layout elements but does not produce semantically optimized Markdown for retrieval workflows.

Q115. You have a Microsoft Foundry project that contains a multi-agent solution. The agents use tool calling to query internal systems.

You need to implement responsible AI auditing to meet the following requirements:

- Capture all the nested operations across the entire agent run.
- Record tool invocation arguments and retuned results as metadata.

What should you use for each requirement?

Options	Answer Area
Hierarchical spans	
A KQL query filter	
Sampling	Capture all the nested operations across the entire agent run:
Tool call attributes	Record tool invocation arguments and results:
Trace sampling policy	

Answer:

Options**Answer Area**☐ Hierarchical spans☐ A KQL query filter☐ Sampling☐ Tool call attributes☐ Trace sampling policy

Capture all the nested operations across the entire agent run:

Hierarchical spans

Record tool invocation arguments and results:

Tool call attributes**Explanation:****Correct Answer:****Capture all the nested operations across the entire agent run → Hierarchical spans**

This is correct because hierarchical spans represent the full execution tree of an agent run, capturing parent-child relationships between steps, sub-steps, and tool calls. This ensures complete visibility into nested operations for auditing and tracing.

Record tool invocation arguments and results → Tool call attributes

This is correct because tool call attributes store metadata about each tool invocation, including input arguments and returned results. This enables detailed auditing of how tools were used and what data was processed.

Incorrect Answers:**A KQL query filter**

Used for querying logs, not for capturing execution traces.

Sampling / Trace sampling policy

Used to limit data collection, not to ensure full capture of operations.

Q116. You have a Microsoft Foundry project that contains an agent.

The agent uses Azure AI Search for Retrieval Augmented Generation (RAG). You plan to ingest and index PDF product manuals.

You need to build a solution that supports semantic similarity matching. The solution must ensure that the agent retrieves relevant data when user questions use different wording than the product manuals.

Which indexing approach should you use?

- A. vector search
- B. semantic ranking
- C. suggesters
- D. analyzers

Correct Answer:

A. vector search

Explanation:**Correct Answer:****A. vector search**

This is correct because vector search enables semantic similarity matching by converting text into embeddings and retrieving content based on meaning rather than exact keywords. This ensures that user queries with different wording still match relevant content from product manuals.

Incorrect Answers:**B. semantic ranking**

This improves ranking of already retrieved results but does not perform semantic retrieval itself.

C. suggesters

Used for autocomplete and search suggestions, not for semantic matching.

D. analyzers

Used for text processing like tokenization, not for semantic similarity search.

Q117. You develop a test method to verify the results retrieved from a call to the Azure Vision in Foundry Tools API. The call is used to analyze the existence of company logos in images. The call returns a collection of brands named brands.

You have the following code segment:

```
1  for brand in image_analysis.brands:
2      if brand.confidence >= 0.75:
3          print(f"\nLogo of {brand.name} between
4              {brand.rectangle.x}, {brand.rectangle.y} and {brand.rectangle.w},
              {brand.rectangle.h}")
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Answer Area

Statements	Yes	No
The code will display the name of each detected brand with a confidence equal to or higher than 75 percent.	☐	☐
The code will display coordinates for the top-left corner of the rectangle that contains the brand logo of the displayed brands.	☐	☐
The code will display coordinates for the bottom-right corner of the rectangle that contains the brand logo of the displayed brands.	☐	☐

Answer:

Answer Area

Statements	Yes	No
The code will display the name of each detected brand with a confidence equal to or higher than 75 percent.	☒	☐
The code will display coordinates for the top-left corner of the rectangle that contains the brand logo of the displayed brands.	☒	☐
The code will display coordinates for the bottom-right corner of the rectangle that contains the brand logo of the displayed brands.	☐	☒

Explanation:

Statement 1 --> Yes

This is correct because the code checks if `brand.confidence >= 0.75`, which ensures that only brands with confidence equal to or greater than 75% are processed. It then prints `brand.name`, so the name of each qualifying detected brand will be displayed.

Statement 2 --> Yes

This is correct because the code prints `{brand.rectangle.x}` and `{brand.rectangle.y}`, which represent the coordinates of the top-left corner of the bounding rectangle around the detected logo.

Statement 3 --> No

This is correct because the code prints width `{brand.rectangle.w}` and height `{brand.rectangle.h}`, not the bottom-right coordinates. Bottom-right coordinates would require calculating `x + w` and `y + h`, which is not done here.

Q118. You have a Microsoft Foundry agent that grounds responses from an Azure Search index that contains the following:

- Searchable text fields for product names and product codes
- A vector field that stores embeddings for product descriptions

You need to ensure that users can query the index by using the following:

- Exact product names or codes
- Natural language descriptions of the products

What should you configure?

- A. vector search only
- B. hybrid search
- C. keyword search only
- D. semantic search only

Correct Answer:

B. hybrid search

Explanation:

Correct Answer:

B. hybrid search

This is correct because hybrid search combines keyword search for exact matches on product names and codes with vector search for semantic similarity on product descriptions. This ensures both precise lookups and natural language queries are handled effectively in a single query.

Incorrect Answers:

A. vector search only

This supports semantic queries but does not handle exact keyword matches efficiently.

C. keyword search only

This works for exact matches but cannot handle natural language or semantic similarity.

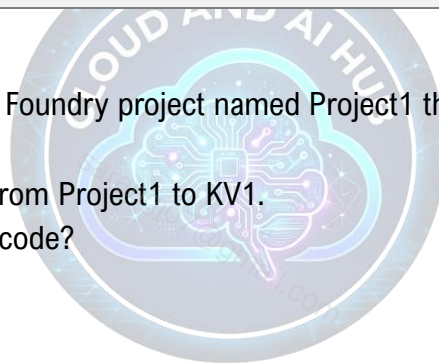
D. semantic search only

Semantic ranking enhances results but does not combine keyword and vector retrieval like hybrid search.

Q119. You plan to create a Microsoft Foundry project named Project1 that will contain an agent and use an Azure key vault named KV1.

You need to configure a connection from Project1 to KV1.

How should you complete the Bicep code?



Sarang Joshi

Answer Area

```
resource existingKeyVault 'Microsoft.KeyVault/vaults@2024-11-01'
existing = {
  name: 'KV1'
  scope: resourceGroup()
}
resource connection
'Microsoft.CognitiveServices/accounts/connections@2025-04-01-
preview' = {
  name: '${aiFoundryName}-keyvault'
  parent: aiFoundry
  properties: {
    category:  ▼
    

'AzureAIService'
      'AzureKeyVault'
      'AzureOpenAI'


  }
  target: existingKeyVault.id
  authType:  ▼
    

'AccountKey'
      'AccountManagedIdentity'
      'ApiKey'


  isSharedToAll:
    metadata: {
      ApiType: 'Azure'
      ResourceId: existingKeyVault.id
      location: existingKeyVault.location
    }
  }
}
```

Answer:

Answer Area

```
resource existingKeyVault 'Microsoft.KeyVault/vaults@2024-11-01'
existing = {
  name: 'KV1'
  scope: resourceGroup()
}
resource connection
'Microsoft.CognitiveServices/accounts/connections@2025-04-01-
preview' = {
  name: '${aiFoundryName}-keyvault'
  parent: aiFoundry
  properties: {
    category:
    target: existingKeyVault.id
    authType:
    isSharedToAll:
      metadata: {
        ApiType: 'Azure'
        ResourceId: existingKeyVault.id
        location: existingKeyVault.location
      }
  }
}
```

▼

'AzureAIService'

'AzureKeyVault'

'AzureOpenAI'

▼

'AccountKey'

'AccountManagedIdentity'

'ApiKey'

Explanation:

Correct Answer:

category → **AzureKeyVault**

This is correct because the connection is being created to an Azure Key Vault resource (KV1). The category must match the target service type so that Foundry correctly identifies and integrates with Key Vault.

authType → **AccountManagedIdentity**

This is correct because managed identity enables secure, keyless authentication between the Foundry project and Key Vault. It aligns with best practices and avoids storing secrets like API keys while supporting least privilege access.

Incorrect Answers:

AzureAIService / AzureOpenAI

These categories are for AI services, not Key Vault connections.

AccountKey / ApiKey

These require storing secrets, which violates secure keyless authentication practices.

Q120. You have an Azure AI Search indexer that ingests PDF policy manuals.

Client applications must display page-level citations that have bounding polygons for both text and images.

You need to add a single built-in multimodal content extraction skill to the Azure AI Search skillset.

The solution must meet the following requirements:

- Provide text and image location metadata.
- Extract tables that span multiple pages.

What should you add?

- A. Document Extraction
- B. Azure Content Understanding in Foundry Tools
- C. GenAI Prompt
- D. Document Layout

Correct Answer:

B. Azure Content Understanding in Foundry Tools

Explanation:

Correct Answer:

B. Azure Content Understanding in Foundry Tools

This is correct because it satisfies both requirements simultaneously. It extracts text and images with bounding polygon metadata, enabling precise page-level citations. More importantly, it can reconstruct tables that span multiple pages as a single coherent structure (Markdown/semantic format), which is explicitly required. Since the solution must handle both multimodal metadata and cross-page table continuity, this is the only option that fully meets the requirements.

Incorrect Answers:

A. Document Extraction

Provides basic text extraction but does not include layout metadata such as bounding polygons or support advanced table structures.

C. GenAI Prompt

Used for generating or transforming content, not for extracting layout, positional metadata, or structured tables from documents.

Side Note (Important Exam Trap)



D. Document Layout

This option is very tempting and commonly chosen. It does provide bounding polygons for text and images, so it partially meets the requirement. However, it fails the critical requirement of handling tables that span multiple pages, as it tends to break them into separate segments or flatten them.

Since the question requires ALL conditions to be met, Document Layout becomes incorrect despite being close.

Q121. You are building an Azure AI Search indexing pipeline named Pipeline1 that ingests invoices stored in Azure Blob Storage. The invoices are stored as scanned images.

You need to enable users to search invoice data across the invoice fields. Which built-in skill should you add to the skillset of Pipeline1?

- A. Text Split
- B. Text Translation
- C. optical character recognition (OCR)
- D. Image Analysis

Correct Answer:

C. optical character recognition (OCR)

Explanation:**Correct Answer:****C. optical character recognition (OCR)**

This is correct because the invoices are stored as **scanned images**, which means they do not contain searchable text. OCR converts the text within images into machine-readable text, enabling Azure AI Search to index invoice fields and make them searchable.

Incorrect Answers:**A. Text Split**

Used to break large text into chunks, but it requires text input and cannot extract text from images.

B. Text Translation

Translates text between languages, but does not extract text from scanned images.

D. Image Analysis

Detects objects, captions, or tags in images, but does not extract structured text required for search indexing.

Q122. You have an invoice-processing application named App1 that uses Azure Constant Understanding in Foundry Tools.

You are building a new Content Understanding pipeline named Pipeline1 that must meet the following requirements:

- Compare an invoice to its related purchase order
- Validate the voice against static vendor contact documents
- Return a single structured output that includes discrepancy findings

You need to configure Pipeline1 and expose the pipeline as a single analyzer endpoint. What should you configure?

- A. a single-file task in standard mode that uses the vendor contract provided as an additional document during analysis.
- B. a single-file task in standard mode that uses confidence scores enabled for the extracted fields.
- C. a multiple-file task in pro mode that uses the vendor contract files as reference data
- D. a multi-file task in standard mode that uses the invoice and purchase order as input to the analyzer

Correct Answer:

C. a multiple-file task in pro mode that uses the vendor contract files as reference data

Explanation:

Correct Answer:

C. a multiple-file task in pro mode that uses the vendor contract files as reference data

This is correct because the requirements involve comparing multiple documents together (invoice vs purchase order and vendor documents) and producing a single structured output with discrepancy findings. Multi-file tasks enable cross-document reasoning, and pro mode supports more advanced scenarios like comparisons and validation against reference documents, making it the only option that satisfies all requirements.

Incorrect Answers:

A. a single-file task in standard mode that uses the vendor contract provided as an additional document during analysis

Single-file tasks cannot properly handle comparisons across multiple documents.

B. a single-file task in standard mode that uses confidence scores enabled for the extracted fields

Confidence scores do not address cross-document comparison or validation requirements.

D. a multi-file task in standard mode that uses the invoice and purchase order as input to the analyzer

Standard mode lacks the advanced reasoning and validation capabilities required for discrepancy detection across multiple documents.

Q123. You have a Microsoft Foundry project.

You need to create a customer support agent that meets the following requirements:

- Grounds responses only in company policy documents stored in curated repositories
- Retains customer preferences across separate chat sessions

How should you configure the agent?

Answer Area

Knowledge grounding:

Configure retrieval from approved data sources.
Upload the policy documents directly to the agent.
Embed the policy documents directly into the agent instructions.

Memory:

Use orchestration-managed session context.
Enable agent memory that uses persistent storage.
Retain user preferences in the state of the client application

Answer:

Answer Area

Knowledge grounding:

Configure retrieval from approved data sources.
Upload the policy documents direct to the agent.
Embed the policy documents directly into the agent instructions.

Memory:

Use orchestration-managed session context.
Enable agent memory that uses persistent storage.
Retain user preferences in the state of the client application

Explanation:

Correct Answer:

Knowledge grounding → Configure retrieval from approved data sources

This is correct because the requirement is to ground responses **only in company policy documents stored in curated repositories**. Retrieval from approved data sources ensures controlled access to trusted content using RAG, preventing the agent from using unverified or embedded information.

Memory → Enable agent memory that uses persistent storage

This is correct because the agent must **retain customer preferences across separate chat sessions**. Persistent memory allows data to be stored beyond a single session, enabling long-term personalization and continuity.

Incorrect Answers:

Upload the policy documents direct to the agent / Embed the policy documents directly into the agent instructions

These approaches are static, not scalable, and do not enforce controlled retrieval from curated repositories.

Use orchestration-managed session context

This only retains information within a single session and does not persist across multiple interactions.

Retain user preferences in the state of the client application

This is not centralized or reliable for enterprise scenarios and does not provide true persistent agent memory.

Q124. You have a Microsoft Foundry project that generates short promotional product videos. After several clips are approved, reviewers notice a small watermark in the top-right corner of some videos. You need to remove the watermark without regenerating the videos.

What should you do?

- A. Modify the original prompt to exclude watermarks.
- B. Crop the video by using the size parameter.
- C. Increase the guidance scale.
- D. Apply a mask-based inpainting edit to the affected part of the video.

Correct Answer:

D. Apply a mask-based inpainting edit to the affected part of the video

Explanation:

Correct Answer:

D. Apply a mask-based inpainting edit to the affected part of the video

This is correct because the requirement is to **remove a watermark from already generated videos without regenerating them**. Mask-based inpainting allows you to selectively edit a specific region (top-right corner) and remove or replace the watermark while preserving the rest of the video content.

Incorrect Answers:

A. Modify the original prompt to exclude watermarks

This would only affect future video generation and does not fix already generated videos.

B. Crop the video by using the size parameter

Cropping would remove part of the video frame and may affect important content. It is not a precise or suitable solution for watermark removal.

C. Increase the guidance scale

Guidance scale influences how closely the model follows the prompt during generation. It has no effect on already generated videos or post-processing edits.

Q125. You have a Microsoft Foundry project that contains an agent. The agent uses threads and file uploads and calls an Azure OpenAI model deployment.

During load testing, calls intermittently fail and return an HTTP 429 rate limit exceeded error. Some user uploads fail and generate an HTTP 400 file size exceeded error.

You need to mitigate the errors and reduce call failures. The solution must remain within the service and model limits.

What should you do to resolve each error?

Actions

Increase tenant-wide quotas.

Move large content to files and use file search.

Use additional agent tools to reduce the message size.

Implement exponential backoff and jitter in the retry logic.

Split content into smaller files before uploading the files.

Answer Area

HTTP 429:

HTTP 400:

Answer:

Actions

Increase tenant-wide quotas.

Move large content to files and use file search.

Use additional agent tools to reduce the message size.

Implement exponential backoff and jitter in the retry logic.

Split content into smaller files before uploading the files.

Answer Area

HTTP 429:

HTTP 400:

Explanation:

Correct Answer:

HTTP 429 → Implement exponential backoff and jitter in the retry logic

This is correct because HTTP 429 indicates **rate limiting**. The proper way to handle rate limits within service constraints is to retry requests using exponential backoff with jitter. This reduces contention and prevents overwhelming the service while staying within allowed quotas.

HTTP 400 → Split content into smaller files before uploading the files

This is correct because HTTP 400 with file size exceeded indicates that uploads are breaching the **maximum file size limit**. Splitting files ensures each upload stays within the allowed limits and succeeds.

Incorrect Answers:

Increase tenant-wide quotas

This violates the requirement to remain within service and model limits.

Move large content to files and use file search

This helps with large prompts but does not directly resolve file upload size limit errors.

Use additional agent tools to reduce the message size

This may reduce prompt size but does not address file upload size constraints or rate limiting directly.

Q126. You have a web app named App1 that sends requests to a multimodal chat model deployment in a Microsoft Foundry project.

User messages can contain both text and images.

Currently, App1 includes image URL: as plain text inside the message content so the model cannot recognize them as images.

Traces show that the requests contain a single text message instead of a multimodal content array.

You need to send the message as a structured array that includes both the text portion and the image reference to ensure that the model can process the image correctly.

What should you do?

- A. Set the user message content array to include items that have type: text and type: image_url.
- B. Encode the image to base64 and include the encoded data inside the content string of the user message.
- C. Add the image URL to the request metadata section, so the model can resolve the processing issue automatically.
- D. Place the image URL inside the System Message and set type to image_url so the model loads the image at initialization.

Correct Answer:

- A. Set the user message content array to include items that have type: text and type: image_url.

Explanation:**Correct Answer:****A. Set the user message content array to include items that have type: text and type: image_url**

This is correct because multimodal chat models expect input in a **structured content array**, where each element explicitly defines its type (for example, text or image_url). This allows the model to correctly interpret and process both text and image inputs within the same message.

Incorrect Answers:**B. Encode the image to base64 and include the encoded data inside the content string of the user message**

Embedding base64 inside a plain text string does not create a multimodal input. The model will treat it as text, not as an image.

C. Add the image URL to the request metadata section

Metadata is not used by the model for content understanding. The model only processes structured message content.

D. Place the image URL inside the System Message and set type to image_url

System messages are meant for instructions and configuration, not for passing user content like images. This will not correctly enable multimodal processing.

Q127. You have a Python application that redacts sensitive information before sending prompt text to a language model. The application has the following code:

```

1  sample_text = "Contact John Doe at 312-555-1234 or
2  john.doe@contoso.com. His SSN is 859-98-0987."
3
4  def redact_for_model(input_text):
5      payload = {
6          "kind": "PiiEntityRecognition",
7          "parameters": {
8              "modelVersion": "latest",
9              "piiCategories": ["Person", "PhoneNumber"],
10             "redactionPolicies": [
11                 {
12                     "policyKind": "entityMask"
13                 }
14             ]
15         },
16         "analysisInput": {
17             "documents": [
18                 {
19                     "id": "1",
20                     "language": "en",
21                     "text": input_text
22                 }
23             ]
24         }
25     }
26     ...
27     return {
28         "text for model": doc.get("redactedText", ""),
29         "audit": [
30             (entity["text"], entity["category"], entity["confidenceScore"])
31             for entity in doc["entities"]
32         ]
33     }
34 result = redact_for_model(sample_text)

```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Answer Area

Statements	Yes	No
For <code>sample_text</code> , audit will include entity records for Contact and SSN.	☐	☐
For <code>sample_text</code> , <code>text_for_model</code> will include <code>john.doe@contoso.com</code> and <code>859-98-0987</code> .	☐	☐
For <code>sample_text</code> , <code>text_for_model</code> will contain entity type masks for John Doe and 312-555-1234.	☐	☐

Answer**Answer Area**

Statements	Yes	No
For <code>sample_text</code> , audit will include entity records for Contact and SSN.	☐	☒
For <code>sample_text</code> , <code>text_for_model</code> will include <code>john.doe@contoso.com</code> and <code>859-98-0987</code> .	☐	☒
For <code>sample_text</code> , <code>text_for_model</code> will contain entity type masks for John Doe and 312-555-1234.	☒	☐

Explanation:**Statement 1 --> No**

This is correct because the configuration restricts detection to "piiCategories": ["Person", "PhoneNumber"]. The terms "Contact" and "SSN" are not included in these categories, so they will not be detected as entities and therefore will not appear in the audit output.

Statement 2 --> No

This is correct because `text_for_model` uses `redactedText`, which applies masking (`entityMask`) to detected entities. Even if email and SSN are present in the text, they will either be masked (if detected) or excluded (if not part of configured categories). Hence, the original values like `john.doe@contoso.com` and `859-98-0987` will not appear.

Statement 3 --> Yes

This is correct because the configuration includes "Person" and "PhoneNumber" with `entityMask` redaction. Therefore, "John Doe" and "312-555-1234" will be detected and replaced with masked placeholders in `text_for_model`.

Q128. You are building a customer support web app named App1 in Microsoft Foundry that uses a GPT realtime model.

App1 must support:

- Live, low-latency voice conversations that use Azure OpenAI
- Streaming audio input from users and playback audio responses

You need to configure a connection method that supports real-time audio streaming in client application and targets approximately 100 ms latency.

Which connection method should you use?

- A. RTMP
- B. WebRTC
- C. SIP
- D. WebSocket

Correct Answer:

B. WebRTC

Explanation:

Correct Answer:

B. WebRTC

This is correct because WebRTC is designed for **real-time, low-latency audio and video streaming**, typically achieving latency around ~100 ms. It supports **bi-directional streaming**, making it ideal for live voice conversations where users send audio input and receive audio responses instantly.

Incorrect Answers:

A. RTMP

This is primarily used for live video streaming (e.g., broadcasting) and has higher latency. It is not suitable for interactive real-time conversations.

C. SIP

SIP is used for telephony signaling but does not handle real-time media streaming efficiently for modern web-based applications like GPT realtime models.

D. WebSocket

WebSocket supports real-time communication but is not optimized for **low-latency audio streaming** like WebRTC. It typically results in higher latency compared to WebRTC for media scenarios.

Q129. You have an Azure subscription.

You need to create a new resource that will generate fictional stores in response to user prompts. The solution must ensure that the resource uses a customer-managed key to protect data.

How should you complete the script?

Answer Area

```
az cognitiveservices account create -n myresource -g
myResourceGroup --kind  --sku S -l WestEurope
```

AI Services
 LanguageAuthoring
 OpenAI

```
`{
  --api-properties
  --assign-identity
  --encryption
  "keySource": "Microsoft.KeyVault",
  "keyVaultProperties": {
    "keyName": "KeyName",
    "keyVersion": "secretVersion",
    "keyVaultUri": "https://issue23056kv.vault.azure.net/"
  }
}'
```

Answer:

Answer Area

```
az cognitiveservices account create -n myresource -g
myResourceGroup --kind  --sku S -l WestEurope
```

AI Services
 LanguageAuthoring
 OpenAI

```
`{
  --api-properties
  --assign-identity
  --encryption
  "keySource": "Microsoft.KeyVault",
  "keyVaultProperties": {
    "keyName": "KeyName",
    "keyVersion": "secretVersion",
    "keyVaultUri": "https://issue23056kv.vault.azure.net/"
  }
}'
```

Explanation:

Correct Answer:

kind → OpenAI

This is correct because the requirement is to generate responses to user prompts (fictional stores), which requires an Azure OpenAI resource. Therefore, the resource type must be set to OpenAI.

parameter → --encryption

This is correct because customer-managed keys (CMK) are configured using the encryption property. The provided JSON block specifies "keySource": "Microsoft.KeyVault" and Key Vault details, which are part of encryption settings for CMK.

Incorrect Answers:**AI Services / LanguageAuthoring**

These resource kinds are used for other AI capabilities (vision, language authoring) and do not support OpenAI prompt-based generation.

--api-properties / --assign-identity

These parameters do not configure customer-managed keys. CMK specifically requires the encryption configuration.

Q130. You have a Microsoft Foundry project that contains an agent and uses a GitHub repository. The repository contains a YAM file named File1 that defines the evaluation settings of the agent. You need to create a GitHub Actions workflow that runs the evaluation defined in File1 when a pull request (PR) is opened. How should you configure the workflow?

- A. Set project-endpoint to the endpoint of the project.
- B. Set evaluation-config to the path of the YAML file.
- C. Set model-deployment-name to the deployed model.
- D. Set tenant-id to the Microsoft Entra tenant ID

Correct Answer:

B. Set evaluation-config to the path of the YAML file.

Explanation:**Correct Answer:****B. Set evaluation-config to the path of the YAML file.**

The GitHub Actions workflow for running evaluations in a Microsoft Foundry project relies on referencing the evaluation configuration file that defines how the agent should be tested. Since File1 is a YAML file containing the evaluation settings, the workflow must point to this file using the evaluation-config parameter. This ensures that when a pull request is opened, the workflow executes the evaluation exactly as defined in the YAML configuration, enabling automated validation of the agent's behavior.

Incorrect answers:**A. Set project-endpoint to the endpoint of the project.**

This is not required for triggering evaluation from a YAML configuration in GitHub Actions and does not define which evaluation to run.

C. Set model-deployment-name to the deployed model.

This is unrelated to running evaluations from a YAML file and is only used when directly referencing a specific model deployment.

D. Set tenant-id to the Microsoft Entra tenant ID

Tenant ID is used for authentication context but does not specify or trigger the evaluation configuration defined in the YAML file.

Q131. You have a Microsoft Foundry project.

You need to deploy a model from the model catalog to support a search solution for internal policy documents. The model must generate vector representations of the text in the documents and of user queries.

Which type of model should you use?

- A. an embedding model
- B. an image generation model
- C. a large language model (LLM)
- D. a small language model (SLM)

Correct Answer:

A. an embedding model

Explanation:**Correct Answer:****A. an embedding model**

Embedding models are specifically designed to convert text into numerical vector representations. These vectors capture the semantic meaning of the text, enabling similarity-based search. In a search solution for internal policy documents, both the documents and user queries must be converted into vectors so that relevant matches can be identified using vector similarity. This is exactly the purpose of embedding models in Azure AI Foundry and related services.

Incorrect answers:**B. an image generation model**

This type of model is used to generate images from text prompts and is not suitable for text vectorization or search.

C. a large language model (LLM)

LLMs are primarily used for text generation, summarization, and reasoning, not for generating vector embeddings for search indexing.

D. a small language model (SLM)

SLMs are optimized for lightweight text processing tasks but are not designed specifically for generating high-quality vector embeddings for semantic search.

Q132. You have a Microsoft Foundry project.

You need to deploy a model from the model catalog to support real-time inference.

The solution must meet the following requirements:

- Use key-based authentication.
- Support real-time REST API access.
- NOT consume the vCPU quota of the virtual machines in the Azure subscription.

Which type of deployment should you use?

- A. serverless API
- B. batch
- C. self-hosted container
- D. standard

Correct Answer:

A. serverless API

Explanation:

Correct Answer:

A. serverless API

A serverless API deployment in Microsoft Foundry allows you to expose the model through a real-time REST endpoint without managing underlying infrastructure. It supports key-based authentication and provides instant scalability for real-time inference. Most importantly, it does not consume the vCPU quota of your Azure subscription because the compute is fully managed by the service, making it ideal for this requirement.

Incorrect answers:

B. batch

Batch deployments are designed for asynchronous, large-scale processing and do not support real-time REST API inference.

C. self-hosted container

This option requires you to run the model on your own infrastructure, which would consume vCPU resources in your Azure subscription.

D. standard

Standard deployments run on dedicated compute resources within your subscription, which means they consume vCPU quota and do not meet the requirement.

Q133. You have a Python application collects customer comments before posting them to a public forum.

You need to send a text comment to Azure AI Content Safety and return the self-harm severity from the response. How should you complete the code?

Answer Area

```
def get_self_harm_severity (comment: str) -> int:
    key = os.environ["CONTENT_SAFETY_KEY"]
    endpoint = os.environ["CONTENT_SAFETY_ENDPOINT"]
    client = ContentSafetyClient(endpoint,
    AzureKeyCredential(key))
    request = 
    response = 
    result = next(
        item for item in response.categories_analysis
        if item.category == TextCategory.SELF_HARM
    )
    return result.severity
```

Answer:

Answer Area

```
def get_self_harm_severity (comment: str) -> int:
    key = os.environ["CONTENT_SAFETY_KEY"]
    endpoint = os.environ["CONTENT_SAFETY_ENDPOINT"]
    client = ContentSafetyClient(endpoint,
    AzureKeyCredential(key))
    request = 
    response = 
    result = next(
        item for item in response.categories_analysis
        if item.category == TextCategory.SELF_HARM
    )
    return result.severity
```

Explanation:

Correct Answer:

request = AnalyzeTextOptions(text=comment)

To analyze text using Azure AI Content Safety, you must create a request using AnalyzeTextOptions and pass the actual text content. The parameter text=comment correctly sends the input string for analysis.

response = client.analyze_text(request)

The correct method to send the text analysis request using the ContentSafetyClient is analyze_text(). This method processes the request and returns categorized analysis, including severity levels such as self-harm.

Incorrect answers:

AnalyzeTextOptions(categories=comment)

This is incorrect because categories is not used to pass input text; it is used to specify which categories to evaluate.

AnalyzeTextOptions(text=[comment])

This is incorrect because the method expects a string, not a list of strings.

TextCategory.SELF_HARM(comment)

This is invalid syntax and TextCategory is used for filtering results, not for creating requests.

client.analyze_image(request)

This method is used for image analysis, not text.

client.moderate_text(request)

This is not the correct SDK method for Content Safety text analysis.

client.path("/text:analyze").post(request)

This is a low-level REST call pattern and not the recommended SDK method for this scenario.

Q134. You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.

You need to improve response completeness. The solution must be implemented in the logic of the application code before responses are returned.

What should you do?

- A. Add a retry evaluation before the responses are returned.
- B. Decrease the value of the temperature parameter.
- C. Increase the value of the presence_penalty parameter
- D. Replace the model with a smaller deployment.

Correct Answer:

A. Add a retry evaluation before the responses are returned.

Explanation:

Correct Answer:

A. Add a retry evaluation before the responses are returned.

To improve response completeness, implementing a retry or evaluation mechanism in the application logic allows the system to validate the generated response and regenerate it if it is incomplete. This approach ensures higher-quality outputs by programmatically checking whether the response meets completeness criteria before returning it to the user. This aligns directly with the requirement to implement the solution in application code logic.

Incorrect answers:

B. Decrease the value of the temperature parameter.

Lowering temperature makes responses more deterministic but does not specifically improve completeness.

C. Increase the value of the presence_penalty parameter

This encourages introducing new topics, which may reduce focus rather than improve completeness.

D. Replace the model with a smaller deployment.

A smaller model typically has lower capability and may reduce response quality and completeness.

Q135. You are developing an application that will detect faulty components produced on a factory production line. The components are specific to your business.

You need to use the Azure Custom Vision API to help detect common faults.

Which three actions should you perform in sequence?

Actions

- Initialize the training dataset.
- Train the classifier model.
- Create a project.
- Upload and tag images.
- Train the object detection model.

Answer Area

- 1
- 2
- 3

Answer:**Actions**

- Initialize the training dataset.
- Train the classifier model.
- Create a project.
- Upload and tag images.
- Train the object detection model.

Answer Area

- 1
- 2
- 3

Create a project.

Upload and tag images.

Train the classifier model.

Explanation:**Correct Answer:****1. Create a project.**

The first step in Azure Custom Vision is to create a project where you define the type of model (classification or object detection) and organize your dataset and training process.

2. Upload and tag images.

After creating the project, you must upload images of the components and tag them appropriately (for example, faulty vs non-faulty). Tagging is critical because it provides labeled data for supervised learning.

3. Train the classifier model.

Once the images are uploaded and tagged, you train the model so it can learn patterns and detect faults based on the labeled dataset.

Incorrect answers:**Initialize the training dataset.**

This is not an explicit step in Azure Custom Vision; dataset preparation happens during image upload and tagging.

Train the object detection model.

Object detection is used when identifying locations of objects in images, not just classifying faults. The scenario focuses on detecting faulty components, which aligns with classification.

Q136. You have an Azure subscription.

You plan to build an app that will use the Azure AI DALL-E model. You need to deploy the model. What should you use?

- A. the Azure SDK for Python and PowerShell cmdlets.
- B. the Azure SDK for JavaScript and Azure Machine Learning Studio.
- C. Microsoft Foundry and the Azure Command Line Interface (CLI)
- D. the Azure portal and Microsoft Graph API

Correct Answer:

C. Microsoft Foundry and the Azure Command Line Interface (CLI)

Explanation:

Correct Answer:

C. Microsoft Foundry and the Azure Command Line Interface (CLI)

Azure AI DALL-E models are part of Azure AI Foundry (formerly Azure AI Studio) and are deployed through model catalog capabilities. You can deploy these models using Microsoft Foundry UI or automate deployments using the Azure CLI. This combination supports proper provisioning, deployment, and management of generative AI models like DALL-E.

Incorrect answers:

A. the Azure SDK for Python and PowerShell cmdlets.

These can interact with deployed models but are not used to deploy DALL-E models from the model catalog.

B. the Azure SDK for JavaScript and Azure Machine Learning Studio.

Azure Machine Learning Studio is not used for deploying Azure AI DALL-E models from the Foundry model catalog.

D. the Azure portal and Microsoft Graph API

Microsoft Graph API is unrelated to deploying AI models and cannot be used for this purpose.

Q137. You have a custom named entity recognition (NER) project in Azure Language in Foundry Tools for support tickets. The schema for the project contains an entity type named ContactInfo.

In tagged training files, ContactInfo is used for phone numbers, email addresses, and social media handles. Model evaluation shows low precision for ContactInfo, including false positives in which nearby text is extracted as ContactInfo.

You need to improve the precision of the project.

What should you do before retraining the model?

- A. Lower the confidence threshold for ContactInfo.

- B. Trigger an auto-labeling job.
- C. Add more support tickets as training data and label more ContactInfo entities.
- D. Replace ContactInfo by using Phone, Email, and SocialMedia entities. Relabel every matching span.

Correct Answer:

D. Replace ContactInfo by using Phone, Email, and SocialMedia entities. Relabel every matching span.

Explanation:

Correct Answer:

D. Replace ContactInfo by using Phone, Email, and SocialMedia entities. Relabel every matching span.

Low precision and false positives indicate that the model is overgeneralizing the ContactInfo entity. By splitting this broad entity into more specific entities such as Phone, Email, and SocialMedia, you provide clearer distinctions and reduce ambiguity. This improves the model's ability to correctly identify only the intended spans and avoid extracting nearby irrelevant text. Fine-grained labeling is a key technique to improve precision in NER models.

Incorrect answers:

A. Lower the confidence threshold for ContactInfo.

Lowering the threshold would increase false positives, worsening precision.

B. Trigger an auto-labeling job.

Auto-labeling may propagate existing labeling issues and does not address the root cause of low precision.

C. Add more support tickets as training data and label more ContactInfo entities.

Adding more data without refining the entity definition may continue the same ambiguity and not significantly improve precision.

Q138. You are building a text-to-speech solution that uses Azure Speech in Foundry Tools to read instructions from the script in a text file.

You discover that the solution often pronounces technical terms incorrectly.

You need to prevent the incorrect pronunciations. The solution must minimize development effort.

What should you do?

- A. From Speech Studio, train a custom neural voice
- B. Use Speech Synthesis Markup Language (SSML) to specify phonemes.
- C. Use Speech Synthesis Markup Language (SSML) to apply say as rules.
- D. Use Speech Synthesis Markup Language (SSML) to adjust the prosody of the voice.
- E. From Azure OpenAI use the Whisper model.

Correct Answer:

B. Use Speech Synthesis Markup Language (SSML) to specify phonemes.

Explanation:**Correct Answer:****B. Use Speech Synthesis Markup Language (SSML) to specify phonemes.**

SSML phoneme tags allow you to explicitly define how specific words should be pronounced by providing their phonetic representation. This is the most effective and low-effort way to correct pronunciation of technical terms without retraining models. It gives precise control over pronunciation and is ideal for handling domain-specific vocabulary.

Incorrect answers:**A. From Speech Studio, train a custom neural voice**

This requires significant effort and is unnecessary for fixing pronunciation of specific terms.

C. Use Speech Synthesis Markup Language (SSML) to apply say as rules.

Say-as is used for formatting (dates, numbers, etc.), not for correcting pronunciation of technical terms.

D. Use Speech Synthesis Markup Language (SSML) to adjust the prosody of the voice.

Prosody controls pitch, rate, and volume, not pronunciation.

E. From Azure OpenAI use the Whisper model.

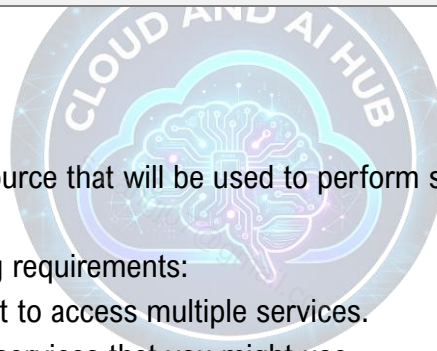
Whisper is used for speech-to-text, not text-to-speech pronunciation control.

Q139. You need to create a new resource that will be used to perform sentiment analysis and optical character recognition (OCR).

The solution must meet the following requirements:

- Use a single key and endpoint to access multiple services.
- Consolidate billing for future services that you might use.
- Support the use of Azure Vision in Foundry Tools in the future.

How should you complete the HTTP request to create the new resource?



Sarang Joshi

Answer Area

▼ PATCH POST PUT	https://management.azure.com/subscriptions/xxxxxxxx-xxxx-xxxx-xxxxxxxxxxxx/resourceGroups/RG1/providers/Microsoft.CognitiveService/accounts/CS1?api-version=2021-04-30
--	---

```
{  
  "location": "West US",  
  "kind": "

▼



CognitiveServices  
ComputerVision  
TextAnalytics

",  
  "sku": {  
    "name": "S0"  
  },  
  "properties": {},  
  "identity": {  
    "type": "SystemAssigned"  
  }  
}
```

Answer:**Answer Area**

▼ PATCH POST PUT	https://management.azure.com/subscriptions/xxxxxxxx-xxxx-xxxx-xxxxxxxxxxxx/resourceGroups/RG1/providers/Microsoft.CognitiveService/accounts/CS1?api-version=2021-04-30
---	---

```
{  
  "location": "West US",  
  "kind": "

▼



CognitiveServices  
ComputerVision  
TextAnalytics

",  
  "sku": {  
    "name": "S0"  
  },  
  "properties": {},  
  "identity": {  
    "type": "SystemAssigned"  
  }  
}
```

Explanation:**Correct Answer:**

HTTP method: PUT

"kind": CognitiveServices

Explanation:**Correct Answer:****HTTP method: PUT**

PUT is the standard Azure Resource Manager verb for creating or fully defining a resource at a known, specific URI (such as `/subscriptions/{subId}/resourceGroups/{rg}/providers/Microsoft.CognitiveServices/accounts/{accountName}`). It is idempotent, meaning sending the same request multiple times results in the same resource state - the standard convention ARM uses when you already know the exact resource name/path you want to create.

"kind": CognitiveServices

Setting "kind" to CognitiveServices creates a multi-service resource, which provides a single key and endpoint to access multiple AI capabilities (sentiment analysis via Language, OCR via Vision) under one resource. This consolidates billing into a single line item and allows additional services, such as Azure Vision in Foundry Tools, to be accessed later without provisioning a new resource.

Incorrect answers:**PATCH**

PATCH performs a partial update to an existing resource, changing only the specified properties while leaving the rest intact. Since this scenario creates a brand-new resource (CS1 doesn't exist yet), there is nothing to partially modify, making PATCH incorrect here.

POST

In Azure Resource Manager, POST is typically used to trigger actions or operations on an existing resource (for example, restart, listKeys, or regenerateKey) rather than to declaratively create a resource at a specific URI with a full desired-state body. ARM's resource-creation pattern is consistently PUT-based, not POST-based.

ComputerVision

Setting "kind" to ComputerVision creates a single-service resource scoped only to the Computer Vision API (image analysis, OCR, etc.). It would satisfy the OCR requirement but provides no access to sentiment analysis under the same key and endpoint, requiring a second separate resource — which violates the requirement for a single key/endpoint and consolidated billing.

TextAnalytics

Setting "kind" to TextAnalytics creates a single-service resource scoped only to Text Analytics (sentiment analysis, language detection, etc.). It would satisfy the sentiment analysis requirement but provides no OCR/vision capability, again requiring a second resource and failing the single key/endpoint and consolidated billing requirements. It would also not support adding Azure Vision in Foundry Tools later without provisioning another resource.

Q140. You are developing a new sales system that will process user-generated video and text from a public-facing website.

You plan to notify users that their data has been processed by the sales system. Which responsible AI principle does this help meet?

- A. fairness
- B. transparency
- C. inclusiveness
- D. reliability and safety

Correct Answer:

B. transparency

Explanation:

Correct Answer:

B. transparency

Transparency in responsible AI refers to informing users how their data is being used and how systems operate. By notifying users that their video and text data are being processed by the sales system, you ensure openness about data usage. This builds trust and aligns with the principle of making AI systems understandable and clear to users.

Incorrect answers:

A. fairness

Fairness focuses on eliminating bias and ensuring equitable treatment, not informing users about data processing.

C. inclusiveness

Inclusiveness ensures accessibility and usability for diverse groups, not transparency of data usage.

D. reliability and safety

This principle relates to consistent and safe system behavior, not user awareness of data processing.

Q141. You are creating an enrichment pipeline that will use Azure AI Search. The knowledge store contains unstructured JSON data and the text from scanned PDF documents.

Which projection type should you use for each data type?

Answer Area

JSON data: ▼

File projection

Object projection

Table projection

Extracted text data: ▼

File projection

Object projection

Table projection

Answer:

Answer Area

JSON data:

▼

File projection
Object projection
Table projection

Extracted text data:

▼

File projection
Object projection
Table projection

Explanation:

Correct Answer:

JSON data: Object projection

Object projection is used for structured or semi-structured data like JSON. It preserves the hierarchical structure of the JSON documents, making it suitable for storing and querying complex objects in the knowledge store.

Extracted text data: Table projection

Table projection is ideal for flattened, structured representations of data such as extracted text from documents. It organizes the data into rows and columns, making it efficient for querying and analysis, especially for OCR-extracted text from scanned PDFs.

Incorrect answers:

File projection

File projection is used to store enriched documents as files (for example, images or PDFs), not for structured JSON or extracted text intended for querying.

Q142. You have an Azure subscription that contains an Azure App Service app named App1. You provision a Microsoft Foundry Service resource named CSAccount1.

You need to configure App1 to access CSAccount1. The solution must minimize administrative effort.

What should you use to configure App1?

- A. the endpoint URI and subscription key
- B. the endpoint URI and an OAuth token
- C. the endpoint URI and a shared access signature (SAS) token
- D. a system assigned managed identity and an X.509 certificate

Correct Answer:

- A. the endpoint URI and subscription key

Explanation:

Correct Answer:**A. the endpoint URI and subscription key**

Using the endpoint URI along with a subscription key is the simplest and most commonly used method to authenticate an App Service app with an Azure AI (Foundry) resource. It requires minimal configuration and administrative effort, making it ideal for quick integration scenarios. The application simply includes the key in the request headers to securely access the service.

Incorrect answers:**B. the endpoint URI and an OAuth token**

This requires additional setup with Microsoft Entra ID and token management, increasing administrative effort.

C. the endpoint URI and a shared access signature (SAS) token

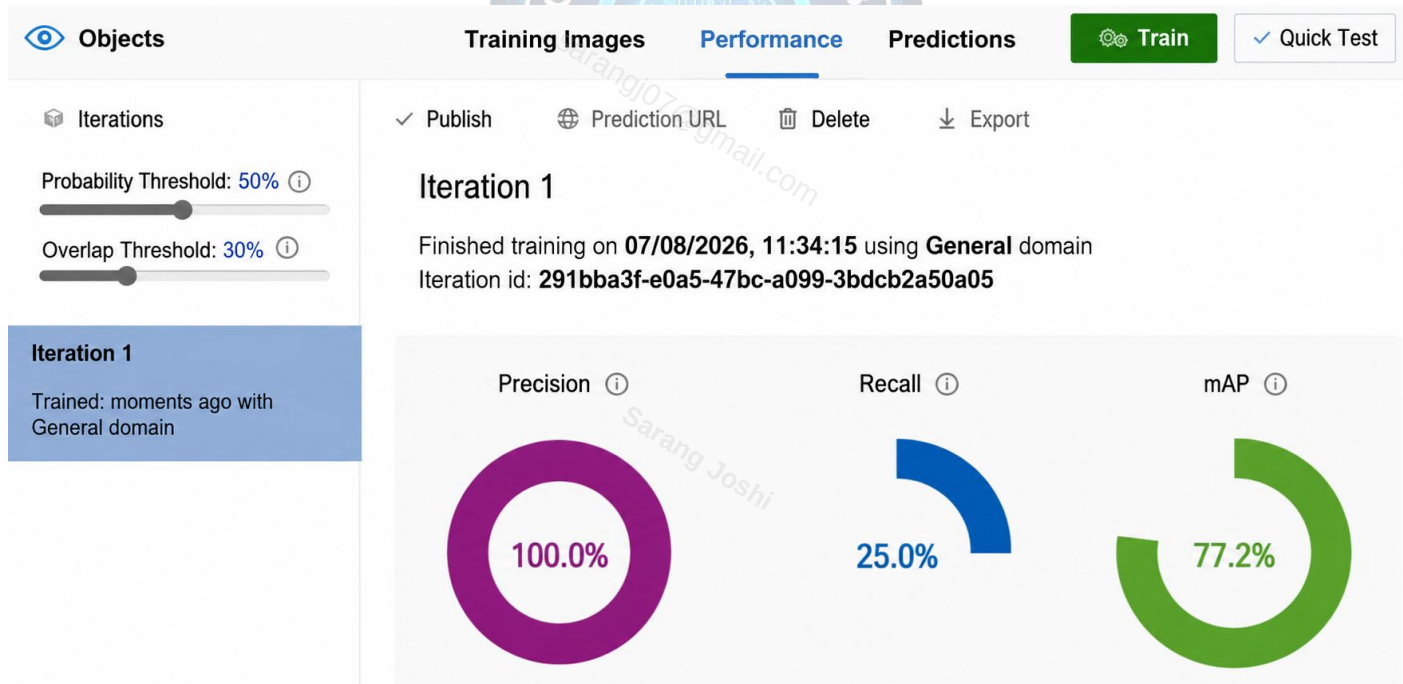
SAS tokens are typically used for Azure Storage, not for accessing Azure AI services.

D. a system assigned managed identity and an X.509 certificate

This involves more complex configuration and certificate management, which does not minimize administrative effort.

Q143. You are building a model to detect objects in images.

The performance of the model based on training data is shown in the following exhibit.



Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the graphic.

Answer Area

The percentage of false positives is:

	▼
0	
25	
30	
50	
100	

The value for the number of true positives divided by the total number of true positives and false negatives is:

	▼
0	
25	
30	
50	
100	

Answer:**Answer Area**

The percentage of false positives is:

	▼
0	
25	
30	
50	
100	

The value for the number of true positives divided by the total number of true positives and false negatives is:

	▼
0	
25	
30	
50	
100	

Explanation:**Correct Answer:****0**

Precision is given as **100%**.

$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$

If precision is 100%, it means **False Positives = 0**.

So, the percentage of false positives is **0%**.

25

Recall is given as **25%**.

$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$

So the required value is directly the recall value, which is **25%**.

Q144. You are developing an application that will use Azure AI Search for internal documents. You need to implement document-level filtering for Azure AI Search.

Which three actions should you include in the solution?

- A. Add allowed groups to each index entry
- B. Create one index per group.
- C. Send access tokens from Microsoft Entra ID, with the search request.
- D. Retrieve all the groups.
- E. Retrieve the group memberships of the user
- F. Supply the groups as a filter for the search requests

Correct Answer:

- A. Add allowed groups to each index entry
- E. Retrieve the group memberships of the user
- F. Supply the groups as a filter for the search requests

Explanation:

Correct Answer:

A. Add allowed groups to each index entry

To enable document-level security, each document in the index must include metadata indicating which groups are allowed to access it. This allows filtering results based on user permissions.

E. Retrieve the group memberships of the user

The application must determine which groups the current user belongs to. This is typically done using Microsoft Entra ID so that access decisions can be enforced dynamically.

F. Supply the groups as a filter for the search requests

When performing a search, the user's group memberships are passed as a filter query. Azure AI Search uses this filter to return only documents that the user is authorized to access.

Incorrect answers:

B. Create one index per group.

This is not scalable and is not how document-level security is implemented in Azure AI Search.

C. Send access tokens from Microsoft Entra ID, with the search request.

While authentication is required, tokens alone do not enforce document-level filtering within the index.

D. Retrieve all the groups.

Retrieving all groups is unnecessary; only the groups relevant to the current user are required.

Q145. You have a web app that uses Azure AI Search.

When reviewing activity you see greater than expected search query volumes. You suspect that the query key is compromised.

You need to prevent unauthorized access to the search endpoint and ensure that users only have read only access to the documents collection. The solution must minimize app downtime.

Which three actions should you perform in sequence?

Actions

Regenerate the primary admin key

Regenerate the secondary admin key

Change the app to use the secondary admin key

Add a new query key

Change the app to use the new key

Delete the compromised key

Answer Area

1

2

3

Answer:

Actions

Regenerate the primary admin key

Regenerate the secondary admin key

Change the app to use the secondary admin key

Add a new query key

Change the app to use the new key

Delete the compromised key

Answer Area

1

2

3

Add a new query key

Change the app to use the new key

Delete the compromised key

Explanation:

Correct Answer:

Add a new query key

Change the app to use the new key

Delete the compromised key

Incorrect answers:

Regenerate the primary admin key

Regenerate the secondary admin key

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Change the app to use the secondary admin key

Using admin keys violates the requirement of read-only access and increases security risk.

Q146. You have an Azure AI Search indexer that ingest PDF policy manuals.

Client applications must display page-level citations that have bounding polygons for both text and images.

You need to add a single built-in multimodal content extraction skill to the Azure AI Search skillset.

The solution must meet the following requirements:

- Provide text and image location metadata.
- Extract tables that span multiple pages.

What should you add?

- A. Document Layout
- B. Document Extraction
- C. Azure Content Understanding
- D. GenAI Prompt

Correct Answer:

C. Azure Content Understanding

Explanation:**Correct Answer:****C. Azure Content Understanding**

Azure Content Understanding is the only built-in multimodal skill that fully satisfies all the requirements in the scenario. It provides **bounding polygon metadata for both text and images**, enabling precise page-level citations. Additionally, it supports **cross-page table extraction**, reconstructing tables that span multiple pages into a single coherent structure. This makes it ideal for processing complex PDF documents such as policy manuals while preserving layout and semantic structure.

Incorrect answers:**A. Document Layout**

Provides text layout and bounding polygons for text, but does not provide bounding polygons for images and does not correctly handle tables that span multiple pages.

B. Document Extraction

Performs basic extraction of content but does not provide detailed layout metadata such as bounding polygons or advanced table reconstruction.

D. GenAI Prompt

Used for generative AI tasks like summarization or transformation, not for extracting structured layout or positional metadata from documents.

Q147. You are designing a content management system.

You need to ensure that the reading experience is optimized for users who have reduced comprehension and learning differences, such as dyslexia. The solution must minimize development effort.

Which Azure service should you include in the solution?

- A. Azure Document Intelligence in Foundry Tools
- B. Azure Language in Foundry Tools
- C. Azure AI Immersive Reader
- D. Azure Translator in Foundry Tools

Correct Answer:

C. Azure AI Immersive Reader

Explanation:

Correct Answer:

C. Azure AI Immersive Reader

Azure AI Immersive Reader is specifically designed to improve reading comprehension and accessibility for users, including those with dyslexia and other learning differences. It provides features such as text-to-speech, syllable breakdown, line focus, and customizable text display. It is easy to integrate and requires minimal development effort, making it the best choice for optimizing the reading experience.

Incorrect answers:

A. Azure Document Intelligence in Foundry Tools

Used for extracting structured data from documents, not for improving reading accessibility.

B. Azure Language in Foundry Tools

Provides NLP capabilities like sentiment analysis and entity recognition, but does not enhance reading experience.

D. Azure Translator in Foundry Tools

Used for language translation, not for improving comprehension or accessibility for users with learning differences.

Q148. You are building a custom vision model that will be deployed as part of an iOS app. You have images of cats and dogs. Each image contains either a cat or a dog.

You need to use the Azure Custom Vision service to detect whether the image is of a cat or a dog.

How should you configure the project in the Azure Custom Vision portal?

Answer Area

Project Types:

▼

Classification
Object Detection

Classification Types:

▼

Multiclass (Single tag per image)
Multilabel (Multiple tags per image)

Domains:

▼

Adult
Food
General
General (compact)
Landmarks
Landmarks (compact)
Retail
Retail (compact)

Answer:

Answer Area

Project Types:

▼

Classification

Object Detection

Classification Types:

▼

Multiclass (Single tag per image)

Multilabel (Multiple tags per image)

Domains:

▼

Adult

Food

General

General (compact)

Landmarks

Landmarks (compact)

Retail

Retail (compact)

Explanation:

Correct Answer:

Project Type → Classification

This is correct because the requirement is to determine whether each image contains a **cat or a dog**. The model only needs to assign a category to the entire image, rather than locate the animals within the image, so classification is appropriate.

Classification Type → Multiclass (Single tag per image)

This is correct because each image contains **either a cat or a dog**, meaning only one label is applicable to each image. Multiclass classification is designed for mutually exclusive categories where each image receives a single tag.

Domain → General (compact)

This is correct because the model will be **deployed as part of an iOS app**. Compact domains are optimized for export and deployment to devices such as mobile applications, while still supporting image classification.

Incorrect Answers:

Object Detection

Object detection is used when you need to identify and locate objects within an image by using bounding boxes. Here, only the overall image category is required.

Multilabel (Multiple tags per image)

Multilabel is used when an image can belong to multiple categories simultaneously. Here, each image contains either a cat or a dog, so a single tag is sufficient.

General / Retail / Food / Landmarks domains

These domains are not the appropriate choice for an iOS-deployed cat-versus-dog classifier. **General (compact)** is the appropriate compact domain for this scenario.

Q149. You have a Microsoft Foundry project that contains an agent. The agent has a Model Context Protocol (MCP) tool named kbsearch that queries a knowledge base stored in Azure AI Search. Some agent runs return answers from the base model without invoking the knowledge base, which results in responses without grounded citations.

You are provided with the following code snippet that runs the agent.

```
1 run = project_client.agents.runs.create_and_process(
2     thread_id=thread.id,
3     agent_id=agent.id
4 )
```

You need to deterministically force the agent to invoke kbsearch on each run. What should you do?

- A. Add the response_format parameter to the create_and_process() method call.
- B. Replace create_and_process() method with the create_thread_and_process_run() method.
- C. Add the toolset parameter to the create_and_process() method call.
- D. Add the tool_choice parameter to the create_and_process() method call.

Correct Answer:

D. Add the tool_choice parameter to the create_and_process() method call.

Explanation:**Correct Answer:**

D. Add the tool_choice parameter to the create_and_process() method call.

This is correct because tool_choice controls which tool the agent must use during a run. By specifying the kbsearch MCP tool as the required tool, you can prevent the agent from answering directly from the base model and ensure that the knowledge base is invoked on every run, providing grounded responses and citations.

Incorrect Answers:

A. Add the response_format parameter to the create_and_process() method call.

This controls the structure or format of the model's response and does not force a specific tool invocation.

B. Replace create_and_process() with create_thread_and_process_run().

This changes how the thread and run are created but does not force the agent to invoke a particular tool.

C. Add the toolset parameter to the create_and_process() method call.

A toolset defines or provides the tools available to the agent, but it does not deterministically force the agent to invoke kbsearch.

Q150. You are developing a text processing solution. You have the following function.

```

1  def get_key_words(textAnalyticsClient, text):
2      response = textAnalyticsClient.recognize_entities (documents = [text])[0]
3      print("Key Words:")
4      for entity in response.entities:
5          print("\t\t", entity.text)
6  
```

You call the function and use the following string as the second argument. Our tour of London included a visit to Buckingham Palace

What will be the output of the function?

- A. London and Buckingham Palace only
- B. Tour and visit only
- C. Our tour of London included a visit to Buckingham Palace
- D. London and Tour only

Correct Answer:

A. London and Buckingham Palace only

Explanation:**Correct Answer:****A. London and Buckingham Palace only**

This is correct because recognize_entities() identifies named entities in the supplied text and returns them through response.entities. In this sentence, **London** is recognized as a location entity and **Buckingham Palace** as a location/landmark entity. The function then prints the recognized entity text, rather than printing every word in the sentence.

Incorrect Answers:

B. Tour and visit only

These are common words and are not recognized as named entities by the entity recognition service.

C. Our tour of London included a visit to Buckingham Palace

The function does not print the original input text. It only prints the text of entities identified by `recognize_entities()`.

D. London and Tour only

“Tour” is not a named entity in this context, while **Buckingham Palace** is recognized as an entity.

Q151. You are building a solution that students will use to find references for essays. You use the following code to start building the solution.

```

1  from azure.core.credentials import AzureKeyCredential
2  from azure.ai.textanalytics import TextAnalyticsClient
3
4  endpoint = os.environ["AZURE_TEXT_ANALYTICS_ENDPOINT"]
5  key = os.environ["AZURE_TEXT_ANALYTICS_KEY"]
6
7  text_analytics_client = TextAnalyticsClient(endpoint=endpoint,
8                                             credential=AzureKeyCredential(key))
9
10 documents = [
11     """
12     Our tour guide took us up the Space Needle during our trip to Seattle last week.
13     """
14 ]
15
16 result = text_analytics_client.recognize_linked_entities(documents)

```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Answer Area

Statements	Yes	No
The code will detect the language of documents.	☐	☐
The url attribute returned for each linked entity will be a Bing search link.	☐	☐
The matches attribute returned for each linked entity will provide the location in a document where the entity is referenced.	☐	☐

Answer:

Answer Area

Statements	Yes	No
The code will detect the language of documents.	☐	☒
The url attribute returned for each linked entity will be a Bing search link.	☐	☒
The matches attribute returned for each linked entity will provide the location in a document where the entity is referenced.	☒	☐

Statement 1 --> No

This is correct because the `recognize_linked_entities()` method is used to identify and link entities to a knowledge base. Although each linked entity has a language property, this represents the language associated with the linked entity's data source, not language detection of the input document. A separate language detection operation is required to detect the language of the documents.

Statement 2 --> No

This is correct because the `url` property contains a link to the entity's page in the associated knowledge base, such as Wikipedia. It is not necessarily a Bing search link. Microsoft documentation specifically describes the URL as pointing to the entity's page from the data source, with Wikipedia being a common source.

Statement 3 --> Yes

This is correct because the `matches` collection contains the occurrences of each linked entity in the document. Each match includes information such as the matched text, confidence score, offset, and length, allowing the location of the entity within the document to be determined.

Q152. You have a Microsoft Foundry project for a support ticket triage app that reads ticket text from a database.

You need to ensure that the app meets the following requirements:

- Compares candidate models by quality, cost, and throughput
- Prevents hosting model weights in the Azure subscription

What should you configure?

Answer Area

Model evaluation:

Configure private endpoint access.
Use deployment lists and license tabs.
Use tool catalog connections and run traces.
Use model catalog leaderboards and model cards.

Deployment option:

Bring your own model.
Build a vector index.
Use a serverless deployment.
Use a managed compute deployment.

Answer:

Answer Area

Model evaluation:

Configure private endpoint access.
Use deployment lists and license tabs.
Use tool catalog connections and run traces.
Use model catalog leaderboards and model cards.

Deployment option:

Bring your own model.
Build a vector index.
Use a serverless deployment.
Use a managed compute deployment.

Explanation:

Model evaluation → **Use model catalog leaderboards and model cards.**

Model catalog **leaderboards and model cards** provide information that helps compare candidate models based on factors such as **quality, cost, and throughput**. This allows you to select an appropriate model before deploying it.

Deployment option → **Use a serverless deployment.**

A **serverless deployment** provides access to the model through an API without hosting the model weights directly in your Azure subscription. The model is hosted and managed by the model provider, helping satisfy the requirement to prevent model weights from being hosted in your Azure environment.

Incorrect Answers:**Configure private endpoint access.**

This controls network access to resources; it does not provide model comparison information or determine where model weights are hosted.

Use deployment lists and license tabs.

These provide deployment and licensing information but are not the primary mechanism for comparing model quality, cost, and throughput.

Use tool catalog connections and run traces.

These relate to tools and application observability, not model evaluation or serverless model hosting.

Bring your own model.

This involves supplying and hosting your own model, which does not meet the requirement to avoid hosting model weights in the Azure subscription.

Build a vector index.

A vector index is used for semantic/vector search and retrieval, not model evaluation or model deployment.

Use a managed compute deployment.

Managed compute deploys the model to compute resources in your Azure environment, so it does not meet the requirement to prevent model weights from being hosted in the Azure subscription.

Q153. You have an Azure subscription.

You need to build an app that will compare documents for semantic similarity. The solution must meet the following requirements:

- Return numeric vectors that represent the tokens of each document.
- Minimize development effort.

Which Azure OpenAI model should you use?

- A. GPT-3.5
- B. embeddings
- C. GPT-4
- D. DALL-E

Correct Answer:**B. embeddings****Explanation:****Correct Answer:****B. embeddings**

Azure OpenAI **embedding models** convert text into **numeric vectors** that capture the semantic meaning of the text. These vectors can then be compared using similarity measures to determine how semantically similar two documents are. Using the built-in embeddings capability minimizes development effort because you do not need to build your own language representation model.

Incorrect Answers:**A. GPT-3.5**

GPT-3.5 is a generative language model designed primarily for generating and understanding text, not directly producing embedding vectors for semantic similarity.

C. GPT-4

GPT-4 is a generative language model and is not the appropriate model for directly generating document embeddings.

D. DALL-E

DALL-E is designed for image generation and is not intended for converting document text into semantic embedding vectors.

Q154. You are building an app that will provide users with definitions of common AI terms. You create the following Python code.

```
...
openai.api_key = key
openai.api_base = endpoint
response = openai.ChatCompletion.create(
    engine=deployment_name,
    messages=[
        { "role": "system", "content": "You are a helpful assistant." },
        { "role": "user", "content": "What is an LLM?" }
    ]
)

print(response['choices'][0]['message']['content'])
...
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Answer Area**Statements****Yes****No**

The response will contain an explanation of large language models (LLMs) that has a high degree of certainty.

☐☐

Changing "what is an LLM?" to "What is an LLM in the context of AI models?" will produce the intended response.

☐☐

Changing "You are a helpful assistant." to "You must answer only within the context of AI language models." will give a higher likelihood of producing the intended response.

☐☐

Answer:**Answer Area****Statements****Yes****No**

The response will contain an explanation of large language models (LLMs) that has a high degree of certainty.

☐☒

Changing "what is an LLM?" to "What is an LLM in the context of AI models?" will produce the intended response.

☐☒

Changing "You are a helpful assistant." to "You must answer only within the context of AI language models." will give a higher likelihood of producing the intended response.

☒☐**Explanation:****Statement 1 --> No**

The prompt **"What is an LLM?"** is ambiguous because the abbreviation "LLM" can have different meanings. The code does not provide enough contextual instruction to guarantee that the response will explain **Large Language Models** with a high degree of certainty.

Statement 2 --> No

Changing the prompt to **"What is an LLM in the context of AI models?"** adds context and may make the intended meaning clearer. However, the wording **"will produce the intended response"** implies a guarantee. Prompt changes can increase the likelihood of a desired response but cannot guarantee a specific output.

Statement 3 --> Yes

Changing the system message to **"You must answer only within the context of AI language models."** provides a stronger and more specific instruction about the intended domain. System-level instructions have a higher priority than user messages, so this change increases the likelihood that the model will interpret LLM as **Large Language Model** and provide the intended response.

Q155. You have an Azure OpenAI model named AI1.

You are building a web app named App1 by using the Azure OpenAI SDK. You need to configure App1 to connect to AI1.

What information must you provide?

- A. the deployment name, key, and model name
- B. the endpoint, key, and model type
- C. the deployment name, endpoint, and key
- D. the endpoint, key, and model name



Correct Answer:**C. the deployment name, endpoint, and key****C. the deployment name, endpoint, and key**

To connect an application to an Azure OpenAI model, the SDK needs three key pieces of information. The **endpoint** identifies the Azure OpenAI resource that the application should connect to, the **key** authenticates the application to that resource, and the **deployment name** identifies the specific model deployment that the application wants to use. In Azure OpenAI, applications generally call a **deployment**, rather than directly specifying the underlying model name in the API request.

Incorrect Answers:**A. the deployment name, key, and model name**

This is missing the **endpoint**, which tells the SDK where the Azure OpenAI resource is located. The model name is also not a substitute for the deployment name when making a request.

B. the endpoint, key, and model type

Although the **endpoint** and **key** are required, the SDK needs the **deployment name** to identify which deployed model should process the request. The model type alone does not identify the specific deployment.

D. the endpoint, key, and model name

The **model name** describes the underlying model, such as a particular GPT model, but Azure OpenAI applications target the **deployment name** when sending requests. Therefore, the deployment name is required instead of the model name.

Q156. You are building a solution in Azure that will use Azure Language in Foundry Tools service to process sensitive customer data.

You need to ensure that only specific Azure processes can access the Language service. The solution must minimize administrative effort.

What should you include in the solution?

- A. Azure Application Gateway
- B. a virtual network gateway
- C. IPsec rules
- D. virtual network rules

Correct Answer:**D. virtual network rules****Explanation:****Correct Answer:**

D. virtual network rules

Virtual network rules allow you to restrict access to an Azure AI Language resource so that only traffic originating from specified **virtual networks and subnets** can reach the service. This provides network-level isolation for sensitive customer data without requiring you to manage individual IP addresses or complex network infrastructure. It also minimizes administrative effort because access can be controlled by defining which Azure virtual networks are trusted.

Incorrect Answers:**A. Azure Application Gateway**

Azure Application Gateway is primarily used as a **Layer 7 web traffic load balancer and reverse proxy**. It can provide features such as URL-based routing and Web Application Firewall protection, but it is not the appropriate mechanism for restricting which Azure virtual networks can access the Language service.

B. a virtual network gateway

A virtual network gateway is used to connect an Azure virtual network to other networks, such as an **on-premises network through VPN or ExpressRoute**. It does not by itself restrict access to an Azure AI Language resource.

C. IPsec rules

IPsec is used to provide **encrypted and authenticated network communication**, commonly for VPN connections. It is not the Azure AI Language resource access-control mechanism described in the question.

Q157. You have an Azure subscription that contains a Microsoft Foundry Service resource named CSAccount1 and a virtual network named VNet1. CSAccount1 is connected to VNet1.

You need to ensure that only specific resources can access CSAccount1. The solution must meet the following requirements:

- Prevent external access to CSAccount1.
- Minimize administrative effort.

Which two actions should you perform?

- A. In CSAccount1, modify the virtual network settings.
- B. In VNet1, create a virtual subnet.
- C. In VNet1, enable a service endpoint for CSAccount1.
- D. In CSAccount1, configure the Access control (IAM) settings.
- E. In VNet1, modify the virtual network settings.

Correct Answers:

A. In CSAccount1, modify the virtual network settings.

C. In VNet1, enable a service endpoint for CSAccount1.

Explanation:

Correct Answer:**A. In CSAccount1, modify the virtual network settings.**

The Microsoft Foundry Service resource can be configured to **restrict network access** so that requests are allowed only from selected virtual networks. By changing the network settings of **CSAccount1**, you can prevent access from the public/external network and allow access through the approved virtual network. This is the resource-side configuration needed to enforce the restriction.

C. In VNet1, enable a service endpoint for CSAccount1.

A **service endpoint** extends the identity of a virtual network to an Azure service and allows the service to recognize traffic coming from an authorized subnet. Enabling the endpoint for CSAccount1 on VNet1 allows resources in the appropriate subnet to access the service while the resource's network rules restrict access from outside the approved network. This provides the required network isolation without the additional management overhead of more complex networking solutions.

Incorrect Answers:**B. In VNet1, create a virtual subnet.**

A subnet is required for many Azure networking configurations, but simply creating a subnet does not restrict access to CSAccount1. The subnet must be associated with the appropriate service endpoint and the service itself must be configured to allow that network.

D. In CSAccount1, configure the Access control (IAM) settings.

IAM controls **who has permission to perform actions on the Azure resource** through Azure role-based access control. It does not prevent external network access to the service. Network access restrictions should instead be configured through the resource's networking settings.

E. In VNet1, modify the virtual network settings.

Changing general VNet settings does not by itself restrict access to CSAccount1. The required configuration is specifically the **service endpoint** on the VNet/subnet and the corresponding network restriction on CSAccount1.

Q158. You have a collection of press releases stored as PDF files.

You need to extract text from the files and perform sentiment analysis.

Which service should you use for each task?

Answer Area

Extract text:

Azure AI Search
Azure Vision in Foundry Tools
Azure Document Intelligence in Foundry Tools

Perform sentiment analysis:

Azure AI Search
Azure AI Computer Vision
Azure Document Intelligence in Foundry Tools
Azure Language in Foundry Tools

Answer:

Answer Area

Extract text:

Azure AI Search
Azure Vision in Foundry Tools
Azure Document Intelligence in Foundry Tools

Perform sentiment analysis:

Azure AI Search
Azure AI Computer Vision
Azure Document Intelligence in Foundry Tools
Azure Language in Foundry Tools

Explanation:

Correct Answer:

Extract text → Azure Document Intelligence in Foundry Tools

Azure Document Intelligence is designed to **extract text and structured information from documents**, including PDF files. It can process the contents of the press releases and make the extracted text available for further analysis.

Perform sentiment analysis → Azure Language in Foundry Tools

Azure Language provides **sentiment analysis** capabilities that determine whether text expresses positive, negative, neutral, or mixed sentiment. Therefore, once the text has been extracted from the PDFs, Azure Language is the appropriate service for analyzing its sentiment.

Incorrect Answers:

Azure AI Search

Azure AI Search is primarily used to **index and search content**, including extracted document text. It is not the appropriate service for performing sentiment analysis.

Azure Vision in Foundry Tools / Azure AI Computer Vision

Vision services are intended for analyzing **images and visual content**, such as detecting objects, describing images, or extracting text from images. They are not the appropriate choice for sentiment analysis of press-release text.

Q159. You are designing an Azure AI solution to identify defective products on a production line. You have a real-time video feed and an image library of sample products that are approved or rejected manually. You need to recommend a service that meets the following requirements:

- Monitors the video feed and identifies the defective products.
- Can train a new model by using the image library.
- Minimizes development effort.

What should you recommend?

- A. Azure Vision in Foundry Tools
- B. Azure AI Video Indexer
- C. Azure AI Custom Vision
- D. Azure Machine Learning

Correct Answer:

C. Azure AI Custom Vision

Explanation:

Correct Answer:

C. Azure AI Custom Vision

Azure AI Custom Vision is designed to create **custom computer vision models using your own labeled images**. You can use the manually labeled library of approved and rejected products to train an image classification or object detection model that distinguishes defective products from acceptable ones. The trained model can then be integrated into an application that processes frames from the **real-time video**

feed, allowing the application to identify defective products while minimizing development effort because Custom Vision provides the training and prediction infrastructure.

Incorrect Answers:

A. Azure Vision in Foundry Tools

Azure Vision provides prebuilt capabilities for analyzing images, such as image descriptions, object detection, and OCR. However, it does not provide the same straightforward **custom model training from your labeled product image library** that Custom Vision provides.

B. Azure AI Video Indexer

Video Indexer is designed to **analyze and extract insights from video**, such as people, objects, scenes, speech, and other video content. It is not intended for training a custom defect-detection model from a manually labeled image library.

D. Azure Machine Learning

Azure Machine Learning can be used to build and train custom computer vision models, but doing so generally requires **more development, data-science, and model-management effort**. Since the requirement explicitly says to minimize development effort, Custom Vision is the better choice.

Q160. You run the following command.

```
C:\> docker run --rm -it -p 5000:5000 --memory 10g --cpus 2 \
> mcr.microsoft.com/azure-cognitive-services/textanalytics/sentiment \
> Eula=accept \
> Billing={ENDPOINT_URI} \
> Apikey={API_KEY}
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Answer Area

Statements	Yes	No
Going to <code>http://localhost:5000/status</code> will query the Azure endpoint to verify whether the API key used to start the container is valid.	☐	☐
The container logging provider will write log data.	☐	☐
Going to <code>http://localhost:5000/swagger</code> will provide the details to access the documentation for the available endpoints.	☐	☐

Answer:**Answer Area**

Statements	Yes	No
Going to <code>http://localhost:5000/status</code> will query the Azure endpoint to verify whether the API key used to start the container is valid.	☐	☒
The container logging provider will write log data.	☐	☒
Going to <code>http://localhost:5000/swagger</code> will provide the details to access the documentation for the available endpoints.	☒	☐

Explanation:**Statement 1 --> No**

The `/status` endpoint is used to verify whether the **API key used to start the container is valid**. However, it performs this verification **without causing an endpoint query to Azure**. Therefore, the statement is incorrect because it specifically says that `/status` will query the Azure endpoint.

Statement 2 --> No

The container logging provider does not automatically write persistent log data simply because the container is started. Logging providers such as **Disk** or **Fluentd** must be explicitly configured. The command shown only provides the EULA, billing endpoint, and API key, so no additional logging provider has been configured.

Statement 3 --> Yes

The `/swagger` endpoint provides the **Swagger/OpenAPI documentation** for the endpoints exposed by the container. It can be used to view the available APIs and the information required to interact with them.

Q161. You are developing an app that will use the Speech and Language APIs.

You need to provision resources for the app. The solution must ensure that each service is accessed by using a single endpoint and credential.

Which type of resource should you create?

- A. Azure Language in Foundry Tools
- B. Microsoft Foundry service
- C. Azure Speech in Foundry Tools
- D. Content Safety in Foundry Control Plane

Correct Answer:

B. Microsoft Foundry service**Explanation:****Correct Answer:****B. Microsoft Foundry service**

A **Microsoft Foundry service** is the appropriate choice when an application needs to use multiple Foundry Tools, such as **Azure Speech and Azure Language**, through a shared resource. A multi-service resource provides a **single endpoint and credential** that can be used to access multiple supported AI services, reducing the need to provision and manage separate resources and credentials. Microsoft documentation confirms that Azure AI multi-service resources support access to multiple Foundry Tools through a single endpoint and API key.

Incorrect Answers:**A. Azure Language in Foundry Tools**

This is a **single-service resource** intended for Language capabilities. It would not provide the required shared resource for both Speech and Language.

C. Azure Speech in Foundry Tools

Similarly, an Azure Speech resource is focused on Speech capabilities. It does not provide the single shared endpoint and credential for both Speech and Language that the question requires.

D. Content Safety in Foundry Control Plane

Content Safety is used to detect and mitigate potentially harmful content. It is unrelated to provisioning a shared resource for the **Speech and Language APIs**.

Q162. You are building an app that will include one million scanned magazine articles. Each article will be stored as an image file.

You need to configure the app to extract text from the images. The solution must minimize development effort.

What should you include in the solution?

- A. Azure Document Intelligence in Foundry Tools
- B. the Read API in Azure Vision in Foundry Tools
- C. Azure Vision in Foundry Tools Image Analysis
- D. Azure Language in Foundry Tools

Correct Answer:**B. the Read API in Azure Vision in Foundry Tools****Explanation:****Correct Answer:****B. the Read API in Azure Vision in Foundry Tools**

The **Read API** is specifically designed to perform **OCR (optical character recognition)** and extract printed or handwritten text from images. Since the magazine articles are stored as image files, the Read API can process the images and return the recognized text. It is also a prebuilt capability, so it requires **less development effort** than building a custom text-extraction solution.

Incorrect Answers:

A. Azure Document Intelligence in Foundry Tools

Document Intelligence is primarily designed for extracting **text, fields, tables, and structured information from documents** such as forms, invoices, receipts, and other document formats. For a large collection of article images where the primary requirement is simply OCR text extraction, the Read API is the more direct choice.

C. Azure Vision in Foundry Tools Image Analysis

Image Analysis can perform several image-understanding tasks, including OCR capabilities, but the **Read API is specifically optimized for text extraction**. Therefore, when the requirement is simply to extract text from a very large collection of scanned images, Read is the more appropriate service.

D. Azure Language in Foundry Tools

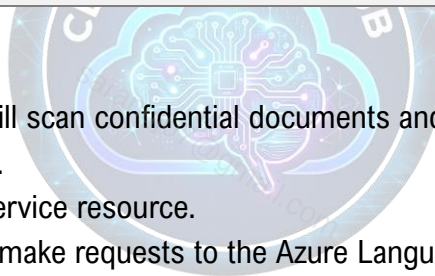
Azure Language is designed for **analyzing text that has already been extracted**, with capabilities such as sentiment analysis, entity recognition, and summarization. It does not perform the initial OCR extraction from image files.

Q163. You are building an app that will scan confidential documents and use the Azure Language in Foundry Tools service to analyze the contents.

You provision a Microsoft Foundry Service resource.

You need to ensure that the app can make requests to the Azure Language in Foundry Tools service endpoint. The solution must ensure that confidential documents remain on-premises.

Which three actions should you perform in sequence?



Sarang Joshi

Actions

Pull an image from Docker Hub.

Run the container and specify an API key and the Endpoint URL of the Azure AI resource.

Provision an on-premises Kubernetes cluster that is isolated from the internet.

Provision an on-premises Kubernetes cluster that has internet connectivity.

Provision an Azure Kubernetes Service (AKS) resource.

Run the container and specify an App ID and Client Secret.

Pull an image from the Microsoft Container Registry (MCR).

Answer Area

1

2

3

Answer:**Actions**

Pull an image from Docker Hub.

Run the container and specify an API key and the Endpoint URL of the Azure AI resource.

Provision an on-premises Kubernetes cluster that is isolated from the internet.

Provision an on-premises Kubernetes cluster that has internet connectivity.

Provision an Azure Kubernetes Service (AKS) resource.

Run the container and specify an App ID and Client Secret.

Pull an image from the Microsoft Container Registry (MCR).

Answer Area

1

Provision an on-premises Kubernetes cluster that has internet connectivity.

2

Pull an image from the Microsoft Container Registry (MCR).

3

Run the container and specify an API key and the Endpoint URL of the Azure AI resource.

Explanation:**1. Provision an on-premises Kubernetes cluster that has internet connectivity.**

The confidential documents must remain **on-premises**, so the Azure AI Language container should run in an on-premises environment rather than in Azure. The Kubernetes cluster needs **internet connectivity** because the container must communicate with Azure for required billing, authorization, and metering operations. The document processing itself can occur within the local container.

2. Pull an image from the Microsoft Container Registry (MCR).

Azure AI Language container images are distributed through the **Microsoft Container Registry (MCR)**. After provisioning the on-premises Kubernetes environment, pull the required Language container image from MCR.

3. Run the container and specify an API key and the Endpoint URL of the Azure AI resource.

The container is configured with the **API key** and **Endpoint URL** of the provisioned Microsoft Foundry Service resource. The application can then send requests to the locally deployed container, allowing the confidential documents to be processed on-premises rather than sending the document contents directly to the Azure Language service.

Incorrect Answers:**Pull an image from Docker Hub.**

Azure AI Language container images are provided through **MCR**, not Docker Hub.

Provision an on-premises Kubernetes cluster that is isolated from the internet.

The container requires outbound connectivity to Azure for required service operations, so completely isolating the cluster from the internet would prevent the required communication.

Provision an Azure Kubernetes Service (AKS) resource.

AKS runs the container workload in Azure. The requirement specifically states that the confidential documents must remain **on-premises**.

Run the container and specify an App ID and Client Secret.

The required container configuration uses the **API key and Endpoint URL** of the Azure AI resource.

Q164. You are developing an app that will use the Azure Vision in Foundry Tools API to analyze an image. You need configure the request that will be used by the app to identify whether an image is clipart or a line drawing.

How should you complete the request?

Answer Area

GET
PATCH
POST

"https://*.cognitiveservices.azure.com/vision/v3.2/analyze?visualFeatures=

description
imageType
objects
tags

&details={string}&language=e

Answer:

Answer Area

GET
PATCH
POST

"https://*.cognitiveservices.azure.com/vision/v3.2/analyze?visualFeatures=

description
imageType
objects
tags

&details={string}&language=e

Explanation:**Correct Answer:****HTTP method → POST**

The Azure Vision **Analyze Image v3.2 API** uses the **POST** method because the image to be analyzed is provided in the request body, either as binary image data or as a JSON payload containing an image URL. Therefore, POST is required for the /vision/v3.2/analyze operation.

visualFeatures → imageType

The **imageType** visual feature is specifically used to determine the type of image. Its results include information about whether the image is **clip art** and whether it is a **line drawing**, which directly satisfies the requirement.

The completed request is:

POST

https://*.cognitiveservices.azure.com/vision/v3.2/analyze?visualFeatures=imageType&details={string}&language=en

Incorrect Answers:**GET**

GET is used to retrieve resources without supplying an image for analysis in the request body. The Analyze Image v3.2 operation requires **POST**.

PATCH

PATCH is used to partially modify an existing resource. It is not the HTTP method used for image analysis.

description

The description feature generates a natural-language description of the image. It does not specifically identify whether the image is clip art or a line drawing.

objects

The objects feature detects objects within an image and their locations. It does not provide the required clip-art or line-drawing classification.

tags

The tags feature generates keywords describing image content. It does not specifically determine the image type.

Q165. You use the Azure Custom Vision service to build a classifier. After training is complete, you need to evaluate the classifier.

Which two metrics are available for review?

- A. F-score
- B. area under the curve (AUC)
- C. precision
- D. weighted accuracy
- E. recall

Correct Answers:

C. precision

E. recall

Explanation:

Correct Answers:

C. precision

Precision measures how many of the images that the classifier **predicted as belonging to a particular class were actually correct**. For example, if the model identifies 100 products as defective and 90 are actually defective, the precision is 90%. A higher precision means the model produces fewer false-positive predictions.

E. recall

Recall measures how many of the **actual examples of a class were correctly identified** by the classifier. For example, if there are 100 genuinely defective products and the model correctly identifies 80 of them, the recall is 80%. A higher recall means the model produces fewer false negatives.

Incorrect Answers:

A. F-score

F-score is a combined metric derived from **precision and recall**, but it is not one of the primary metrics provided for reviewing a Custom Vision classifier in this context.

B. area under the curve (AUC)

AUC is commonly used to evaluate classification models by measuring performance across different classification thresholds, but it is not the metric being tested as an available Custom Vision classifier evaluation metric here.

D. weighted accuracy

Weighted accuracy is not one of the standard evaluation metrics displayed for an Azure Custom Vision classifier.



Sarang Joshi

Case Study

Overview

Company Information

Contoso, Ltd is a multinational retail company that builds, deploys, and manages generative AI and agent-based solutions by using Microsoft Foundry.

Existing Environment

Identity Environment

- Contoso uses Microsoft Entra ID for identity management, authentication, and authorization capabilities that enable agents to access organizational resources and services.
- Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.
- The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement, monitor, and secure AI applications.
- Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution deployments.

Generative Environment

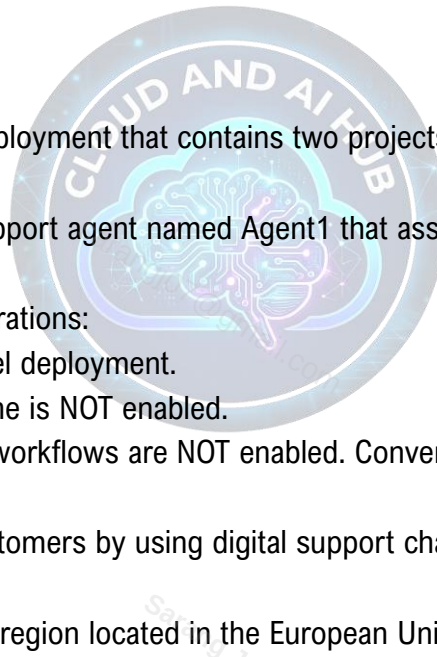
Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.

Project1

- Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.
- Agent1 has the following configurations:
 - Agent1 uses a base model deployment.
 - A safety evaluation pipeline is NOT enabled.
 - Tool invocation approval workflows are NOT enabled. Conversation memory constraints are NOT configured.
 - Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.
- Project1 is deployed to an Azure region located in the European Union (EU).
- Agent1Dev Team will use Project1 to optimize and maintain Agent1.

Project2

- Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.
- Development of the solution is incomplete.



Data Environment

- Contoso stores product-related information in Azure resources that support AI applications.
- The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.
- The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.

Problem Statements

Contoso identifies the following issues:

- Agent1 has only general knowledge of the Contoso products.
- A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.
- Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.
- The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.

Requirements

Planned Changes

Contoso plans to implement the following changes:

- Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.
- Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.
- Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1. Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.
- Complete the development of the video creation solution.

Technical Requirements

Contoso identifies the following technical requirements:

- The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity. The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.
- Responses generated by using the product sheet information must be relevant, complete, and accurate. Agent1 must be able to use the product sheets to answer natural language questions about product details. The model version used by Agent1 must remain consistent to ensure stable responses.
- The data processed by the model must remain within the EU.

Security and Compliance Requirements

Contoso identifies the following security and compliance requirements:

- API keys must NOT be used to access Foundry-deployed models.
- Access to the Azure resources must follow the principle of least privilege.
- The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.
- Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.
- Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.
- Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.
- The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.

Business Requirements

Contoso identifies the following business requirements:

- Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.
- Agent1 must answer questions only about the products sold by Contoso.

Q166. You need to configure the model deployment for Agent1 to meet the technical requirements. What should you configure?

Deployment type:

▼

Standard

Global Standard

Global Provisioned

Version update policy:

▼

Once the current version expires

Opt out of automatic model version upgrades

Upgrade once a new default version becomes available

Answer:

Deployment type: ▼

- Standard**
- Global Standard
- Global Provisioned

Version update policy: ▼

- Once the current version expires
- Opt out of automatic model version upgrades**
- Upgrade once a new default version becomes available

Explanation**Correct Answers:****Deployment type: Standard**

The technical requirements specify **Scalable, high-throughput generative AI workloads, Dynamic scaling** for variable customer support traffic. **No reserved throughput capacity** and **Data must remain within the EU**.

A **Standard** deployment is the correct choice because it provides **serverless, pay-as-you-go inference**, automatically scales with demand, and is deployed within the selected Azure region (EU in this case). It does **not** require provisioning throughput in advance.

Global Standard and **Global Provisioned** can route requests outside the deployment region, which conflicts with the requirement that **data processed by the model must remain within the EU**.

Version update policy: Opt out of automatic model version upgrades

The requirements state that the model version must be **standardized**, the model version must **remain consistent** and responses must be **stable and predictable**.

Opting out of automatic model version upgrades ensures that Microsoft does not automatically switch the deployment to a newer model version, preserving consistent behavior until Contoso explicitly chooses to upgrade.

Incorrect Answers:**Global Standard**

Although it provides dynamic scaling, it may process requests outside the EU, violating the data residency requirement.

Global Provisioned

Requires reserved throughput capacity and may process requests globally, violating both the scalability and EU data residency requirements.

Once the current version expires

This allows automatic version changes when the deployed version reaches end of life, which does not guarantee long-term consistency.

Upgrade once a new default version becomes available

This automatically upgrades the deployment whenever Microsoft publishes a new default version, potentially changing model behavior and violating the requirement for consistent responses.

Q167. You need to configure Agent1 to meet the security and compliance requirements.

What should you use?

- A. self-harm content filtering
- B. prompt shields
- C. Personally identifiable information (PII) Detection
- D. violence content filtering

Answer:

B. Prompt shields

Explanation

Correct Answer:

B. Prompt shields

One of the security and compliance requirements states that **the product sheets might contain images with embedded text**, and **Agent1 must be protected from malicious instructions hidden within those images**.

This is a classic **indirect prompt injection** scenario. When images are processed through OCR, hidden instructions can become part of the prompt sent to the language model. **Prompt Shields for documents** are specifically designed to detect and mitigate these embedded instructions before they influence the model. Therefore, **Prompt Shields** are the correct solution.

Incorrect Answers:

A. Self-harm content filtering

This filters self-harm-related content and does not protect against hidden prompt injection attacks.

C. Personally Identifiable Information (PII) Detection

PII Detection identifies and redacts sensitive personal information, but it does not prevent malicious instructions embedded in documents or images from influencing the model.

D. Violence content filtering

This filters violent content and is unrelated to protecting the model from prompt injection attacks.

Q168. You need to recommend a solution to assess the responses generated by Agent1 when the agent uses the product information stored in storage1. The solution must meet the technical requirements.

What should you include in the recommendation?

- A. a Retrieval Augmented Generation (RAG) evaluator
- B. a custom guardrail
- C. model fine-tuning
- D. a groundedness evaluator

Answer:**A. A Retrieval Augmented Generation (RAG) evaluator****Explanation****Correct Answer:****A. A Retrieval Augmented Generation (RAG) evaluator**

The scenario requires assessing the quality of **Agent1's responses after it retrieves information from the product sheets stored in storage1**.

The technical requirements specify that the responses must be **relevant, complete and accurate**

A **RAG evaluator** is designed specifically to evaluate the overall performance of a Retrieval-Augmented Generation pipeline. It assesses both the **retrieval** and **generation** stages using metrics such as relevance, groundedness, completeness, and other quality measures.

Since Contoso is implementing a RAG solution over indexed product sheets and wants to assess the responses against the technical requirements, a **RAG evaluator** is the most appropriate recommendation.

Incorrect Answers:**B. A custom guardrail**

Guardrails enforce safety and policy constraints, such as blocking unsafe outputs or restricting topics. They do not evaluate the quality of RAG-generated responses.

C. Model fine-tuning

Fine-tuning changes the behavior of the model but does not provide a mechanism to evaluate response quality.

D. A groundedness evaluator

A groundedness evaluator measures whether a response is supported by the retrieved documents. While groundedness is an important metric within a RAG system, it evaluates only one aspect of response quality. The requirements also call for responses to be **relevant** and **complete**, making a **RAG evaluator** the better choice because it evaluates the overall RAG pipeline.

Q169. You need to configure Agent1 to answer customer questions about only the Contoso products. The solution must meet the business requirements.

What should you do?

- A. Modify the system message instructions.
- B. Add few-shot examples.
- C. Apply top-p sampling.
- D. Increase the value of the temperature parameter.

Answer:**A. Modify the system message instructions.****Explanation****Correct Answer:****A. Modify the system message instructions.**

One of the business requirements states that **Agent1 must answer questions only about the products sold by Contoso.**

The **system message** defines the agent's behavior, scope, and constraints. By updating the system instructions, you can explicitly instruct Agent1 to answer only questions related to Contoso products, use the product information retrieved from the indexed product sheets and politely decline questions that are outside the scope of Contoso products.

This is the recommended approach for constraining an agent's behavior and ensuring it remains focused on a specific domain.

Incorrect Answers:

B. Add few-shot examples.

Few-shot examples help guide the model by providing sample interactions, but they do not enforce strict behavioral boundaries. The model may still answer questions outside the intended scope.

C. Apply top-p sampling.

Top-p sampling controls the diversity of generated text. It has no effect on restricting the topics the agent is allowed to answer.

D. Increase the value of the temperature parameter.

Increasing the temperature makes responses more creative and less deterministic. It does not limit the agent to answering only questions about Contoso products.

Q170. You need to recommend an invoice review solution that resolves the issue reported by the finance department.

What should you include in the recommendation?

- A. chat completions
- B. Azure Document Intelligence in Foundry Tools
- C. Azure Content Understanding in Foundry Tools
- D. Image Analysis

Answer:

C. Azure Content Understanding in Foundry Tools

Explanation

Correct Answer:

C. Azure Content Understanding in Foundry Tools

The finance department needs a solution that can analyze **vendor invoices with varied layouts**, evaluate both the **visual layout** and the **textual content**, process **tables, logos**, and other document elements and verify invoice details against vendor contract terms.

Azure Content Understanding in Foundry Tools is specifically designed for this type of intelligent document processing. It combines OCR, layout analysis, and AI-powered field extraction to understand complex business documents across varying templates. It can extract structured information from invoices while preserving the document structure, making it ideal for downstream verification against vendor contracts.

Incorrect Answers:**A. Chat completions**

Chat completions generate natural language responses but do not perform document analysis, OCR, or structured extraction from invoices.

B. Azure Document Intelligence in Foundry Tools

Azure Document Intelligence performs OCR and document analysis. However, **Azure Content Understanding** is the newer Foundry capability that builds on these capabilities and is specifically intended for extracting structured business information across varying document layouts with minimal custom development. It is the recommended choice in Microsoft Foundry.

D. Image Analysis

Image Analysis can detect objects, text, and visual features within images but is not designed to extract structured business fields from complex invoices or analyze document layouts for invoice processing.

Q171. You need to configure an indexing pipeline for Agent1 to retrieve the relevant product information in storage1. The solution must meet the technical requirement.

Which two built-in skills should you use?

- A. Azure OpenAI Embedding
- B. Entity Recognition
- C. Text Split
- D. Merge
- E. Language Detection
- F. key phrase extraction

**Answer:****A. Azure OpenAI Embedding****C. Text Split****Explanation****Correct Answers:****A. Azure OpenAI Embedding**

The technical requirements specify that the indexing pipeline must support **semantic search and vector search**. The **Azure OpenAI Embedding** skill generates vector embeddings for the product sheet content. These embeddings are stored in the Azure AI Search index and enable semantic similarity matching, allowing Agent1 to retrieve the most relevant product information for natural language queries.

C. Text Split

The product sheets are stored as **PDF documents**, which can be lengthy. The **Text Split** skill divides the extracted text into smaller chunks before embeddings are generated. Chunking improves retrieval quality because embeddings are created for manageable sections of text rather than entire documents, resulting in more accurate semantic and vector search.

Incorrect Answers:

B. Entity Recognition

Entity Recognition extracts named entities such as people, organizations, and locations. It does not support semantic or vector search.

D. Merge

The Merge skill combines text from multiple sources, such as OCR results and extracted document text. It is not required to build a semantic and vector search pipeline.

E. Language Detection

Language Detection identifies the language of the document but does not contribute to semantic indexing or vector search.

F. Key Phrase Extraction

Key Phrase Extraction identifies important phrases within documents. While useful for text analytics, it is not required for implementing semantic and vector search.

Q172. You need to recommend a solution to support the planned changes and technical requirements for Agent1 to use the product information stored in storage1.

What should you include in the recommendation?

- A. Azure Translator in Foundry Tools
- B. Grounding with Bing Search
- C. Azure AI Search
- D. Azure Document intelligence in Foundry Tools

Answer:

C. Azure AI Search

Explanation

Correct Answer:

C. Azure AI Search

The planned changes require Agent1 to retrieve product information stored in **Azure Blob Storage (storage1)**, answer **natural language questions** about Contoso products, use an **indexing pipeline** that supports **semantic search and vector search** and generate **relevant, complete, and accurate** responses. **Azure AI Search** is the core service used to implement a Retrieval-Augmented Generation (RAG) solution. It indexes the product sheets, supports both **semantic search** and **vector search**, and retrieves the most relevant content for Agent1 to ground its responses.

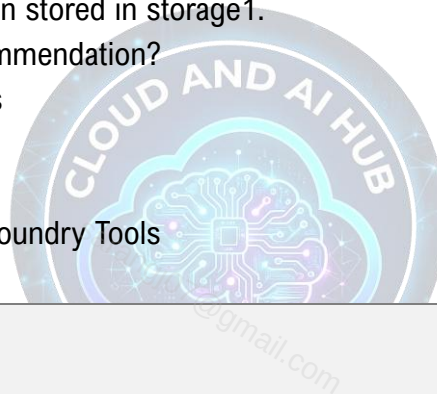
This directly satisfies both the planned changes and the technical requirements.

Incorrect Answers:

A. Azure Translator in Foundry Tools

Azure Translator is used for language translation. The scenario does not require translating the product sheets or customer questions.

B. Grounding with Bing Search



Grounding with Bing Search retrieves **public web content**. The requirement is to retrieve information from **internal product sheets stored in Azure Blob Storage**, not from the internet.

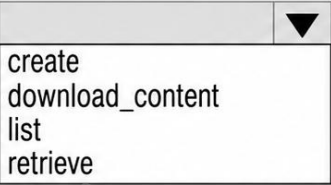
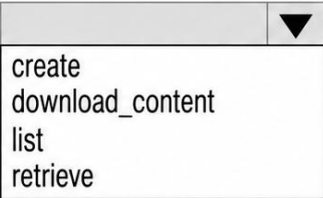
D. Azure Document Intelligence in Foundry Tools

Azure Document Intelligence extracts text and structure from documents, but it is **not** the retrieval engine used for RAG. While it can help process PDFs, the technical requirements specifically call for an indexing solution that supports **semantic and vector search**, which is provided by **Azure AI Search**.

Q173. You need to ensure that the marketing department can generate videos by using the model deployed to Project2.

How should you complete the Python code?

Answer Area

```
import time
from openai import OpenAI
client = OpenAI(
    base_url= "https://Contoso.openai.azure.com/openai/v1/",
    api_key=...,
)
video = client.videos.  (
    model=deployment_name,
    prompt= "A video of our products",
)
while video.status not in ["completed", "failed", "cancelled"]:
    time.sleep(20)
    video = client.videos.  (video.id)

    print(video.status)
```

Answer:

Answer Area

```
import time
from openai import OpenAI
client = OpenAI(
    base_url= "https://Contoso.openai.azure.com/openai/v1/",
    api_key=...,
)
video = client.videos.  (
    model=deployment_name,
    prompt= "A video of our products",
)
while video.status not in ["completed", "failed", "cancelled"]:
    time.sleep(20)
    video = client.videos.  (video.id)

    print(video.status)
```

Explanation:

Correct Answer:

create

To generate a new video using the deployed model, you must call the create method. This initializes the video generation job using the provided prompt and model deployment.

Retrieve

Video generation is an asynchronous operation. The retrieve method is used to check the status of the video generation job until it is completed, failed, or canceled.

Incorrect answers:

download_content

Used to download generated video content after completion, not to create or monitor the job.

list

Used to list existing video jobs, not to create or track a specific one.

retrieve (first dropdown)

Retrieve cannot be used to initiate video generation.

create (second dropdown)

Create is only used to start the job, not to poll or check status.

Q174. You need to configure personalized user interactions for Agent1. The solution must meet the business requirements.

What should you include in the solution?

- A. knowledge
- B. memory
- C. guardrails
- D. tools

Correct Answer:

B. memory

Explanation:

Correct Answer:

B. memory

Memory enables the agent to retain conversation context and recall relevant information from previous interactions. This directly satisfies the business requirement of providing personalized user experiences across sessions. By using memory, Agent1 can remember past preferences, previous queries, and context, allowing more relevant and tailored responses.

Incorrect answers:

A. knowledge

Knowledge refers to external data sources (like product documents), not personalization or remembering user interactions.

C. guardrails

Guardrails are used for safety and policy enforcement, not for personalization.

D. tools

Tools enable the agent to perform actions or call external services, but do not provide conversational memory or personalization.

Q175. You need to ensure that Agent1Dev Team can access Agent1. The solution must meet the security and compliance requirements.

How should you complete the Python code?

Answer Area

```

from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
from azure.core.credentials import AzureKeyCredential
myEndpoint = "https://contoso.services.ai.azure.com/api/projects/project1"
project_client = AIProjectClient(
    endpoint=myEndpoint,
    credential=
)
myAgent = "Agent1"
agent = project_client.agents. (agent_name=myAgent)

print(f "Retrieved agent: {agent.name}")

```

Answer:

Answer Area

```

from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
from azure.core.credentials import AzureKeyCredential
myEndpoint = "https://contoso.services.ai.azure.com/api/projects/project1"
project_client = AIProjectClient(
    endpoint=myEndpoint,
    credential=
)
myAgent = "Agent1"
agent = project_client.agents. (agent_name=myAgent)

print(f "Retrieved agent: {agent.name}")

```

Explanation:

Correct Answer:

credential = DefaultAzureCredential()

The security requirement explicitly states that API keys must NOT be used and authentication must be done using Microsoft Entra ID. DefaultAzureCredential() enables secure, identity-based authentication using Entra ID and follows best practices like managed identities, thereby meeting compliance requirements.

agent = project_client.agents.get(agent_name=myAgent)

To allow Agent1Dev Team to access Agent1, the code must retrieve the existing agent. The get method is used to fetch the agent by name, ensuring proper access without modifying or creating new versions.

Incorrect answers:

AzureKeyCredential()

Uses API keys, which violates the requirement that API keys must NOT be used.

None

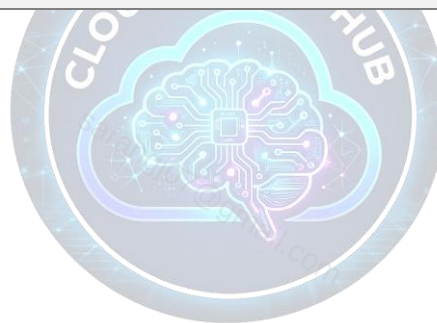
No authentication mechanism is provided, which is insecure and does not meet requirements.

create_version

Creates a new version of the agent, which is not required for accessing an existing agent.

get_version

Retrieves a specific version, not the agent itself, and is unnecessary for basic access.



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